What the map module needs from SettlementForge

As of 2026-10-18. The owner's document; its live copy is the Claude doc https://claude.ai/code/artifact/ce27743a-1e15-4f51-9c10-066d70a665bb (J391 (6)). Drafted by the workflow session/host-requirements (Opus/high readers, Fable/high drafters, a Fable/xhigh completeness critic; the critic's fourteen corrections and twenty-one additions applied; J390), reviewed and placed by the session. Sources and working files: docs/validation/host-requirements-2026-10-18/.

Scope

This document lists what SettlementForge must build, fix or supply so the map module's world, sites, lenses and encounter maps work as designed, as of 2026-10-18, the day the owner confirmed that world facts may flow to the host.

- **Who it is for.** The person or session working in the settlement-engine repository. Every item names the map-side source it comes from, so it can be checked there.
- **What "the host" means.** SettlementForge's engine: its campaign, settlement and simulation code. The map module reads it through dated receipts and exported functions and never changes it itself.
- **Which design state it reflects.** Design volumes V13 (world) and V14 (sites) and amendment files j through q as applied on 2026-10-17, plus the door-1 answer of 2026-10-18 (file r item 926). A Fable re-check of the 2026-10-17 work is running; any correction it makes will be carried into this document.
- **How to read it.** Sections 3 and 4 are the lists the host team acts on. Sections 5 and 6 go deeper on the two areas the owner asked about: patrols inside a territory, and encounter maps. Section 7 states the data rules every item obeys. Section 8 holds everything else worth knowing before the host work is scheduled.

The door-1 answer and what it lights

On 2026-10-18 the owner answered yes: the world canon stays outside door 1, so world-scale facts may reach SettlementForge's simulation while a town's own geometry still never does.

- **What door 1 is.** An early rule of the program: the host's simulator never reads the map's geometry, meaning no length, bearing, extent or coordinate of a settlement frame. That rule stands unchanged.
- **What the world canon is.** A per-settlement record derived from the generated world: travel effort between seats, water access and site facts. It carries no frame or arena geometry. It reaches the simulation only through the host's own receipted canonize: at a campaign's birth, through the additive re-canonize when the host binds a new seat later, or through a DM-confirmed re-canonize.
- **What the answer lights.** H5, the host's second builder of its frozen record over the canon, can now be built with milestone M-W1. H22, every new campaign on a world from this module canonized at birth, is no longer gated; until today the texts said new campaigns are born uncanonized. The canon-gated lanes therefore run from a campaign's first week: the crisis columns, the plague front and the routed supply links. Every lens's regime starts on the canon instead of the uncanonized regime.
- **What stays closed.** The settlement frame's geometry. The export bundle call keeps returning opaque blobs only.
- **Where it is recorded.** Amendments file r, item 926, records the answer and replaces every "until Q4 is answered" sentence in volume 13, the volumes index, the owner brief and the base design. A Fable seat checked the application (20 fragments across six files) and it is committed on the working branch.

The host-change list, by milestone

SettlementForge (the host) makes thirty changes, H1 to H30, in seven groups, and builds them in the numbered order (V13 §8.5, lines 3717–3718). One rule binds every row: a change starts switched off and switches on only when its data exists; absent data leaves the host's output byte-for-byte as before (V13 §8.5, lines 3718–3719). This is the design's only host-change list; the realm review's list and the proposal's older items are replaced by it (V13 §8.5, line 3717).

A few words used below, in plain terms. The *fork* is the host's own copy of the old map generator (V13 §8.5, H1, line 3739). The *canon* is the map's frozen geography, handed to the host once and only changed by a receipted *re-canonize* (V13 §11 Q4, lines 4301–4304). The *port* is the message channel between the map and the host (V13 §8.5, H4, line 3742). A *seat* is a settlement as the map sees it; an *advance* is one simulation time step; a *window* is the packet of facts the host sends after each advance (V13 §8.5, H9, line 3747). A *lens* is a themed map view (plague, sway, trade and so on), and the *DM* is the game master running a campaign (V13 §8.5, H23, line 3761).

Every row also carries, in the host's own files, its tier line, its degraded arm (what happens when data or hardware is missing) and its coupling row; its first-visible line; measured spreads or the word UNMEASURED for every numeric band it creates; and its felt sentence, what the player notices (V13 §8.5, lines 3718–3721). Every row lands as a lazy leaf (code loaded only when first needed) under the host's first-paint ratchet, the rule that the first screen may never load slower (V13 §8.5, line 3721).

What every row also obliges the host to add

The host's own constitution (its coherence fold's requirements 15 to 24 and its war program's laws) binds every row below as it binds every host wave. In plain words, each row also owes these, beside its code.

Obligation	What it is, in plain words	The host rule and pointer
A switch, and proof the switch is off	Every optional read lands behind a virtual flag (absent from the defaults, read only as strictly true) or behind a datum that is absent until written. The flag joins the host's gated-key manifest in the same commit as its first read, with a certification row, and the walker test asserts that one-key change. Four proofs ship with it: a golden captured before wiring, byte-identical with the flag absent; an absent-versus-false differential; a call-path spy proving the gated entry is never invoked dark; a source-scan census that every read is strictly true; plus the lit-mutant control. A display-only export (the lens window, H23) takes an exempt-keys rationale row instead of a flag.	research/sf/fp-simulator.md §3.1 (lines 644–665), §3.2 (666–687), §3.4 item 1 (line 709); host `src/domain/worldPulse/simulationRules.js:230`, `tests/lint/`; the data-gated `kmScale` precedent, `src/domain/spatial/spatialDigest.js:571-576`

Obligation	What it is, in plain words	The host rule and pointer
A lifecycle for every new stored field	Every field a row persists (H8's time-delta receipt; H13's strength band, force band and garrison presence if stored; H26's shipment path and departure week; H27's column route, carriage and toll; H28's share history; H29's band-change receipt; H10's world record) needs, before its writer is built: a canonical-model entry; exactly one writer; the ten lifecycle paths stated (create, read, persist, regenerate, undo, import, migrate, dark-after-lit, death, veil); a normalizer taught the field in the same commit; "absent, never null"; and a shrink-only source-scan census with an executed third-writer plant.	research/sf/fp-simulator.md §2.1 row 22 (line 472), §2.6 (613–641), §3.4 item 4 (line 721)
The news joins	Every news kind a row adds (caravan arrivals, starvations and entrepôt foundings in H26; abolition and founding news with the presser named in H28) registers its five Herald joins and carries an id, a full address chain, a typed action, the affected settlements by name and the recorded reason.	research/sf/fp-simulator.md §2.2 law 5 (line 495), §2.6 (line 633)
The borrow census and the vocabulary register	Every band a row mints (strength in fifths and culture distance in fifths in H13, the force band 0 to 4, the patrol basis words) passes the host's borrow census, no new ladder where an existing one serves, and enters its vocabulary register with a homonym check: "gate", "toll", "sway", "patrol", "territory", "fighting" and "prosperity" each keep one meaning per surface.	research/sf/fp-simulator.md §2.2 J-WR-10/10-B (line 515), §2.1 row 21 (line 471); V13 §5.8 (6) (line 3183)
A coupling row and a cause	Every cross-layer read lands its coupling-registry row in the same commit, and every map-originated event with downstream effects (a canonize receipt, a `physicalChange`, a route charter) is a named `causedBy` parent at mint so the cause walk reaches it.	research/sf/fp-simulator.md §2.6 (lines 627–636), §2.1 row 20 (line 470); host `src/domain/certification/couplingRegistry*.js`; V13 §8.5 row H16 (line 3754)
A fork row	Every new seeded choice registers in the habit-fork registry (the H24 plague offers and the H27 dispatch offer add draws).	research/sf/fp-simulator.md §2.6 (line 633); host `src/domain/worldPulse/habitForkRegistry.js:120`
The felt sentence and its latency	A mechanism's first-visible sign must surface within a one-month advance or carry a planting line; DONE means the DM-felt sentence fires in a lit preset on a seeded world, not a green fixture.	research/sf/fp-simulator.md §2.1 rows 16 and 19 (lines 466, 469); V13 §8.5 preamble (lines 3717–3721)

Now, on the fork

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H1	Save, with every map capture, the map's width, height, seed, kind, distance scale and unit; add one read-only command that returns the fork's full JSON export for the loaded snapshot; record the seed when a world is regenerated. The command should return the generator's Full JSON export, not a `.map` file: the Full JSON carries `cells.g`, the only link from the packed mesh back to the grid, which a `.map` omits (research/fmg/notes/fmg-data-model-and-io.md §7b, lines 341–347; V13 §2.8, lines 1323–1325). Host: the fork's capture, its export command, world regenerate. V13 §8.5 H1 (line 3739).	Without these saved, an old campaign's map can never be frozen and read by the map's import; H18 waits on this.	Nothing.	Transition, phase 0 (line 3739)
H2	Derive every number band that governs movers (travellers, armies, caravans) from the realm's measured spread of values, or justify it; the reinforcement band first. Host: its simulation's band constants. V13 §8.5 H2 (line 3740).	The absolute distance for new worlds (H3) and the scale stamp (H7) rest on this measurement; unmeasured bands would set a false scale.	Nothing.	Required (line 3740)
H3	Documents only: give the host's register rows that mirror this volume's wording; date the source of the world scale, so a map-module world declares its own scale and the old generator's scale is read only by the import; the absolute distance lit for new worlds waits on H2's per-map band census. Host: its register, its scale ruling (V13 §3.11.6 (9)). V13 §8.5 H3 (line 3741).	Keeps the host's written rules matching the map and gives each world exactly one source of scale.	Nothing; its distance bracket waits on H2.	Required, documents (line 3741)
H21	Supply what the map's audit could not see: the modified upstream source; the missing design documents (the reference-map weave volume, the map program's generation specification and fabric charter, the open-decisions register); the production frame size; other snapshot versions. V13 §8.5 H21 (line 3759).	Map decisions that rest on these stay unverified until they are read.	Nothing; any time.	Information request (line 3759)

The host can launch on its fork with H1 to H3 done, and the standalone world ships either way (V13 §11 Q2, lines 4290–4291).

One trap in the order. The host lights its own flags in dependency order as separate lighting units, only when a layer's instrument reads satisfied, and a flag lit out of order is an invalid configuration its convergence walker rejects; flag-minting commits serialize across the host's volumes, so a row here that mints a flag cannot be built in the same worktree at the same time as one of the host's own flag waves. Two rows sit inside that order: H12's genesis mount at world birth comes before the host's `routeLifecycleEnabled` and `demographicsEnabled` (its ruling OD-FC-18), and H22's birth canon lights every canon-gated lane at once on the host's birth defaults, which the host's population and plague reads call a behaviour switch to record with the owner (research/sf/fp-simulator.md §3.3 (lines 688–706), §3.1 CQ5 (line 660), §3.4 item 7 (line 730); research/sf/fp-physical-integration.md §6 row I41 (line 345)).

With M-W1 (the world)

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H4	Put the port's messages into the host client: removing a settlement is sent as `unbindSeat` (a "drop this seat" message), offered only to the premium re-canonize, never the additive one; the consumption law in code: no port call inside a simulation tick, answers stored with their time stamps, the client inside the host's own import scan for courts. Host: its client (V13 §3.11.6). V13 §8.5 H4 (line 3742).	Every row that talks to the map (H6 to H17, H23) goes through this channel.	M-SF; M-W1.	Required (line 3742)
H5	Build the host's frozen geography record a second way, from the map's canon, with the shape its readers see unchanged and every sub-record sourced or declared absent; add a conformance fixture (a test file that proves both builds agree); move cell-level readers to per-seat queries and points; make a satellite's site a world point; read each ground point's terrain class and step cost from the canon's sites slot. Host: its frozen-record builder. V13 §8.5 H5 (line 3743).	Without it the host keeps reading its own grid cells and never uses the map's geography; H6, H7, H11, H12, H15, H19 and H22 rest on it.	M-W1. Its other wait, V13 §11 Q4, was answered "yes" by the owner on 2026-10-18 (line 3743; §11 Q4, lines 4309–4315; file r item 926), so H5 is lit.	Required (line 3743)
H6	Write the host's travel-cost rule ("cost law 2") in this volume's schema; add a cost-units register row; stamp every native record with its cost-law version and the scale from the world header (valid only within one geometry version); add one coordinate adapter between the map's frame and the host's render frame. Host: cost law 2, its register, its render frame. V13 §8.5 H6 (line 3744).	Route costs are then stated in one schema and one unit, each record says which law and scale made it, and positions line up between the two frames.	M-W1; H4; H5.	Required (line 3744)
H7	Create a host-born world's seed at the host's storage layer and keep it with the campaign; the first canonize stamps the scale only after H2; until then the normalized arm (relative distances over the canon's nearest neighbours) is used, its calibration sent with the travel law (H8). Host: its store layer. V13 §8.5 H7 (line 3745).	The seed makes a world reproducible, and no absolute scale rests on unmeasured bands; H22 stamps its scale by this rule.	H2; M-W1; H4; H5.	Required (line 3745)

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H8	At every canon adopted (H22's birth canon and every re-canonize, the additive one included) send the travel law as data with its calibration in weeks per cost unit, normalized or absolute; send `canonized` per canon; store the time-delta receipt; freeze leg weeks for a migrated campaign. Host: its travel law. V13 §8.5 H8 (line 3746).	The map turns route costs into weeks only through this calibration; H22 waits on it.	M-W1; H4.	Required (line 3746)
H9	After every advance, undo and resume, send an events window: committed facts only, keyed by the advance nonce where the advance epoch is lit, else by pulse-history sequence, each fact with who may see it; the same audience rule applied to the conditions a frame draws. Host: its advance, undo and resume paths. V13 §8.5 H9 (line 3747).	The map's event layers and every lens read dated events here; H13 and H23 wait on it.	M-W1; H4.	Required (line 3747)
H10	Store a per-campaign world record beside the settlement map records (snapshot, log, backups, quota, lifecycle); build the players' world view from the party's picture set. Host: its campaign storage. V13 §8.5 H10 (line 3748).	The campaign's world needs a place to live, and players need a view limited to what their party has seen.	M-W1; H4.	Required (line 3748)
H11	Read the host's harbour verdict ("port verdict") from the canon's water access and return it as a view attribute; send its frozen per-settlement water verdicts as constraints; lay its water lanes over the canon's candidate lanes under its own law; add a lake vocabulary that admits salt and a climate-band version over the settlement's physical record. Host: port verdict, water lanes, lake vocabulary. V13 §8.5 H11 (line 3749).	Harbours and sea lanes then sit on the map's actual water, not on the host's own grid.	M-W1; M-W2; H4; H5.	Required (line 3749)
H14	For a settlement generated from a world charter (the map's facts for that spot), take its access word from the charter's road and water facts and its terrain word from relief, biome and water, by the inverse of DR1's relief-class row, before generation; retire the terrain-agreement notice, its role taken by `refuse.charter.dossier`; existing campaigns unchanged. Host: its settlement generator, DR1. V13 §8.5 H14 (line 3752).	A generated settlement then matches the ground it sits on; H16's flood exposure reads the terrain word this derives.	M-W1; M-W2; H4.	Required (line 3752)

H11 and H14 start with M-W1 and finish with M-W2; if the host's schedule slips past M-W1, the host side of the port is mocked and the standalone world ships without the host's settlements (V13 §8.4, line 3712).

With M-W2 (routes, polity, siting stage 2)

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H12	Lay the first road network ("genesis") at world birth over the canon, founding members placed first, sent as one frozen record even while dark; retire render-tier roads; send route changes and grades per window; run genesis when a member joins; make the charter command reachable from the world editor; check the quiet-quay rule for a decayed water lane; name the move-lock's release per seat. Host: its genesis mount, its world editor. V13 §8.5 H12 (line 3750).	The map's roads come from this genesis; H16, H19's road re-pointing and H22 depend on it.	M-W2; H4; H5; the host's genesis mount at world birth (its own schedule puts that after its release).	Required (line 3750)
H13	Send the political facts every window: each seat's tier, size third, prosperity rung, strength order (its own comparator, banded in fifths), force band and garrison presence; vassal links and treaty terms (tribute, compelled alliance, treaty-held occupation); its occupations, the ledger's and the stressor's records resolved to one; pairwise relations; held sites; standing claims; the per-link continuity verdict whenever its sovereignty market is lit (optional while that market is dark); satellites with parent, tier and site point; forced tributes; culture distance in fifths; water-lane state and adopted lanes; road closures and wardens' terms; the DM's christenings; optionally per-tick political deltas during an advance. Host: its relationship-state plane (J382; J384). V13 §8.5 H13 (line 3738).	The map's seat territory, sway, occupation, fighting and patrol layers draw from these, and so does the sway-and-patrols lens; H16 and H25 wait on it.	M-W2; H4; H9.	Required (line 3751)
H15	Hold world edits while an advance runs or is parked; apply a physical map edit (`physicalChange`) at the next tick head with the DM's name on it; send `consumed` back; adopt geometry shifts and lay versions only by a receipted re-canonize after a dry run of the canon's diff. Host: its advance loop, its re-canonize. V13 §8.5 H15 (line 3753).	The world never changes inside a running simulation step, and every geometry change carries a receipt.	M-W2; H4; H5.	Required (line 3753)
H16	Each part optional. Who keeps a gate is ruled (J385): the settlement whose territory a path crosses keeps it until the sway passes its 300-year soak test (T-V13-41 (d)); then the crossing's political holder is offered gate by gate, each with its own register row. Also: patrol coverage re-priced through one distance reader, read only by outcome rolls, and by any route choice only through the host's belief (its known world), the crossing holder likewise; sway as an advance-head overlay (fixed for one advance, at most one advance stale); a border segment as a hearing card's "where"; seasonal sets; route facts as overlays, with a water edge's limits on request; flood exposure from the terrain word H14 derives; aerial legs; canon causes as receipt parents; a hemisphere lag. The soak that unlocks the per-gate offer is the map's own test T-V13-41 (d), 300 simulated years under the B-POLITY corpus with the polity cells measured; it cannot run before M-W2 and its measurement, so plan the gate offer as the last of H16's parts, not the first (V13 §11, line 4317; §12, line 4355). V13 §8.5 H16 (line 3754).	Lets the map's patrol, sway and route facts reach the host's outcomes; until a part is adopted it stays a map-feed reading only.	M-W2; H12; H13.	Offer, each part optional (line 3754)
H17	Send presentation messages: movers and embattlement phases, projected for the audience; the DM's believed view where belief is recorded; optionally an audience's known political events. Host: its presentation layer. V13 §8.5 H17 (line 3755).	The map's mover and fighting layers, and its belief and audience inputs, draw from these.	M-W2; H4.	Required (line 3755)
H22	Every new campaign on a map-module world, whatever its tier, adopts the canon as it is created: scale stamped by H7's rule, roads laid at birth by H12, so the canon-gated lanes (crisis columns, the plague front, routed supply links) run from week one and every lens starts on the canon. A settlement founded later joins by an automatic, receipted *additive re-canonize* the host runs itself between advances, adding only the new seats and the territory and water they touch, every travel-band crossing reported, never silent. The host builds its record over the additive canon carrying forward, from the record it replaces, its own geometry, cost-law and overlay versions, the slots and sub-records that record holds, and the existing seats' port and teleport sets (as it already carries its scale forward), so the additive re-canonize adopts no host law, overlay, slot or other seat's port newer than that record; the derivations the added territories feed (all-pairs costs, tiers, gates, the calibration) re-run under the law carried forward and are reported in the time delta; the H5 conformance fixture holds a record so built to the prior record's version stamps. A seat's removal is never additive (`unbindSeat` goes to the premium re-canonize, H4); everything not yet adopted is marked `W_HOST_CANON_BEHIND` (V13 §3.11.6, lines 2331–2356). Existing campaigns untouched. V13 §8.5 H22 (line 3760).	Without it new campaigns are born uncanonized, lenses open uncanonized, and a later settlement stays off the map until the DM pays to re-canonize.	H5; H7; H8; H12. Its other wait, V13 §11 Q4, was answered "yes" by the owner on 2026-10-18 (line 3760; §11, lines 4326–4335). Lands on the owner's schedule (V13 §11 Q2).	**Product ruling the host implements, not an offer** (J386; J387; J388); the premium canonize entitlement never gates it (lines 3729–3730, 3760)

H22 is for new campaigns only, a declared shift; until it lands a lens opens in its uncanonized regime, so no lens waits on it (V13 §8.5, lines 3732–3733).

Where H22's premium gate lives today. The host's canonize body is tier-blind and lazily loaded (`src/store/campaignSpatialCanonize.js:2-16`); the three synchronous gates are read at the store call site: in-flight, entitlement (`if (state.auth?.tier !== 'premium') return { ok: false, reason: 'not_entitled' }`, `src/store/campaignWorldPulseSlice.js:359-391`) and generated-map (today only a generated reference-map pack is admitted). H22's birth canon and additive re-canonize must pass around the entitlement gate and replace the generated-map gate with "a canon from the map module"; `unbindSeat` (H4), the geometry-shifting re-canonize, a retcon and H19's rebase keep the entitlement gate. Campaigns are themselves created only under the premium tier or by staff

(`src/store/campaignSlice.js:586`; docs/PROGRESS.md row J389), so the split bites staff accounts and any future lower tier. Two entitlement questions no source answers: whether the lens window (H23) is itself a premium surface ("the simulator is premium", research/sf/fp-simulator.md §2.1 row 17, line 467), and whether H28's command adopting a DM-placed holy site is gated like a DM authoring act (today no deity exists until a premium DM authors one).

With M-W3 (labels, exports, the adapter)

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H18	Phase 1: freeze every saved map's capture with its export, on the owner's next open and by a headless background batch to a published deadline, public gallery rows first; each verdict confirmed, absent with a witness, or changed; a legacy re-canonize reads the stored capture; satellites' cells translated once to world points under a confirmed verdict. Host: its saved maps, its gallery. V13 §8.5 H18 (line 3756).	Every old map must be in a form the map's import can read before the fork retires (H20).	M-W3 (the 1.114 adapter); H1.	Transition, phase 1 (line 3756)

M-W3 itself carries the retirement of the host's reference fork on the host's track (V13 §8.2, line 3678).

With M-W5 (the lenses)

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H23	After every advance, undo and resume, beside the events window, send the lens window (`lensWindow`): the ledger rows the host already keeps (plague, faith, economy, rites, supply, travellers, transfers, columns, head counts, its alignment reading), each with who may see it; where it chooses, values its own laws compute, each labelled with its law; its resolved regime with every window; the interior weeks of a multi-week advance or a per-week delta log; once, at first bind, its kept pulse records; and in the events window (H9) the dated events each lens reads. Host: its ledgers, its regime. V13 §8.5 H23 (line 3761).	Every lens reads its data here; H24 to H30 wait on it. No lens waits on it to open.	M-W5 (its first step); H4; H9.	Export, changes no simulated byte (line 3761)
H24	Exports: the source seat's key on each plague record and arrival event; a bounded front history, one receipt per hop; the road season read in its weeks-away display. Offers: an arrived plague writes its condition where it lands, with its loop capped as the famine loop is; a settlement missing from the frozen record read as "no spatial delay known", never one week; the ledger's plague obligation mapped to its one plague type; each crisis served as transfers or as columns, never both. V13 §8.5 H24 (line 3762).	The plague lens shows where a plague came from and how it spread; the offers close gaps the review found.	H23.	Export plus offers the host may decline (lines 3724–3728, 3762)
H25	Exports: strength order in the polity window (H13) and, where it chooses, the stronger side per adversarial pair; tier changes as dated facts; a DM-cut road's closer named; the sea-lanes switch copy corrected; "same governing name" defined. Offer: a raid's outcome word and, where known, its place, the word that makes a raid a verdict (V13 §3.9 (11)). Not asked (the owner's decision, 2026-10-18): a host meaning for patrol coverage — patrols stay a map-only picture for now, and the host's own programme may give the coverage a meaning later, when its route estate lights or OD-FC-2 is decided (H16's optional door stays open). Also: whether a DM-confirmed war front can embattle a settlement with the war layer dark. V13 §8.5 H25 (line 3763).	The sway-and-patrols lens needs real outcomes; patrol coverage means nothing to the host today, and by the owner's decision of 2026-10-18 the host is not asked to give it one now.	H13; H16; H23.	Export plus one offer (lines 3724–3728, 3763)
H26	Store a shipment's path and departure week; emit caravan arrivals, starvations and entrepôt foundings as news; record each seizure at a route's worst intermediary settlement (gate, good, take) as a dated DM-only events-window row, not world news; put settlement keys on purchase news (the prosperity lens reads them too). Host: its trade, its news. V13 §8.5 H26 (line 3764).	The trade lens draws shipments, caravans and seizures from stored paths and dated rows.	H23.	Export, changes no simulated byte (line 3764)
H27	Exports: store each column's (migrant group's) route, carriage (land, sea lane or circle) and toll (departures, deaths at origin and on the road); name the origin on arrival rows; keep each population change's reason. Offer: dispatch columns when a queued emigration, a terminal death or a forced abandonment is approved. Host: its population columns. V13 §8.5 H27 (line 3765).	The populations lens draws who moved, how, and at what cost.	H23.	Export plus one offer (lines 3724–3728, 3765)
H28	Put the deity key and the strongest carrier's source on each faith-front mint and conversion receipt, and retire a mint when its carrier ends; keep a share history, or a dated share-change receipt, per settlement pantheon; send prescription rows (presser, target, weight) and the abolition and founding news of moral institutions with the presser named; add a host command adopting a DM-placed holy or hidden cult site (V14 §2.2 `devotion`; J384 (10)). V13 §8.5 H28 (line 3753).	The religion lens draws where faiths spread from and how their shares changed.	H23; V14 §11 Q2 (its site record).	Export, changes no simulated byte (line 3766)
H29	Export: a band-change receipt (settlement, from, to, writer, week), a boom's sustained rungs included. Offers: a distance term in the resettlement donor pool, or the honest wording "the realm's richest"; the founding band frozen for town drawing (today the town layout and the district wealth word re-read the live band). Host: its prosperity bands, resettlement, town drawing. V13 §8.5 H29 (line 3767).	The prosperity lens draws dated band changes.	H23.	Export plus offers (lines 3724–3728, 3767)
H30	Export: a per-advance alignment reading in the lens window (both axes and their bands by its own law, the aggression chip word, the top causes), published as its faith projection is; a week on each tolerance offset's causes. Offer: fix the moral drift's compounding decay and its freeze outside belief worlds, a defect the alignment read found. Host: its alignment, its moral drift. V13 §8.5 H30 (line 3768).	The alignment lens draws per-advance readings; lenses scrub stored plates and replay nothing (V13 §5.8 (8)).	H23.	Export plus one offer (lines 3724–3728, 3768)

Within H23 to H30 the exports (the lens window, keys, receipts, news, stored paths) change no simulated byte; each offer is a declared shift for new campaigns or sits behind its own rules flag, and no lens waits on any of them (V13 §8.5, lines 3723–3729).

After M-W3

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
H19	Phase 4: an elective, premium, receipted rebase per campaign with its forecast diff shown first (V13 §11 Q1); share codes gain a geography-source axis; custom-backdrop campaigns get an elective trace-a-world path; the host's map screens re-pointed (the territory lens stays on seat territory; sway drawn by a new "allegiance" lens; roads to the map feed's paths; travel rings to its own bands; travellers to the overlay; the charter road card to its edge's path). V13 §8.5 H19 (line 3757).	An existing campaign can move onto the map's geography by choice, and the host's screens then read the map's data.	M-W3; H5; H6; H18.	Transition, phase 4 (line 3757)
H20	Phase 5: retire the fork, its content-security policy, origin, hash manifests and validation gate; keep blobs and exports read-only for a stated retention; keep every geometry-1 record forever. Host: the fork. V13 §8.5 H20 (line 3758).	Removes the old generator once every map is frozen.	H18 (at H18's deadline).	Transition, phase 5 (line 3758)

The transition for existing campaigns

Existing campaigns move over in phases 0 to 5; phases 0 to 3 change no byte of a campaign's simulated state (V13 §8.5, lines 3733–3735). The rows above already hold the host's phases; this table is the reading order of the move.

#	What the host changes (plain words, then the host program or law)	Why the map needs it	Waits on	Kind
Phase 0	H1: save map size, seed and export on the fork (line 3739).	The import later needs these to freeze an old map.	Nothing.	Transition (line 3739)
Phase 1	H18: freeze every saved map's capture with its export (line 3756).	Every old map readable by the import before the fork goes.	M-W3; H1.	Transition (line 3756)
Phases 2–3	Nothing in the host; these phases are the map's. The map imports only on request, never on load; a legacy campaign's world view writes nothing to its simulated state, its overlays converted through the stored capture and the import's transform (lines 3770–3771).	An old campaign can be shown on the map without its simulation ever changing.	H18's stored captures.	Map's own work (lines 3770–3771)
Phase 4	H19: the elective, premium, receipted rebase per campaign (line 3757).	Moves a living campaign onto the map's geography only by the DM's choice.	M-W3; H5; H6; H18.	Transition (line 3757)
Phase 5	H20: retire the fork (line 3758).	The old generator is gone once nothing depends on it.	H18.	Transition (line 3758)

The schedule of this whole track against M-W1 to M-W3 and M-W5 is the owner's open question (V13 §11 Q2, lines 4286–4292; §8.5, lines 3771–3772).

The milestones named above

Milestone	What it delivers
M-SF	The SettlementForge connection (SF0 to SF7): the host client that the port, the per-campaign charter and the host's seats ride on; the world milestones follow it (V13 §8.1, lines 3662–3664).
M-W0	Seven pass-or-fail feasibility tests (SP-W1 to SP-W7) before M-W1, after M0 closes, with an eighth (SP-W8, the canon's budget) in §8.4; 3 person-weeks (V13 §8.2, line 3674; §8.4, line 3713).
M-W0h	The water state at the local map (surfaces, groundwater, water class, the two edit verbs and their cascade); no host dependency; about 4 person-weeks (V13 §8.1, lines 3664–3667; §8.2, line 3675).
M-W1	The world: terrain, climate, hydrology, biomes, resources and first-pass siting; the charter both ways; the port's canon, charter, proposals, cost and travel laws, seat binding, events windows; the world dialog and placement tool; 14 person-weeks (V13 §8.2, line 3676).
M-W2	Routes under the host's cost law; polity with the patrol step and border-post proposals; second-pass siting; the port's routes, map feed, patrol offer, border segment, polity windows and presentation inputs; the territory, sway, occupation, fighting, patrol and event layers; 8 person-weeks (V13 §8.2, line 3677).
M-W3	Labels and their placement; the four exporters; the map-import adapter with its schema file; retirement of the host's reference fork on the host's track; 6 person-weeks (V13 §8.2, line 3678).
M-W5	The lens mechanism and the port's lens window, then each lens in order: plague; sway and patrols; trade and populations; religion; prosperity and alignment; after M-W2, beside M-W3; 7 person-weeks (V13 §8.2, line 3679).
M-W4	Hardening: budgets measured, memory test, low-tier refusal, migration of older frames at scale, clean-room lint; every T-V13 test green nightly; 3 person-weeks (V13 §8.2, line 3680). M-W0 to M-W5 together are one staged row of 37 to 45 person-weeks (lines 3682–3683).

Host defects found by the system reads

Six host systems were read whole: faith, alignment, trade, population, prosperity and plague. The reads found 262 host-side items: 100 defects, 48 contradictions between a document and the code, 59 stale documents or comments, and 55 open questions. The finding that matters most: a freshly born campaign carries the default preset's name but none of its switches. So nothing described below as "lit in the default preset" runs until a DM applies a preset by hand (trade-2, prosperity-C14 and trade-OQ5; host `src/domain/worldPulse/simulationRules.js:31`, `:886-892`, `:1552-1558`).

Every row below comes from the closing sections of the six research notes under `research/sf/`, each read whole in the line ranges the notes give. Nothing was run; where a note says a finding is inferred or plausible until executed, the row says so. Where two reads found the same defect it appears under each system with a cross-reference, so the counts include those pairs twice (alignment-30 and trade-39; faith-17 and alignment-31a; faith-9.2 and alignment-17e; population-10.1-6 and prosperity-C9).

The "what the host should do" column reports only what the note recommends. Where the note names no fix, the cell says so; this document adds no remedy of its own. The last column gives the kind and the source line in the note.

Words used in the tables

- **Host**: SettlementForge, the app that runs the world simulation. The map module reads its facts.
- **Preset**: one of the host's seven shipped rule sets. RR is Realistic Regional, the default. The others are QL Quiet Local, DC Dramatic Campaign, SC Static Campaign, NC Narrative Campaign, LR Living Realm and FS Full Simulation.
- **Lit / dark**: a feature switched on / built but switched off.
- **Canon, canonized**: the DM has frozen the map geography so the simulation can read travel data. The **digest** is that frozen travel table, with territories and **gates** (the cheapest crossing between two territories). Many movement features run only on canonized worlds.
- **Tick**: one simulated week. **Same-tick** or **teleport**: people or effects move instantly, with no journey.
- **Patron**: the god a settlement seats as its chief deity. **Pantheon**: the set of gods a settlement holds.
- **Faith front**: a confirmed link along which faith pressure moves between two towns, drawn as a purple dashed line on the host map. **Carrier**: any link (diplomatic, garrison, trade or war) that carries faith pressure.
- **Spread lane / local lane**: faith conversion between towns, which needs a switch, versus faith change inside one town, which always runs.
- **Column**: a refugee or migrant group that walks a route over weeks (the host's M4 system). **Netted transfer**: an instant move of a number of people.
- **Steading**: a small satellite farmstead seeded by a parent town.
- **Band / rung**: prosperity is a seven-word ladder from Subsistence upward; a rung is one step.
- **Plague front**: the system that moves plague between mapped towns along trade and shipment links, with delays.
- **Embattlement**: the host's record of how fought-over a place is. It drives road danger.
- **Lens**: a dated map layer the map module plans (territory, religion, plague, patrols).
- **Header / comment**: explanatory text inside a code file. **Receipt**: a record the simulation writes to explain what happened in a tick.

- **Ruling** (OD-FC-n, J-n): a numbered decision by the owner recorded in the design documents.
- **Kinds**: *defect* means the code does something wrong or nothing. *Contradiction* means a document, comment or screen disagrees with the code and the note does not say which is right. *Doc fix* means the code is right and the text is stale. *Open question* means a decision nobody has made.

Which host change answers which defect

The lens asks H24 to H30 were written from these reads; this cross-index says which rows close which defects, so the host can see what a row buys.

H-row	The defect or open question it answers
H24 plague	plague-9-9 and plague-10.2-5 (a front history the prune destroys); plague-10.1-17 (one-week hops for unmapped settlements); plague-10.1-1 and plague-10.2-4 (the condition gap when a plague arrives); plague-10.1-25 (the uncapped loop); plague-10.1-23 and plague-10.2-7 (the ledger's second plague); population-10.2-4 (a crisis served twice); plague-10.1-13 (weeks-away ignores the season)
H25 sway and patrols	alignment-27 and alignment-25f ("same governing name"); trade-38 (the sea-lanes switch copy); trade-7 (a DM-cut road stops no caravan); trade-OQ4 and trade-OQ6 (patrol coverage's host meaning, left to the host's own programme by the owner's decision of 2026-10-18; the unowned wardenship slot); trade-OQ7 (a war front with the war layer dark); J384 (3) (the strength order)
H26 trade	trade-5 and trade-OQ3 (no stored path or departure tick); trade-36 (arrivals and starvations are not news); prosperity-D9 (purchase news names no settlement)
H27 populations	population-10.2-2 (a DM-approved exodus never travels); population-10.1-1 (road tolls dropped); population-10.2-5 (the chosen route versus the stamped time)
H28 religion	faith-25b, faith-25d and faith-OQ3 (deity key and source on a mint; fronts never retired); faith-OQ9 and J384 (10) (holy sites); alignment-22 (prescription "converts"); alignment-21 and alignment-26 (institution founding and abolition)
H29 prosperity	prosperity-D8 and prosperity-9-12 (no band history); population-10.1-6 and prosperity-C9 (the realm-wide donor pool); prosperity-9-15 and prosperity-Q-b (the town plan re-reads the live band)
H30 alignment	alignment-25b (a per-pulse export); alignment-9a, alignment-9b and alignment-25e (drift decay)
No row yet	population-10.2-13 (the host timelapse reads population deltas as a map, so the map's scrub cannot reuse its frames until it is fixed); trade-2 and trade-OQ5 (the null birth door, which the regime banner of every lens depends on); prosperity-D1 (the prosperity sort)

Faith

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
faith-1	The religion design still describes a "chief" deity and a 1.15 hold multiplier. The code uses patron fields and resists 0.35 of each downward step. It also adds legitimacy, a sink, patron falls, suppression ticks, a push margin and a dead band against flicker. Host `docs/RELIGION_REWORK.md:93-101`, `:204-207` versus `src/domain/worldPulse/religionState.js:45-85` and `worldState.js:272-280`.	Anyone building from the document, the map included, gets wrong field names and wrong numbers.	Bring the document up to the code.	doc fix; faith-system.md §10.1 item 1 (line 1285)
faith-2	The document ranks deities by number of believers. The code counts only the settlements that seat the deity as patron. Host `docs/RELIGION_REWORK.md` §2.7 versus the code cited in the note's §4.2.	A legend ordered by "number of believers" would show a number the host never computes.	Correct the document to "patron seats".	doc fix; faith-system.md §10.1 item 2 (line 1289)
faith-3	The document says the leading deity is the top share among the top three. The code takes the top share overall; the top three appear only in the contest. Host `docs/RELIGION_REWORK.md:32-34`.	Small; a reader expects a rule the code does not run.	Correct the document.	doc fix; faith-system.md §10.1 item 3 (line 1290)
faith-4	The document derives a deity's niche from a stored temperament field. The code derives temper from the deity's two axes, and two code comments still say "stored verbatim". Host `docs/RELIGION_REWORK.md:29`, `:37`; `src/domain/worldPulse/cultImpositionApply.js:22-26`; `src/domain/worldPulse/religiousContest.js:224`.	A reader would trust a stored value that no engine read uses.	Fix the document and the two comments.	doc fix; faith-system.md §10.1 item 4 (line 1292)
faith-5	Six places still say every world is born with a pantheon. Generation no longer creates one; the latent pantheon was deleted. Host `docs/RELIGION_REWORK.md:180-185`; `docs/PHASE4_FAITH_DELTA.md` §4; `src/components/compendium/PantheonActivationStrip.jsx:6-9`; `src/copy/deityAuthoring.js:74`, `:105-106`; `src/components/settlement/faithPanelModel.js:12-14`; `src/domain/display/publicSafe.js:73-75`; the header of supabase migration 128.	Users and developers are told a fresh world has gods when it has none. A faith lens on a born world is empty.	Remove the claim from the listed files.	doc fix; faith-system.md §10.1 item 5 (line 1294)
faith-6	The religion document presents carriers, prevalence and occupation entry as the whole model, without saying they run only in the spread lane. The Phase 4 document says piety is absent (it is built) and puts realm-wide deity politics in the spread lane, while the ranking advances on local activity. Host `docs/RELIGION_REWORK.md` §2.4-§2.6; `docs/PHASE4_FAITH_DELTA.md` §1.3, §1.9; `src/domain/worldPulse/pulseKernel.js:1704`.	A reader would think spread runs everywhere and that piety does not exist.	Mark those passages spread-only and update §1.3 and §1.9.	doc fix; faith-system.md §10.1 item 6 (line 1298)
faith-7	Two comments say faith never crosses a settlement boundary unless spread is on. In the default preset, faith prescription, the fracture spread, tolerance drift, faith-weighted relief and (on canon) faith-weighted contraband all cross boundaries. The per-campaign Living-World switch can light spread itself. Host `src/domain/worldPulse/simulationRules.js:90-100`; `src/domain/worldPulse/religiousContest.js:24-33`.	A map built on the comment would draw the default world as having no faith pressure between towns.	Correct both comments; the note's §6.1, §6.10, §6.11, §7.4 and §7.10 carry the evidence.	contradiction; faith-system.md §10.1 item 7 (line 1301)
faith-8	A comment says the fracture spreads along the religious-authority channel; the code declares three channels. "Conversion spreads along religious_authority" confuses the drawn faith front (a record) with the creed's real path. Host `src/domain/worldPulse/religiousContest.js:10-12`.	The map would draw the wrong link type as the path of a schism or conversion.	Correct the comment.	doc fix; faith-system.md §10.1 item 8 (line 1305)
faith-9	The spatial design says conversion inherits both distance and travel delay; the code applies distance only. The couplings document's "faith spreads at carrier reach" omits the distance weakening altogether. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md` §3; `docs/DESIGN_FP_COUPLINGS.md`; `src/domain/worldPulse/religiousContest.js:365-376`.	A map showing faith arriving "after N weeks" would invent a delay the host does not have.	Correct both documents to "distance only".	doc fix; faith-system.md §10.1 item 9 (line 1308)
faith-10	Three faith files say they are unused; they are wired. `faithField.js` says nothing calls it (called since F4c); `customContentSchema.js` says "no mechanical consumer"; `faithWitnessSource.js` says its funnel is absent though the catalog admits it. Host `faithField.js` header (folder not given in the note); `customContentSchema.js:241-242`, `:297-298`; `faithWitnessSource.js:16-30`.	Developers may delete or ignore live code.	Correct the three headers.	doc fix; faith-system.md §10.1 item 10 (line 1310)
faith-11a	Three design documents say a patron can fall for five causes; the code has four. Host `docs/DESIGN_FP_ARCH_WF.md:270`, `:418`; `docs/DESIGN_FP_FAITH.md` J-WF-5 `:1498`; `docs/DESIGN_FP_ARCHITECTURE.md:1991`; `src/domain/worldPulse/patronFall.js:76`.	A legend of fall causes would list one that never happens.	Correct the documents to four.	doc fix; faith-system.md §10.1 item 11 (line 1313)
faith-11b	The faith design says a Schism realm arc exists; it does not. Host `docs/DESIGN_FP_FAITH.md:1327`.	A time scrub would look for a dated event the host never makes.	Correct the document.	doc fix; faith-system.md §10.1 item 11 (line 1314)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
faith-11c	The owner's ruling OD-FC-8 renamed "God's Abandonment"; the old text still ships.	Players see wording the owner ruled out.	Apply OD-FC-8 (listed again as awaiting application in the note's §10.3).	contradiction; faith-system.md §10.1 item 11 (line 1314)
faith-11d	The Ascendancy headline says the faithful walk the roads; nothing in the code moves them.	Players read about travellers who are not simulated. An encounter map would have no host fact to show.	The note records the gap; no fix given.	contradiction; faith-system.md §10.1 item 11 (line 1315)
faith-12	Ruling OD-FC-7 lights believed devotion in three presets; it is not applied at revision 81f1cd96.	The player-side "what people believe about faith" stays dark where the owner said it should be lit.	Apply OD-FC-7.	contradiction; faith-system.md §10.1 item 12 (line 1316)
faith-13	Which lane carries pilgrims and legates is recorded three ways. The faith design says WF-2 (the designed pilgrim feature) rides SPREAD plus SP-1; the WF architecture says SPREAD plus the errand spine; a slice brief put pilgrims in the Wayfare document, which never mentions them. Host `docs/DESIGN_FP_FAITH.md` §3; `docs/DESIGN_FP_ARCH_WF.md:190`; `docs/DESIGN_FP_ARCH_WY.md`; `src/domain/worldPulse/envoyErrandVocabulary.js:322-335`.	Encounter maps that place pilgrims depend on which system owns them; today the documents disagree.	Reconcile the documents on one lane and one home.	contradiction; faith-system.md §10.1 item 13 (line 1317)
faith-14a	The WF architecture says the pilgrimage_right grant cannot form without WF-2; the draft lens drafts it with no gate. Host `docs/DESIGN_FP_ARCH_WF.md:499`.	A peace or treaty view could offer a grant the engine cannot produce.	Gate the draft on WF-2.	contradiction; faith-system.md §10.1 item 14 (line 1320)
faith-14b	The grammar document said missionary_access "moves believer-share bands"; the faith design says it is a gate, never a share. Already clarified at `docs/DESIGN_FP_GRAMMAR.md:868`.	Settled; listed for completeness.	Nothing further.	doc fix; faith-system.md §10.1 item 14 (line 1321)
faith-14c	Under the one-finder law the GR-6 temple arm can never be reached. Host `docs/DESIGN_FP_GRAMMAR.md:1290`.	Designed behaviour that cannot happen.	The note records it; no fix given.	defect; faith-system.md §10.1 item 14 (line 1322)
faith-15	A ruling says one writer family owns a value; the traditions nudge still writes it. Host `docs/DESIGN_FP_COHERENCE_FOLD.md:267`.	Two writers can make a value drift silently.	Apply the code cure the fold says is owed.	defect; faith-system.md §10.1 item 15 (line 1323)
faith-16	The tithe has four sources, ruled to be one flow with two ends. The static tithe written at generation is covered by no ruling (note §7.4).	An undecided money flow; a map showing tithe would not know which source is true.	Rule on the generation-time tithe.	open question; faith-system.md §10.1 item 16 (line 1324)
faith-17	A normaliser comment about the "full" information mode is stale. Same lines as alignment-31a.	Misleads a reader about whether beliefs run in FS.	Correct the comment.	doc fix; faith-system.md §10.1 item 17 (line 1326)
faith-18a	The WF architecture (R4) and the couplings document cite old line numbers for the kernel mounts, the carriers and mediation. Host `docs/DESIGN_FP_ARCH_WF.md` R4; `docs/DESIGN_FP_COUPLINGS.md`.	Readers land on the wrong code.	Refresh the line citations.	doc fix; faith-system.md §10.1 item 18 (line 1327)
faith-18b	The spatial design says the Living-World gate writes only the legacy key; since CL-0 it writes both keys. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md:1588-1589`.	Misleads anyone deciding which flag to read.	Correct the document.	doc fix; faith-system.md §10.1 item 18 (line 1328)
faith-18c	The regional graph header says active propagation is for a future tier; it already happens (note §7.10.3). Host `regionalGraph.js:19-22`.	Readers think the graph is passive.	Correct the header.	doc fix; faith-system.md §10.1 item 18 (line 1329)
faith-19	Turning off the "War & faith" layer leaves the faith fronts drawn, because they belong to the Relationships layer. The Herald's Faith door offers no map-lens chip. Host `src/components/map/WarFaithMapOverlay.jsx:1-26`; `src/components/map/LayersPanel.jsx:238-242`; `src/components/map/HeraldBody.jsx:282-291`.	The switch's name promises control the user does not get.	Owner question (faith-OQ8): should the switch govern the faith front.	contradiction; faith-system.md §10.2 item 19 (line 1333)
faith-20	The pantheon panel's strength bar uses the earned ranking tier. The engine converts at the higher of authored and earned tier. So an authored-major deity with one seat shows as a 35% cult but converts at 0.95. Host `src/components/map/PantheonPanel.jsx:185-186`, `:216`.	Wrong number on screen; a lens copying the panel would repeat it.	Display the tier the engine uses (the fix the note implies).	defect; faith-system.md §10.2 item 20 (line 1334)
faith-21	The edge style comment says the faith front lies along an allied or trade edge; it can lie along a war front. Host `src/components/map/relationshipEdgeStyle.js:60-61`.	Misleads the map designer about what the purple line means.	Correct the comment.	doc fix; faith-system.md §10.2 item 21 (line 1337)
faith-22	The pantheon activation strip's "dynamics" milestone demands the legacy spread flag, though the local lane and the ranking need no flag. Host `src/components/compendium/PantheonActivationStrip.jsx:45-49`.	The DM is told dynamics are off when local dynamics run.	The note records it; no fix given.	defect; faith-system.md §10.2 item 22 (line 1338)
faith-23	The "patron-less town" text (DS-FTH-2) is mounted only for a premium viewer with no embed, who gets no Faith tab. The desk's header admits an untested direction. Host `src/components/new/tabs/WarFaithDesk.jsx:272-292`, `:32-43`; `FaithTab.jsx:227`, `:284-309`; `src/components/OutputContainer.jsx:511-514`.	Authored content nobody sees.	The note records it; no fix given.	defect; faith-system.md §10.2 item 23 (line 1340)
faith-24a	The assign-deity screen says the war verbs are coming soon; they are not. Host `src/components/map/AssignDeityFromMap.jsx:14-16`.	A stale promise to users.	Correct the copy.	doc fix; faith-system.md §10.2 item 24 (line 1343)
faith-24b	One component says the faith tab hides itself; another says no door does. Host `src/components/map/PantheonPanel.jsx:9-11`; `RealmInspector.jsx:83-87`.	Developers cannot tell which behaviour is intended.	Reconcile the two headers.	contradiction; faith-system.md §10.2 item 24 (line 1343)
faith-24c	The faith panel falls back to a "portfolio" field that deity snapshots no longer carry. Host `src/components/settlement/faithPanelModel.js`.	The fallback silently yields nothing.	The note records it; no fix given.	defect; faith-system.md §10.2 item 24 (line 1344)
faith-25a	A cult imposed by the DM after seeding is never admitted to the pantheon (note §3.11).	A DM action appears to do nothing.	Owner to decide whether intended (faith-OQ10).	defect; faith-system.md §10.2 item 25 (line 1346)
faith-25b	A conversion's receipt names the convert as its own source (note §6.7).	The map cannot say where a conversion came from.	Put the strongest carrier source on the receipt (the host ask in the note's §9.6 item 5).	defect; faith-system.md §10.2 item 25 (line 1346)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
faith-25c	When a DM re-assigns a patron, no win or loss is banked to the ranking (note §3.6).	The global ranking misses DM-made changes.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1347)
faith-25d	Once a faith front is minted it stays, even after its carrier ends (note §6.3).	The map shows old fronts as if live; a lens must style them as historic.	Retire fronts when the carrier ends (host ask in the note's §9.6 item 5; owner question faith-OQ3).	defect; faith-system.md §10.2 item 25 (line 1347)
faith-25e	War-front links carry faith at 0.72, stronger than any friendly link (note §6.2).	Enemies convert each other fastest; a map drawing faith along friendly links would be wrong.	Owner question (faith-OQ4).	defect; faith-system.md §10.2 item 25 (line 1348)
faith-25f	The neighbour finder counts any link, including rival, criminal and dormant ones, as a prevalence neighbour (note §6.5).	Faith prevalence leaks along links that should not carry it.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1348)
faith-25g	The spread lane's reach and the separate faith-reach builder disagree; one is gated, one is not (note §6.11, §9.5 item 3).	Two "reach" numbers for the same deity; a lens must pick one and say which.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1349)
faith-25h	History text still uses generic deity names and named Christian orders (note §2.5).	Breaks the setting's own deity names.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1350)
faith-25i	Clearing a patron leaves the settlement's pantheon and ledgers stale (note §3.11).	The map would show a faith mix for a town with no patron.	Owner question (faith-OQ7): should a clear empty the pantheon.	defect; faith-system.md §10.2 item 25 (line 1350)
faith-25j	A demotion leaves a pantheon holding more deities than its tier allows (note §3.11).	More deities shown than the tier allows.	Owner question (faith-OQ7): should a demotion trim it.	defect; faith-system.md §10.2 item 25 (line 1351)
faith-25k	Rival-cult blame names cults the pantheon never admitted (note §3.11).	Receipts name a cult the town does not hold.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1352)
faith-25l	A suggested discovery candidate could pin a faith front at "suggested" status; the note marks this inferred (note §6.3).	A front could stay unconfirmed and undrawn.	The note records it as inferred; no fix given.	defect; faith-system.md §10.2 item 25 (line 1352)
faith-25m	The realm dashboard's "Dominant faith" tile names the deity by the tail of its id, not its name (note §7.10.11). Host `src/components/map/RealmDashboard.jsx:65-77`, `:368-371`.	Users see a code string instead of a god's name.	The note records it; no fix given.	defect; faith-system.md §10.2 item 25 (line 1353)
faith-9.1	The Wayfare document ruled the deployment arrows deleted; the "War & faith" overlay still draws them. Host `docs/DESIGN_FP_ARCH_WY.md:56-58`, `:1653-1660`; `src/components/map/WarFaithMapOverlay.jsx:98-105`.	The world map draws something the owner ruled out.	Apply the Wayfare ruling.	defect; faith-system.md §9.1 (line 1130)
faith-9.2	Town buildings reserve a patron look, but both alignments are hard-coded to 0.5 and the emblem comes from the scene, not any deity. Host `src/domain/townScene/buildingProfiles.js:331-336`; `src/domain/townMap/arch/params.js:18-31`. Same seam as alignment-17e.	Town maps cannot show a temple's god.	The note records it; the town worship ladder is designed only and ruled out of FP.	defect; faith-system.md §9.2 (line 1155)
faith-9.6-7	Pilgrims, missionaries and inquisitors are not simulated as travellers. The only pilgrim maths is a deity-blind lift that discards who came; "missionize" is a trust nudge; WF-2's season and named movers are unbuilt and their errand spine is dark.	Faith encounters on an encounter map would be invention until WF-2 lands.	Build WF-2 if faith travellers are wanted on encounter maps; until then the map shows none.	open question; faith-system.md §9.6 item 7 (line 1260)
faith-OQ1	No deity exists until a premium DM authors one (FPQ-62).	Every faith lens is empty on fresh worlds.	Owner to decide whether a new world is born with any deity.	open question; faith-system.md §10.3 (line 1360)
faith-OQ2	Faith spread is off in the default preset unless a DM lights it per campaign.	Without it, conversion between settlements does not exist in the preset worlds are born into.	Owner to decide whether the default preset turns on spread.	open question; faith-system.md §10.3 (line 1361)
faith-OQ3	Faith fronts carry no deity id and no source, and are never retired.	Decides whether a lens can draw present fronts or only historic ones, and whether it can name a conversion's source.	Owner and host to decide; the note's §9.6 item 5 frames the host ask.	open question; faith-system.md §10.3 (line 1363); §9.6 item 5 (line 1251)
faith-OQ4	Hostile relationships carry faith at 0.72, the strongest two-way carrier (an overlord's authority over its vassal reaches 0.82, one way; see faith-25e).	Shapes every faith-pressure map.	Owner to decide.	open question; faith-system.md §10.3 (line 1365)
faith-OQ5	Only the parked ripple and moral founding read a season; realm piety is not size- or distance-weighted. A sea route as a cheap faith carrier would be new host simulation (note §9.6 item 9).	Decides whether faith movement can ever follow geography.	Owner and host to decide.	open question; faith-system.md §10.3 (line 1366); §9.6 item 9 (line 1269)
faith-OQ6	The map profile's religious-hierarchy output (religious-superior and pilgrimage-centre links) says "the map should render" it; nothing consumes it.	A host fact the map is asked to draw and does not.	Owner to decide whether the map module's lens renders it.	open question; faith-system.md §10.3 (line 1368); §9.3 (line 1181)
faith-OQ7	Should clearing a patron clear the pantheon, and should a demotion trim it (faith-25i, faith-25j)?	Stale faith mixes on the map.	Owner and host to decide.	open question; faith-system.md §10.3 (line 1370)
faith-OQ8	Fronts are DM-only by default and dropped from published maps (see faith-19).	Decides the audience and switch behaviour of the purple lines.	Owner to decide whether "War & faith" governs the front and whether players ever see DM-only fronts.	open question; faith-system.md §10.3 (line 1372)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
faith-OQ9	The host keeps no pilgrim catchment and no positioned holy sites; settlement labels exist. Positioned sites would be map-side facts needing an owner ruling and host adoption (note §9.6 item 6).	Decides whether encounter and religion maps may show pilgrim areas and shrines.	Owner to decide.	open question; faith-system.md §10.3 (line 1373); §9.6 item 6 (line 1254)
faith-OQ10	A cult imposed after seeding is never admitted (see faith-25a).	A DM action appears to do nothing.	Owner to decide whether intended.	open question; faith-system.md §10.3 (line 1375)
faith-OQ11	Two rulings await application: OD-FC-7 (believed devotion) and OD-FC-8 (the rename). See faith-12 and faith-11c.	Ruled behaviour not shipped.	Apply both.	open question; faith-system.md §10.3 (line 1376)

Alignment

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
alignment-1	Headers say the alignment reads have no live consumer "this wave"; about thirty modules import them, several lit in the default preset. Host `src/domain/worldPulse/settlementAlignment.js:19-21`; `src/domain/worldPulse/disposition.js:365-366`, `:530`, `:562`; `src/domain/spatial/moralDrift.js:27-30`; `src/domain/worldPulse/pulseKernel.js:2294-2295`.	Developers may change alignment thinking nothing depends on it.	Correct the headers; the grep list is in the note's §11.3.	doc fix; alignment-system.md §10 item 1 (line 962)
alignment-2	The disposition header says "three blended terms" and lists four, and says the deity term reads the retired temperament field. The code reads the derived temper. Host `src/domain/worldPulse/disposition.js:9-27`, `:208-212` versus `src/domain/worldPulse/deityAxes.js:26-33`.	Misleads anyone tracing how a town's character is computed.	Correct the header.	doc fix; alignment-system.md §10 item 2 (line 966)
alignment-3	Each NPC's weight is importance times the governing faction's power times 1.5. The last two factors are the same for every NPC and cancel in the average. So the seat of power does not colour the town more than a back-bench guild. Host `src/domain/worldPulse/disposition.js:151-181`, `:418-435`.	A town's alignment ignores who rules it; a map "why" card would mis-explain the colour.	The note records it as a no-op; no fix given.	defect; alignment-system.md §10 item 3 (line 969)
alignment-4	NPCs with no scored word are skipped, so a single scored NPC sets the whole conscience or aggression term. Host `src/domain/worldPulse/disposition.js:171`, `:428`.	A hamlet can read "malicious" from one person; a colour-by-region map would show it as fact.	The note calls it a design question, not a bug.	open question; alignment-system.md §10 item 4 (line 972)
alignment-5a	The cohesion design says the faith quadrant comes from alignment closeness via culture distance; the owner's ruling 3 says faith war is distance across both axes. The code reads only the difference in the two patrons' evil score, so "schism" can never occur. Host `docs/DESIGN_COHESION_WEAVE.md:25-28`; `docs/DESIGN_REALM_DIRECTIVES.md:166-172`; `src/domain/worldPulse/sacredClaim.js:121`; `src/domain/worldPulse/generosityKernel.js:333`.	A religion lens must draw three classes, not four; the documents and the ruling describe something the code does not run.	Owner to decide which reading is meant (alignment-25d).	contradiction; alignment-system.md §10 item 5 (line 974)
alignment-5b	Two towns with no gods read as "respectable rivals" in generosity, which scales every conscience term by 0.95 on a born world. The sacred-claim code guards against this; generosity does not. Host `src/domain/spatial/cohesionWeave.js:53`; `src/domain/worldPulse/generosityKernel.js:414`, `:700`; `src/domain/worldPulse/sacredClaim.js:39-49`.	A silent 5% damping of every relief decision on fresh worlds.	The note records it; the guard in `sacredClaim.js` is the model it points to.	defect; alignment-system.md §10 item 5 (line 977)
alignment-6	The refuse threshold is 0.12 times hysteresis divided by the gate. Natural enemies (gate 0.55) need a larger withhold (0.218) to refuse aid, and "brothers" (1.25) a smaller one (0.096); the comment says the opposite. Bites only with seated deities. Host `src/domain/spatial/generosityEV.js:479-494`; `src/domain/spatial/cohesionWeave.js:49-54`.	Relief arcs on the map go the wrong way between enemy and allied faiths.	The note records it as inverted; no fix given.	defect; alignment-system.md §10 item 6 (line 980)
alignment-7	Alignment and faith read the governing seat as ruler; generosity takes the highest-power faction. The faith code measured that as the wrong faction on 36.7% of fresh settlements. Host `src/domain/worldPulse/generosityKernel.js:303-313`; `src/domain/worldPulse/religionLegitimacy.js:226-238`.	Two systems disagree on who rules; a map naming "the ruler" would contradict one of them.	The note records it; no fix given.	defect; alignment-system.md §10 item 7 (line 984)
alignment-8	The economic-base lens's three source keys have no writer, so every settlement reads "mixed". Host `src/domain/spatial/cohesionWeave.js:258-269`.	A cohesion input that never varies.	The note records it; no fix given.	defect; alignment-system.md §10 item 8 (line 987)
alignment-9a	Each tick the already-faded drift is multiplied again by 0.92 to the power of all ticks since the last act, so fading compounds. Half-life is about 3.6 weeks instead of about 8.3; gone after 8 silent weeks instead of about 32 (measured by running the host function). Host `src/domain/spatial/moralDrift.js:218-224`, `:251-254`.	A town's bad deeds vanish from its character in two months; a time scrub replaying the designed curve would be wrong.	Owner question (alignment-25e): keep the value at the last act so it does not compound.	defect; alignment-system.md §10 item 9 (line 989)
alignment-9b	The drift ledger advances only when beliefs are active or a web-war atrocity happened this tick. Only RR among raiding presets is omniscient. So in RR on canon with war lit, a raid-born drift does not fade between raids. It stays forever if war is turned off. Host `src/domain/worldPulse/pulseKernel.js:2296`, `:2341-2350`; `src/domain/worldPulse/beliefMap.js:204-209`; `src/domain/worldPulse/simulationRules.js:953`, `:996`, `:1073`, `:1404-1408`, `:720-730`.	A permanent stain from one raid in the default preset.	Owner question (alignment-25e): should drift decay every tick regardless of the belief gate.	defect; alignment-system.md §10 item 9 (line 998)
alignment-10	The spatial design says a deceived good actor drifts toward the manipulator's alignment; the code drifts every unjust instigator toward malice and lawlessness. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md:514-517`; `src/domain/spatial/moralDrift.js:145-156`.	Document and code disagree on direction.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 10 (line 1006)
alignment-11	The spine requires a town's character to move through NPC alignment, roster and institutions. The derived read never reads the NPC alignment word or institutions, and moral drift is a separate accumulator the spine itself calls a defect. Host `docs/DESIGN_FP_SPINE.md:84-92`; `src/domain/worldPulse/disposition.js:112-128`, `:568-593`; `src/domain/worldPulse/moralMartialLean.js:143-163`.	The design's own rule is broken; edits a DM makes to NPC alignment do not show on the map.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 11 (line 1009)
alignment-12	Corruption is listed as an alignment input but is not read (note §3.6).	A "why" card would cite a cause that has no effect.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 12 (line 1015)
alignment-13	Three ways compute a town's character (the W0 axes, the faith conduct plane, and the war seat books' fallback), and four archetype tables disagree; nothing reconciles them, though requirement 13 names W0 as the read. Host `src/domain/worldPulse/warSeatBooks.js:298-300`.	Different systems see different characters; a fourth on the map side would make it worse.	The note records it; the map must call the host selector, not reimplement (note §9.3 item 14).	defect; alignment-system.md §10 item 13 (line 1016)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
alignment-14	The peace press, the quadrant and fidelity noise read patron axes. The spine requires the town's own two-axis read, and the selector header promises a widening ("Wave A") not done. Host `src/domain/worldPulse/peaceTermsAppraisal.js:86-104`; `src/domain/worldPulse/fidelityNoise.js:65-72`; `src/domain/worldPulse/settlementAlignment.js:20-21`; `docs/DESIGN_FP_SPINE.md:79-88`.	A godless town's character never reaches these systems.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 14 (line 1019)
alignment-15	The war-convenience document calls the lawfulness price in war termination and momentum "ALREADY LIVE". The momentum call is live in DC and FS on canon; the war-termination call is dark. Host `docs/DESIGN_WAR_CONVENIENCE_AND_TRIBUTE.md:646-651`; `src/domain/worldPulse/momentum.js:1265`; `src/domain/worldPulse/warTermination.js:745`, `:144`.	Overstates what runs.	Correct the document.	doc fix; alignment-system.md §10 item 15 (line 1023)
alignment-16	No lit engine reads the seeded NPC alignment word; its vocabulary lacks chaotic_good though towns and deities can be chaotic-good. Host `src/domain/worldPulse/npcAgency.js:161-170`; `src/domain/npc/characterConsumers.js:518-524`.	Authored data with no effect and a missing value.	The note records it; no fix given.	defect; alignment-system.md §10 item 16 (line 1027)
alignment-17a	Army and fleet route risk is passed as null but presented as a mechanism (note §5-§7).	Armies and fleets do not choose routes by danger; a patrol or encounter map cannot show alignment-shaped army routes.	The note records it as a seam (a reserved slot with nothing behind it).	defect; alignment-system.md §10 item 17 (line 1030)
alignment-17b	The good giver's charity exception is hard-coded off.	Designed behaviour that never fires.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1031)
alignment-17c	The triage function is never called.	Designed behaviour that never fires.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1031)
alignment-17d	Nothing reads the humanitarian evidence door.	Designed behaviour that never fires.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1031)
alignment-17e	Town statues are hard-coded 0.5, read an unwritten `settlement.moralDrift`, and their note sources patron alignment from the deleted latent pantheon. Host `src/domain/townMap/arch/params.js:16-30`. Same seam as faith-9.2.	Town maps cannot show a town's moral character in its statues.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1031)
alignment-17f	NPC replacement reads `settlement.alignment` / `config.alignment`, which nothing writes, so every lean is neutral.	Replacement NPCs never match the town.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1033)
alignment-17g	The faction-alignment read for NPC circulation exists; no caller supplies it.	NPC circulation ignores faction character.	The note records it as a seam.	defect; alignment-system.md §10 item 17 (line 1034)
alignment-18	The conquest-intent module says its bands are copied from the war seat books. The seat books say "merciful"; the intent and razing lanes say "benevolent". Both lanes dark. Host `src/domain/worldPulse/conquestIntent.js:64-71`; `src/domain/worldPulse/warSeatBooks.js:321-326`.	Matching on the word would fail when lit.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 18 (line 1036)
alignment-19	The paradigm catalog says it has no consumers; four modules import it. Host `src/domain/npc/paradigmAxisCatalog.js:4-9`, `:78`.	Misleads developers.	Correct the header.	doc fix; alignment-system.md §10 item 19 (line 1039)
alignment-20	The deity temperament axis is still authored and saved but no engine read uses it; stored and derived values can diverge silently. Host `src/domain/worldPulse/deityAxes.js:26-33`; `docs/SETTLEMENT_CAPABILITY_ATLAS.md:9502`.	A DM can author a value with no effect.	The note records it; no fix given.	defect; alignment-system.md §10 item 20 (line 1041)
alignment-21	Founding severity peaks at 0.64, below the 0.78 proposal threshold, so a founding always applies itself even when major changes require proposals. Abolition reaches the threshold only at 0.95 or more. Host `src/domain/worldPulse/moralInstitutionPressure.js:347`, `:361`, `:508`, `:527`.	The DM's "ask me first" setting does not cover foundings.	The note leaves open whether this is intended.	open question; alignment-system.md §10 item 21 (line 1043)
alignment-22	"Converts" in faith prescription implies converted towns; the code presses every carrier-linked neighbour (note §5.1).	A map label using the host's word would mislead.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 22 (line 1047)
alignment-23	The coherence fold says the migration menu is empty because nothing writes the route network. The M4 dispatch ranks destinations by the digest's path cost, not the route network. Host `docs/DESIGN_FP_COHERENCE_FOLD.md:339`; `src/domain/worldPulse/migrationKernel.js:469-474`.	Unclear whether refugees move; matters to any map of population movement.	Open question for the population seat whether the fold's sentence covers M4.	open question; alignment-system.md §10 item 23 (line 1048)
alignment-24	Word clashes found in the note's §2.7 are not in the fold's §5 register. Host `docs/DESIGN_FP_COHERENCE_FOLD.md` §5.	Same word, different meanings, unrecorded.	Add them to the register.	doc fix; alignment-system.md §10 item 24 (line 1052)
alignment-25b	Nothing about alignment is stored, so a time scrub needs either a map-side snapshot or a host export: both axes, bands and top causes per pulse, published like the faith projection (note §9.3 item 13).	Without it, the map must cache its own history.	Owner and host to decide whether the host exports a dated per-pulse alignment digest.	open question; alignment-system.md §10 item 25(b) (line 1054); §9.3 item 13 (line 939)
alignment-25d	Is the quadrant meant to read town alignment or patron axes, one axis or both (see alignment-5a)?	Decides how many classes a religion lens draws.	Owner to decide.	open question; alignment-system.md §10 item 25(d) (line 1056)
alignment-25e	Should moral drift decay every tick, and without compounding (see alignment-9a, 9b)?	Decides how long a town's deeds stain its character.	Owner to decide.	open question; alignment-system.md §10 item 25(e) (line 1057)
alignment-25f	Should the culture identity term compare a real overlord (see alignment-27)?	Decides whether vassals feel closer to their overlord on territory maps.	Owner to decide.	open question; alignment-system.md §10 item 25(f) (line 1058)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
alignment-25g	Is the born roster's malicious and warlike lean intended (see alignment-29)?	Decides whether fresh worlds should look tilted.	Owner to decide.	open question; alignment-system.md §10 item 25(g) (line 1059)
alignment-26	The pressure function reads the name-matched lean only, while the loop and the embargo read the stamped lean. So an institution with an unmatched name can never be abolished. Host `src/domain/worldPulse/moralInstitutionPressure.js:119-121`, `:308-315`; `src/domain/worldPulse/institutionTolerance.js:104-110`, `:119-136`.	Permanent institutions by accident.	The note records it; no fix given.	defect; alignment-system.md §10 item 26 (line 1060)
alignment-27	The culture "same sovereign" header says the term checks the same overlord and that conquest flips the label. In code the governing name is a government-type label at generation and an "occupier occupation authority" string after conquest. Host `src/domain/spatial/cultureDistance.js:20-22`; `src/generators/power/rulingStructure.js:251-290`, `:783-792`; `src/domain/worldPulse/warDeployment.js:857`; `src/domain/rulingPower.js:665`.	No realm-wide closing between vassals; a territory map would wrongly expect conquest to make towns culturally closer.	Fix the header; owner to decide whether the term should compare a real overlord (alignment-25f).	contradiction; alignment-system.md §10 item 27 (line 1063)
alignment-28	The roads module gives a bold traveller risk tolerance 0.9 and a coward 0.35, and uses it correctly to decide refusal. It passes the same number to the route chooser, where it multiplies the danger surcharge. So the bold traveller avoids danger most and the coward least. The return leg uses the frozen value too. Lit with roads on canon (RR, DC, LR, FS). Host `src/domain/roads/state.js:248-253`, `:334-343`; `src/domain/worldPulse/roadsKernel.js:1010-1018`, `:506`; `src/domain/spatial/embattlement.js:307-320`, `:347-357`, `:407-423`.	Named travellers' routes on any encounter map would be backwards.	The note records it as inferred unintended; no fix given.	defect; alignment-system.md §10 item 28 (line 1068)
alignment-29	Trait tables score more vices than virtues, and the slot roll keeps the flaw 70% of the time (note §3.5).	Every fresh world's character map looks tilted.	Design question, not a code bug (alignment-25g).	open question; alignment-system.md §10 item 29 (line 1075)
alignment-30	The slave contraband ban has no catalog good. Host `src/domain/spatial/smuggle.js:136-138`; `src/domain/region/goodsCatalog.js:130-245`. Same as trade-39.	A "bans slave cargo" gate glyph would show a rule that almost never fires.	The note records it; no fix given.	defect; alignment-system.md §10 item 30 (line 1077)
alignment-31a	A comment says the "full" information mode falls back to omniscient; the mode reader accepts "full", so FS beliefs are live. Host `src/domain/worldPulse/simulationRules.js:1509-1511` versus `:1404-1408`.	Misleads anyone deciding whether beliefs run in FS.	Correct the comment.	doc fix; alignment-system.md §10 item 31 (line 1079)
alignment-31b	The military strength header says it is mounted nowhere; eight modules call it. Host `src/domain/worldPulse/militaryStrength.js:21-22`.	Misleads developers about patrol and war strength reads.	Correct the header.	doc fix; alignment-system.md §10 item 31 (line 1081)
alignment-32	The twenty-year decay of disposition history belongs only to the dark WR-2 arm, so as lit the history never fades. Host `src/domain/worldPulse/dispositionLedger.js:604-607`.	Old grudges never fade in lit presets.	The note records it; no fix given.	contradiction; alignment-system.md §10 item 32 (line 1082)

Trade

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
trade-1	The trade design's original sentence says commodity flow (M6a, the finite, conserved, tolled trade model) is lit only in FS. A correction bracket on the same line says RR and FS, absent from the birth door, which matches the code. Host `docs/DESIGN_FP_TRADE.md:200`; `src/domain/worldPulse/simulationRules.js:896`, `:1081`.	Stale only for a reader who skips the bracket.	Nothing further; the bracket already corrects it.	doc fix; trade-system.md §10.1 item 1 (line 988)
trade-2	The birth door is null: a plainly born campaign carries the default preset's id without the switches that preset turns on. Host `src/domain/worldPulse/simulationRules.js:31`, `:1552-1558`.	Notes and maps that say "lit in RR" are wrong for a freshly born campaign until a preset is applied.	See trade-OQ5: whether the birth door will be lit.	contradiction; trade-system.md §10.1 item 2 (line 992)
trade-3	The coherence fold's "TR: LIT -" line is right for FP, but M2, M6a-M6d, M7, generosity and boom/bust also run in lit presets on canon. Host `docs/DESIGN_FP_COHERENCE_FOLD.md:340`.	Readers underestimate what moves.	Correct the status line.	doc fix; trade-system.md §10.1 item 3 (line 995)
trade-4	The trade tally's header says it never feeds the engine; upswing, pestilence and the urban fabric read it. Host `src/domain/spatial/tradeFlow.js:13-18`.	A developer could change it believing nothing depends on it.	Correct the header.	doc fix; trade-system.md §10.1 item 4 (line 997)
trade-5	The spatial design says caravans are invisible; the Wayfare design makes them a map sub-layer; the record cannot support positions (no path, no departure tick). Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md:76-80`; `docs/DESIGN_FP_ARCH_WY.md:111-131`, `:734`.	An encounter map cannot place a caravan on a road today.	Store a path and departure tick (WY-3) if positions are wanted; see trade-OQ3.	contradiction; trade-system.md §10.1 item 5 (line 999)
trade-6	Two designs treat siege, blockade, embargo and interception as one model. In code only war fronts and blockades cut caravans, embargo paths are modifiers, tariffs do not exist, and web-war embargo is a recorded pressure. Host `docs/DESIGN_SUPPLY_WEB_WARFARE.md` §4; `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md` §7.	A map drawing an embargo as a cut road would be wrong.	The note records it; no fix given.	contradiction; trade-system.md §10.1 item 6 (line 1001)
trade-7	The spatial design promises a DM can force a road that bypasses gates. Caravans read only digest gates, sea lanes and teleport edges; a DM-chartered route adds only a neutral neighbour link, and a DM-cut route stops no caravan. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md:54-55`.	A DM placing or closing a road on the map changes no traffic; patrols on such a road guard nothing.	The note records it; the map legend must say so (note §9 item 20).	contradiction; trade-system.md §10.1 item 7 (line 1004)
trade-8	The design says water routes have enforced high capacity. Sea-lane capacity is enforced only for naval convoys; land has no capacity at all. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md` §4j.	Overstates a limit.	Correct the document.	doc fix; trade-system.md §10.1 item 8 (line 1008)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
trade-9	"Shipments have no quantity" is true of M2 only; M6a records carry the load. Host `docs/DESIGN_FP_COHERENCE_FOLD.md` (WC row); also the map module's `research/sf/fp-simulator.md:413`.	Readers think quantities are unavailable.	Correct both texts.	doc fix; trade-system.md §10.1 item 9 (line 1010)
trade-10	The trade design says truth-side scarcity derives from physical stocks; SP-B's ground truth is the generation lists plus the food ratio. Host `docs/DESIGN_FP_TRADE.md` T1.	A "scarcity" map would not know which truth it shows.	The note records it; no fix given.	contradiction; trade-system.md §10.1 item 10 (line 1012)
trade-11a	The owner reversed the "no coin" rule (T2, W-COIN, dark); the headers of `treatyTransfer.js` and `generosityEV.js` still state it.	Misleads developers.	Correct the headers.	doc fix; trade-system.md §10.1 item 11 (line 1013)
trade-11b	The entrepot toll has "no owner" in code while EC-6 designs tolls into the treasury.	An undecided money flow.	The note records it; no fix given.	open question; trade-system.md §10.1 item 11 (line 1014)
trade-12	Food is the flagship good but is never shipped; EC's dossier sentences feature salt caravans and sea-borne salt. Host `docs/DESIGN_FP_ARCH_EC.md:2261`, `:2268`, `:2310`.	A food-flow arrow would draw something the host does not move.	The note records it; no fix given.	contradiction; trade-system.md §10.1 item 12 (line 1015)
trade-13	FS, "the ceiling" preset, declares route lifecycle and brokerage off; sovereignty, brokerage and the reputation race are dark everywhere. Host `src/domain/worldPulse/simulationRules.js:1119`, `:1153`.	The fullest preset is not full.	The note records it; no fix given.	contradiction; trade-system.md §10.1 item 13 (line 1017)
trade-14	M6a adds raw route weeks to the pulse tick, where M2 divides by the tick span. Harmless today because every committed advance runs one-week ticks; a direct coarse call would expose it. Host `src/domain/spatial/commodityFlow.js:783` versus `src/domain/spatial/supplyShipments.js:389-399`; `src/domain/worldPulse/advanceInterval.js:97-105`, `:511`.	Arrival times would be wrong if the host ever ran month-long ticks.	The note records it as latent; no fix given.	defect; trade-system.md §10.2 item 14 (line 1022)
trade-15	A town with a single supplier of a critical input can never starve; two- and three-sourced links can. The starvation delay is receipt text only. Host `src/domain/spatial/commodityFlow.js:927-932`.	The most fragile supply line is the safest; a map of fragile links would be inverted.	The note records it; no fix given.	defect; trade-system.md §10.2 item 15 (line 1024)
trade-16	Despite its comment, the shortage band does not feed the impairment; only total starvation does. Host `src/domain/spatial/commodityFlow.js:51-56`; `src/domain/worldPulse/supplyKernel.js:532-563`.	Shortage on the map has no consequence in the town.	The note records it; no fix given.	defect; trade-system.md §10.2 item 16 (line 1025)
trade-17	The "no smuggle context" branch, which would cut M6a trade at a hostile gate, is unreachable because the kernel always builds the context. Host `src/domain/worldPulse/supplyKernel.js:476`, `:486`.	A patrol or hostile gate on the map never stops a caravan outright.	The note records it; no fix given.	defect; trade-system.md §10.2 item 17 (line 1027)
trade-18	The ally-relief override is documented but never returned. Host `src/domain/spatial/dispatchEV.js:65-68`; `src/domain/worldPulse/supplyKernel.js:668`.	Allies do not get the promised priority.	The note records it; no fix given.	defect; trade-system.md §10.2 item 18 (line 1029)
trade-19	Supply-web warfare reads buffers M6a never writes, so in RR and FS buffer thinness reads the default. Its veto prose names only DC and FS though RR and LR carry the wave once war is lit. Host `src/domain/worldPulse/supplyWebWarfare.js:340-341`; `src/domain/worldPulse/supplyKernel.js:495-497`.	Raid targeting ignores real stock levels.	The note records it; no fix given.	defect; trade-system.md §10.2 item 19 (line 1030)
trade-20	A trade-war flip re-points a channel, but caravan origins ignore channels and the prize. Discovery proposes partners at any distance while caravans use only the three nearest producers. Host `src/domain/worldPulse/supplyKernel.js:155-180`.	A trade-war prize on the map may be contested by a supplier that never ships to the buyer.	The note records it; no fix given.	defect; trade-system.md §10.2 item 20 (line 1033)
trade-21	Route risk for relief shipments is computed with the hostile-gate term fixed false. Host `src/domain/worldPulse/generosityKernel.js:691`.	Relief cannot be intercepted; an encounter map could show no relief convoy at risk.	The note records it; no fix given.	defect; trade-system.md §10.2 item 21 (line 1036)
trade-22	The tally tests "is a port" by route words or a harbour in the roster; routing tests it by the digest's eligible ports. Host `src/domain/spatial/tradeFlow.js:112-140`; `src/domain/spatial/distanceRead.js:299-316`.	A town can be a port for one and not the other.	The note records it; map footholds should use the digest's ports (note §9 item 7).	contradiction; trade-system.md §10.2 item 22 (line 1037)
trade-23	The brokerage distance surcharge is always zero. Host `src/domain/worldPulse/brokerageServicesRules.js:276`.	Distance never prices brokerage.	The note records it; no fix given.	defect; trade-system.md §10.2 item 23 (line 1039)
trade-24	"Caravan" is a traveller kind in the lived network, but the lived network's genesis and flow accrual have no caller, so every consumer reads empty. Caravans route on the digest instead.	Any encounter or road-use map built on the lived network shows nothing.	The note records it; no fix given.	defect; trade-system.md §10.2 item 24 (line 1040)
trade-25	Fuzzy goods matching mislabels producers and goods: bakeries index as iron producers; "Toll income (transit)" becomes a shippable good; "Medicinal herbs" becomes a service, so plague demand never ships; "Scholarly services" becomes financial services (note §2.2).	Wrong goods move on the trade map.	The note records it; no fix given.	defect; trade-system.md §10.2 item 25 (line 1042)
trade-26	The resource reconcile updates chains, exports and local production but never imports, so a depleted mine never gains its shortage import. Host `src/domain/worldPulse/resourceDynamicsKernel.js:624-680`.	Depletion never creates new trade.	The note records it; no fix given.	defect; trade-system.md §10.2 item 26 (line 1047)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
trade-27	Per-tier export probabilities draw randomness and gate income rows but never reach the export list; the Trade Dynamics panel presents them as the tier's goods. Host `src/generators/economy/economicState.js:802`, `:808`.	The panel shows goods the town does not trade.	The note records it; no fix given.	defect; trade-system.md §10.2 item 27 (line 1049)
trade-28	Exportable entrepot outputs survive without the "(transit)" mark, and the cap of 12 truncates transit first. Host `src/generators/computeActiveChains.js:744-797`.	Pass-through goods look like local goods.	The note records it; no fix given.	defect; trade-system.md §10.2 item 28 (line 1052)
trade-29	Three definitions of entrepot and three sets of isolation tiers exist. Host `src/generators/economy/tradeGoods.js:54-56`; `src/generators/computeActiveChains.js:689-692`, `:806-818`; `src/generators/economy/customTradeEndpointIntegration.js:100-110`.	A town can be a trade hub by one test and not another.	The note records it; no fix given.	contradiction; trade-system.md §10.2 item 29 (line 1054)
trade-30	Three encodings of supply-chain dependency disagree: data upstream chains, the hard-coded cascade and upstream import labels. Host `src/generators/computeActiveChains.js:458-492`, `:838-852`.	Which chain breaks first depends on which encoding is read.	The note records it; no fix given.	contradiction; trade-system.md §10.2 item 30 (line 1056)
trade-31	A mountain pass is honoured by capacity, food and connectivity but yields road-identical goods; the import-capacity table has no mountain_pass or coastal row. Host `src/generators/foodGenerator.js:182-189` versus `:309-318`.	Pass towns look like road towns in trade.	The note records it; no fix given.	defect; trade-system.md §10.2 item 31 (line 1058)
trade-32	The prosperity ladder has six labels at generation and seven at runtime. Host `src/data/constants.js:92-94`; `src/domain/autonomy/signalRegistry.js:76`.	A prosperity legend may miss "Subsistence".	The note records it; no fix given.	contradiction; trade-system.md §10.2 item 32 (line 1060)
trade-33	"Has local salt" is true when salt is a necessity import, the opposite of its name. Host `src/generators/economy/tradeGoods.js:123-129`.	Salt-preserved goods are stripped by the wrong test (they are discarded anyway).	The note records it; no fix given.	defect; trade-system.md §10.2 item 33 (line 1062)
trade-34	The service gate `requiredTradeRoute` is set by no data row. Host `src/generators/services/institutionServices.js:178`.	A dead gate.	The note records it; no fix given.	defect; trade-system.md §10.2 item 34 (line 1064)
trade-35	The chain status "captured" is produced nowhere. Host `src/domain/supplyChainState.js:161-164`.	A map state that can never appear.	The note records it; no fix given.	defect; trade-system.md §10.2 item 35 (line 1065)
trade-36	Arrivals, starvations, "emboldened" receipts and entrepot foundings are returned and then dropped; none becomes news.	The map's time scrub cannot replay them; it must snapshot per pulse (note §9 item 15).	The note records it; no fix given.	defect; trade-system.md §10.2 item 36 (line 1066)
trade-37	The host map's "Supply chains" layer is a fourth unrelated derivation of six resource-and-consumer pairs; it draws a grain chain although grain never ships. Host `src/lib/supplyChains.js:11-60`, `:125-150`.	The existing world map already shows a flow that does not happen.	The note records it; no fix given.	contradiction; trade-system.md §10.2 item 37 (line 1067)
trade-38	The "Sea lanes" switch text says it governs trade. Caravans use sea lanes by digest data regardless; the switch governs naval war and blockade only. Host `src/components/map/SimulationRulesAxes.jsx:370`; `src/domain/spatial/distanceRead.js:286-292`.	A DM turning sea lanes off still has sea trade.	Correct the copy.	doc fix; trade-system.md §10.2 item 38 (line 1069)
trade-39	No catalog good has the slave category. So the rule fires only when both ends' slave labels slug to the same custom good (inferred, unprobed). Host `src/domain/spatial/smuggle.js:143-148`; `src/domain/region/goodsCatalog.js:466-476`. Same as alignment-30.	A patrol or gate glyph for it would show a rule, not traffic.	The note records it; no fix given.	defect; trade-system.md §10.2 item 39 (line 1072)
trade-40	Route danger starts only above 0.55 after security relief (at most 0.25). A war-front siege (0.85) always clears it; an occupation only while contested or unstable; war exhaustion only with high crime or the trade-hub premium; crime plus premium never. The tuning comment says siege or occupation alone always clears it, which the occupation weight contradicts. All drivers are war-layer ledgers. In a realm that has never held a live war front, dispatch refusal, banditry, interception and smuggling spill do not fire. Host `src/domain/spatial/embattlement.js:84-98`, `:113-115`, `:196-206`, `:237-245`; `src/domain/worldPulse/pulseKernel.js:2366-2389`.	In peacetime worlds, road encounters and patrol effects have no host basis; the comment misleads.	Correct the tuning comment; whether the host adds a peacetime danger or patrol term is open (trade-OQ4).	defect; trade-system.md §10.2 item 40 (line 1075)
trade-42	The food writer gives roads +0.15 on mountain, desert or hills and grants the minor-route trickle only when "isolated"; the view has no terrain boost and grants it on any disconnected route. A comment claims both read one constant so they cannot drift. The view now reads the writer's split when one exists, so drift touches only older saves and calls without a food record. Host `src/generators/foodGenerator.js:294-316`; `src/generators/economy/foodBalance.js:191-213`, `:308-336`; `src/domain/tradeRouteSemantics.js:66-76`.	Old saves show different food security than the generator computed.	The note records it; no fix given.	contradiction; trade-system.md §10.2 item 42 (line 1092)
trade-43	Severance checks only whether the source is under a war front or blockade; interception checks only intermediaries. A besieger next to its target can stay the cheapest supplier; under M2 nothing refuses it; under M6a the siege danger usually does, except for the vassal must-go. Host `src/domain/worldPulse/supplyKernel.js:254-257`, `:447-450`; `src/domain/spatial/supplyShipments.js:200-216`.	A war map would show besiegers feeding the besieged.	The note records it; no fix given.	defect; trade-system.md §10.2 item 43 (line 1099)
trade-44	The naval layer calls the blockade-to-peace-reason link a "seam"; it is consumed. Host `src/domain/spatial/navalLayer.js:533-536` versus `src/domain/worldPulse/peaceReasons.js:564` and `src/domain/display/navalDisplay.js:49`.	Misleads developers.	Correct the note in the code.	doc fix; trade-system.md §10.2 item 44 (line 1104)
trade-OQ1	Today "Map geography" accepts only a generated FMG map's pack; the routing partition, gates, sea lanes and seasonal overlay all come from that capture.	Decides whether any native-map geography ever reaches caravans, patrols or travel times.	Owner and host to decide whether the host canonizes from the map module's own map (see `research/sf/fp-physical-integration.md` §8).	open question; trade-system.md §10.3 (line 1110)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
trade-OQ2	Host gates come from a nearest-by-cost partition with no wild land; the map's W8 territory has unclaimed land and contested marches. Fed to the host as its territory, towns across unclaimed land would lose their gate and the caravan graph could break.	Feeding the wrong product would cut trade routes.	Owner and host to decide: supply a routing partition for the host and keep political territory as a lens, or the host treats wild cells as passable non-gates (a host change).	open question; trade-system.md §10.3 (line 1113); §9 item 1 (line 894)
trade-OQ3	No caravan path or departure tick is stored today; named Roads merchants do carry both.	Encounter maps cannot place caravans without it.	Owner to decide whether positioned caravans (WY-3) come before the map shows motion.	open question; trade-system.md §10.3 (line 1114); §9 item 13 (line 956)
trade-OQ4	Caravan routing has no patrol term. The only nearby slot is the embattlement ramp's security relief (0.4 times security, capped at 0.25), which matters only where war drives danger. Designed but unfed slots: TR-1's route_wardenship / route_predation pairs (registered, no producer) and EC-6's route_wardenship stance term (unowned, OD-FC-2). The note proposes offering J378's patrol coverage per route segment as a producer for those slots, or as a security input. It would be keyed to the host's gates.	Without adoption, patrols drawn on the map change nothing in the simulation.	Host to decide whether to adopt patrol coverage, and in which slot.	open question; trade-system.md §10.3 (line 1115); §9 item 4 (line 909)
trade-OQ5	The new-campaign preset constant is null; the code comment prices lighting it as a declared shift of seven persisted keys per new campaign. Host `src/domain/worldPulse/simulationRules.js:22-30`.	Decides whether newborn worlds run the default preset's systems.	Owner and host to decide whether the RR birth door will be lit.	open question; trade-system.md §10.3 (line 1116)
trade-OQ6	Grain provision, route wardenship and the realm-pressure consumer have no owner (OD-FC-2); EC-15 cannot light while `tradePactsEnabled` has no gate read. Host `docs/DESIGN_FP_ARCH_EC.md:578`.	Route wardenship, the patrol-shaped slot, stays unowned.	Owner to assign.	open question; trade-system.md §10.3 (line 1118)
trade-OQ7	The live-war-front test rejects only pure relationship-label fronts; discovery proposes war_front channels for hostile neighbours. If a DM confirms one it could embattle, sever caravans and damp Roads missions with the war layer off. Not traced. Host `src/domain/worldPulse/warFrontReads.js:1-75`; `src/domain/region/discoverDependencyCandidates.js:83`, `:189-190`.	The only possible path to road danger in a peaceful preset; untested.	Trace it.	open question; trade-system.md §10.3 (line 1120)
trade-9.3	Route danger is embattlement at settlements (siege 0.85, occupation 0.75 times state, exhaustion 0.45, crime 0.3, minus security), not along borders.	A contested-march overlay must not imply caravan danger.	Only if the host adopts a border term; otherwise nothing.	open question; trade-system.md §9 item 3 (line 906)

Population

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
population-9.1-6	Refugee columns cross marches and die by embattlement and season, but nothing the map calls a patrol enters that sum. Road deaths read only the M1 embattlement ledger (note §5.2 C).	The map's patrol layer (J378) cannot claim patrols make roads safer for refugees unless the host consumes a patrol fact.	Host to decide whether road danger should consume a patrol fact; until then the map must not imply patrols lower road deaths.	open question; population-system.md §9.1 item 6 (line 792)
population-9.3-1	Setting the canon marker lights refugee columns, the plague front, traditions influx, refuge steering, steading sites and (in DC, LR, FS) belief axes. Whether a native-map world counts as canonized at birth is not decided. Map `docs/design/proposal-world-native.md` §6 item 1.	Shipping the native map could silently switch on a whole set of simulation behaviours for every new campaign.	Make this the host's explicit choice, recorded with the owner (also population-10.3-1).	open question; population-system.md §9.3 item 1 (line 816)
population-10.1-1	The demographic design says every person is accounted for (law 4). The code loses or invents people. Shedding on a map-less world loses 55%, a promotion creates people, and migrants from an unmapped origin vanish. Road deaths are dropped and a dead parent's steadings are lost. Host `docs/DESIGN_DEMOGRAPHIC_ENGINE.md` law 4 against the note's §4.	Host totals leak and mint; a lens that balances people in against people out will not add up.	The cure exists but is dark; the host must light it or restate law 4.	contradiction; population-system.md §10.1 item 1 (line 843)
population-10.1-2	The population volume says no wave teleports people and the M4 header says the displaced pool no longer teleports. Stressor refugee flows, resettlement, steading transfers and the war sack still move people in the same tick on canonized realms. Host `src/domain/worldPulse/flows.js:58-109`; `src/domain/spatial/migration.js:4-8`; `docs/DESIGN_FP_POPULATIONS.md:76-79`.	The map cannot treat every population move as a walking column; some arrive instantly with no road.	The Wayfare architecture (`docs/DESIGN_FP_ARCH_WY.md`) calls the flow case a live violation that wave WY-4 repairs.	contradiction; population-system.md §10.1 item 2 (line 845)
population-10.1-3	The demographic design says the balancing mechanism runs on machinery that exists today. The route network it walks is never written, so lit it would move nobody; lighting routes beside it would cut every settlement's imports to a quarter. Host `src/domain/worldPulse/demographicsRates.js:255`, `:563-577`; `docs/DESIGN_FP_COHERENCE_FOLD.md` §0 lines 66-70.	Switching it on as documented would do nothing or quarter every settlement's imports.	The note names no remedy; it records the gap before any lighting.	contradiction; population-system.md §10.1 item 3 (line 849)
population-10.1-4	The spatial design says people beyond a ~20% food-deficit tolerance migrate. In code M4's tolerance governs only whether the origin survives; the number who leave still comes from the legacy rate. Host `src/domain/spatial/migration.js:11-16`; `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md` §4c.	A map explaining why people left cannot quote the design's famine rule.	The note names no remedy; a document-versus-code mismatch for the host to resolve.	contradiction; population-system.md §10.1 item 4 (line 852)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
population-10.1-5	Realm Directives amendment R says a share of a razed town flees as refugees; the code counts 12% "escapees" and gives them to no settlement (the lane is dark). Host `src/domain/worldPulse/razing.js:597`; `docs/DESIGN_REALM_DIRECTIVES.md` amendment R.	Survivors of a razing simply disappear; the map has no column to draw.	The note names no remedy beyond recording that the lane is dark.	contradiction; population-system.md §10.1 item 5 (line 854)
population-10.1-6	The tuning comment says resettlement takes people from neighbours; the code takes donors realm-wide with no distance term (note §3.5). Same finding as prosperity-C9.	A resettlement arc may join two far-apart towns; any "nearby" wording would mislead.	Fix the comment or add a distance term (host's choice).	contradiction; population-system.md §10.1 item 6 (line 856)
population-10.1-7	The Wayfare architecture's "positioned from depart Tick" is the planned cure, not today's state. The overlay places a column from its arrival minus a re-computed hop time, along the cheapest route, with direction chevrons the design forbids. Host `src/components/map/TravelersLayer.jsx:93-108`; `docs/DESIGN_FP_ARCH_WY.md` §0 C1, §3a.	Columns on the host map can sit on the wrong road at the wrong place.	The note treats it as the planned WY cure; no date given.	contradiction; population-system.md §10.1 item 7 (line 857)
population-10.1-8	The owner's size-scaled column speed has no code. The Wayfare architecture says it lands with WY-4; ruling OD-FC-13 makes it the population volume's row that WY-4 only reads. Host `docs/DESIGN_FP_ARCH_WY.md`; OD-FC-13.	Until ownership is settled nobody builds it, and the map cannot time columns by size.	Settle which volume owns the row.	open question; population-system.md §10.1 item 8 (line 860)
population-10.1-9	The demographic design's farm-village example contradicts the town-or-larger cap table; the ruling went to the cap table. Host `docs/DESIGN_DEMOGRAPHIC_ENGINE.md` §5 and §11b; `src/domain/worldPulse/settlementLifecycleKernel.js:196`.	Readers of §5 expect villages to seed new places; they cannot.	Correct the §5 example (the ruling remains vetoable by the owner).	doc fix; population-system.md §10.1 item 9 (line 862)
population-10.1-10	The steading header says charters are executed by the parked V2 machinery; that executor is dark, so no steading ever becomes a settlement (note §3.4).	Steadings stay cosmetic dots forever and can never become claim points on the map.	The note names no remedy; the header overstates.	contradiction; population-system.md §10.1 item 10 (line 864)
population-10.1-11	Wave P's headers say P4 brings road mortality as a named cause. P4 landed, P2 still hard-codes zero, and the population volume now says zero until the owner prices one. Host `src/domain/worldPulse/demographicsMigration.js:36-43`, `:588-598`; `docs/DESIGN_FP_POPULATIONS.md:100-104`.	The headers promise a cause class the map might try to show; it does not exist.	Bring the Wave P headers into line with the population volume.	doc fix; population-system.md §10.1 item 11 (line 866)
population-10.1-12	A comment says a lit world only moves people conservatively. On a lit world without a canon, mass emigration hands on only 45% and the rest vanishes with no death record (inferred). Host `src/domain/worldPulse/populationDynamics.js:420-481`.	Population quietly leaks where the comment promises it is conserved.	Correct the comment or close the leak.	contradiction; population-system.md §10.1 item 12 (line 869)
population-10.1-13	The migration-reason module calls a shed settlement "the engine's one migration cause"; columns now also come from calamity, terminal death and (dark) P2 classes. Host `src/domain/roads/migrationReason.js:30-33`.	Migration explanations shown to the DM can name the wrong cause.	Update the comment and any explanation built on it.	doc fix; population-system.md §10.1 item 13 (line 871)
population-10.1-14	Both population volumes and the fold's §2 say nothing in population is lit. The older movers that are lit (P-2 to P-10) are missing from the fold's table. Host `docs/DESIGN_FP_COHERENCE_FOLD.md` §2.	The status of record understates what moves people today.	Add the lit pre-FP movers to the fold's §2 table.	doc fix; population-system.md §10.1 item 14 (line 873)
population-10.1-15	Several file-and-line addresses in the population volumes no longer hold. Host `docs/DESIGN_FP_POPULATIONS.md` §2; `docs/DESIGN_FP_ARCH_POP.md` §1a.	Readers following those addresses land on the wrong code.	Refresh or drop the stale addresses.	doc fix; population-system.md §10.1 item 15 (line 875)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
population-10.2-1	The seeding drive looks for a history row whose reason starts "Displaced population", but the shared writer stamps the outcome headline instead. Map-less dispersal never adds its 0.20 inflow term, and the receiving town's history shows the origin's "population is falling" headline. Host `src/domain/worldPulse/settlementLifecycleKernel.js:307-314`; `src/domain/worldPulse/populationDynamics.js:599-607`.	Steadings seed less than designed on map-less worlds; the receiving town's history reads wrongly.	The note names no fix; found by reading, not executed.	defect; population-system.md §10.2 item 1 (line 880)
population-10.2-2	In presets that need DM approval (RR, LR), a major mass emigration or terminal death is queued as a proposal. M4 sends columns only for outcomes debited in the same tick. The approval path applies the outcome with no dispatch step, so only the origin's loss is applied. The DM's FORCE_ABANDON on a canonized realm meets the same fate. Host `src/domain/worldPulse/migrationKernel.js:398-415` (comment at `:383-392` promises release on approval); `src/domain/worldPulse/applyWorldPulse.js:1176-1230`, `:1305-1340`, `:524-538`; `src/domain/worldPulse/realmVerbExecution.js:899-910`; `src/domain/worldPulse/settlementLifecycleFirstClass.js:199-210`; `dispatchMigrations` callers at `pulseKernel.js:2418`, `calamityKernel.js:735`, `realmVerbExecution.js:824`.	In the default preset the largest DM-approved population movements vanish instead of walking; the map has no columns to draw.	Rated plausible until executed; run it to confirm before fixing.	defect; population-system.md §10.2 item 2 (line 886)
population-10.2-3	All of a source's flow candidates share one severity, and only one per source per tick is admitted. Ties break by candidate id, which contains the destination id. A source's refugees go to the neighbour whose id sorts first, tick after tick. Host `src/domain/worldPulse/flows.js:85-111`; `src/domain/worldPulse/candidateEvents.js:385-438`.	Refugee arcs on the map would all point one way from each source, an artefact of id spelling.	The note names no fix.	defect; population-system.md §10.2 item 3 (line 901)
population-10.2-4	The instant refugee flow runs beside M4 columns, so one crisis sends both (confirmed). Host `src/domain/worldPulse/candidateEvents.js:599-605`.	Double-counted displacement on the host and on any map lens.	The note names no fix; WY-4 is the planned repair (population-10.1-2).	defect; population-system.md §10.2 item 4 (line 906)
population-10.2-5	M4 picks a route by danger and seasonal cost but stamps the pair's standard hop time; the overlay then moves its marker along a third path. Host `src/domain/worldPulse/migrationKernel.js:220-223`, `:262`; `src/components/map/TravelersLayer.jsx:93-108`.	Arrival dates and drawn paths disagree with the route the simulation chose.	The note names no fix; the map-side answer is the stored route path (note §8.6).	defect; population-system.md §10.2 item 5 (line 907)
population-10.2-6	Dispatch skips an origin with no map cell, with no receipt, after the origin has already been debited. Host `src/domain/worldPulse/migrationKernel.js:460`.	People are lost with no record; settlements founded after the canon leak population.	The note names no fix.	defect; population-system.md §10.2 item 6 (line 910)
population-10.2-7	A crisis shed in QL and NC is labelled emigration and credits nobody. Host `src/domain/worldPulse/populationDynamics.js:445-485`.	The label promises migrants who never arrive anywhere.	The note names no fix.	defect; population-system.md §10.2 item 7 (line 912)
population-10.2-8	The travellers overlay drops port and teleport-circle columns, exactly the ones M4 may have priced by sea or circle. It also repositions every marker by the current season. Host `src/components/map/TravelersLayer.jsx:93-108`; `src/domain/spatial/distanceRead.js:653-680`.	The host map under-shows movement and moves markers when the season turns.	The note names no host fix; the native map should not copy it.	defect; population-system.md §10.2 item 8 (line 914)
population-10.2-9	A column counts as "on the road" if its origin or destination lies on the party's route. So a crossing column is missed, and an endpoint match may not travel that road. Host `src/domain/briefs/roadScene.js:158-163`.	Travel briefs show the wrong migrants on the road.	The note names no fix.	defect; population-system.md §10.2 item 9 (line 916)
population-10.2-10	The pact lane signs `migration_right`, `labor_compact` and `settlement_provision` grants that no code consumes. Host `docs/DESIGN_FP_POPULATIONS.md` POP-5b; `docs/DESIGN_FP_COHERENCE_FOLD.md` §2 GR row.	Treaties promise movement rights that change nothing.	The note names no fix.	defect; population-system.md §10.2 item 10 (line 919)
population-10.2-11	Wave P's tuning record reads `signed: false, lit: null`. Host `src/domain/worldPulse/demographicsRates.js:375-404`.	The demographic rates have no owner sign-off, so they cannot be lit as tuned.	Owner sign-off at the tuning review.	defect; population-system.md §10.2 item 11 (line 921)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
population-10.2-12	The demographic Herald writes "were turned back" for people who never departed (an open R-28-class finding). Host `src/domain/worldPulse/demographicsHerald.js`.	Players and DM read news of a journey that did not happen.	Open in the fold; no fix stated.	defect; population-system.md §10.2 item 12 (line 922)
population-10.2-13	The timeline track sums `populationDeltas` as an id-to-number map, but every producer writes an array of rows, so each entry is skipped. The track never reads `mechanicalOutcomes` where minor drift is recorded. Host `src/domain/display/timelineTrack.js:81-86`; producers `src/domain/worldPulse/populationDynamics.js:451`, `flows.js:106`, `deploymentReturn.js:415-423`, `pulseHelpers.js:144`; `tests/domain/timelineTrack.test.js:18`, `:60`.	The host timelapse shows no population change; the map's time scrub was meant to reuse its frames.	Rated plausible until executed; confirm by running, then fix before the map reuses the frames.	defect; population-system.md §10.2 item 13 (line 924)
population-10.2-14	Markers carry their text as SVG titles inside groups that ignore pointer events, so the browser probably never shows them on hover. Host `src/components/map/TravelersLayer.jsx:140-145`; `src/components/map/PlacementsLayer.jsx:320-331`.	Users cannot see what a marker is by pointing at it.	Inferred, not executed; confirm before fixing.	defect; population-system.md §10.2 item 14 (line 932)
population-10.3-1	Should a native world count as canonized at birth, lighting M4 columns and the rest by default? Or should the premium canonize stay the act that makes people walk?	Decides whether shipping the native map changes simulation behaviour for every new campaign.	Owner and host to decide and record.	open question; population-system.md §10.3 item 1 (line 939)

Prosperity

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
prosperity-9-12	Boom rungs reach no news, mover receipts are dropped, news is capped at 240 world-wide, and purchase news carries no ids. The map cannot replay band changes.	The map must run its own per-advance snapshot service and cannot show anything before it started watching.	Offer the host change in the note's §8.6 item 1 (a host-kept history); the map snapshots meanwhile.	open question; prosperity-system.md §9.3 item 12 (line 828)
prosperity-9-15	The v2 town layout re-reads the live prosperity band at each build. It reads it again through the district wealth word that decides which outer quarters the wall takes in. A town map meant never to change can shift after a boom or bust.	A town's drawn plan could change in play.	Either the map module or the host must freeze the genesis band for town drawing, covering the district wealth read; who freezes it is open (prosperity-Q-b).	open question; prosperity-system.md §9.3 item 15 (line 839)
prosperity-D1	Rows carry the lower-case band word; both sorts rank it with the case-sensitive rank function, where a lower-case word ranks -1. Every comparison ties and the order stays as inserted. Host `src/components/map/heraldRegister.js:103-106`, `:171`; `src/components/map/HeraldGazetteer.jsx:122`; `src/components/map/RealmComparisons.jsx:91`.	The DM's prosperity sort and histogram order look broken.	The note names no fix (re-checked by probe).	defect; prosperity-system.md §10.1 D1 (line 858)
prosperity-D2	The 0.4 food-surplus prosperity bonus is rounded to 0. Host `src/generators/economy/prosperity.js:263`; `src/generators/foodGenerator.js:377`.	Food-rich settlements never get the designed boost.	The note names no fix.	defect; prosperity-system.md §10.1 D2 (line 862)
prosperity-D3	Rounding drops the grain-purchase fraction each tick with no carry-over, so Poor and Moderate buyers never pay. The code's note covers only the seller side. Host `src/domain/worldPulse/generosityUpdates.js:121-150`.	Buying grain is free for two bands; the economy is cheaper than designed.	Open question prosperity-Q-d: carry the fraction across ticks as the boom accrual does.	defect; prosperity-system.md §10.1 D3 (line 864)
prosperity-D4	The comment says a flourishing town is about Prosperous or better; Comfortable already qualifies. Host `src/domain/worldPulse/upswingKernel.js:129`.	Readers misjudge which towns count as flourishing.	Correct the comment.	doc fix; prosperity-system.md §10.1 D4 (line 866)
prosperity-D5	Unknown band words read as 0, 0.4, 0.5 or 50 depending on the reader. A {tier} object reads neutral to economic capacity and a {label} reads -1 to the ladder; the condition-severity modifier takes only an exact-case string, so a {tier} object gets no modifier. Host `src/domain/districtProfile.js:326-334`; `src/domain/conditionPromotion.js:121-123`.	The same settlement can be read as rich by one system and neutral or poor by another.	The note names no fix.	defect; prosperity-system.md §10.1 D5 (line 867)
prosperity-D6	The module calls itself "THE ONE ladder", but two numeric ladders and a seed disagree on every rung (Comfortable 0.65, 0.667 or 62; Struggling 0.10, 0.167 or 30). The town scene and cartography hold private ladders too. Host `src/domain/prosperityRank.js`.	The map cannot take "the host's number" for a band; it must use the rank.	The note names no fix.	defect; prosperity-system.md §10.1 D6 (line 872)
prosperity-D7	`prosperity01Of` is rank divided by 6 in upswing, generosity, lifecycle and the dark statecraft modules, but economic capacity divided by 100 in demographics; the PG design calls the latter the existing function. Host `src/domain/worldPulse/demographicsPushPull.js:347-350`; `docs/DESIGN_FP_ARCH_PG.md:210-211`.	Design work built on the name will read the wrong quantity.	The note names no fix.	defect; prosperity-system.md §10.1 D7 (line 874)
prosperity-D8	The boom sustain branch emits only a receipt, and receipts are dropped at the pulse mover, although the upswing design promises receipted, receipt-visible steps (its §0 and B2) and the code's own tuning comment says "receipted". Host `docs/DESIGN_UPSWING_RELATIONSHIP_FLOWS.md:12`, `:103`; `src/domain/worldPulse/upswingKernel.js:121` [Fable-reviewed 2026-10-18].	No history of boom climbs exists for the DM or for a map time scrub.	The note names no fix beyond the host-history offer in its §8.6 item 1.	defect; prosperity-system.md §10.1 D8 (line 877)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
prosperity-D9	Purchase news has an empty settlement-id list, while relief news in the same module carries both ids. Host `src/domain/worldPulse/generosityNews.js:157` (relief at `:40`).	Purchases cannot be placed on the map or linked to the towns involved.	The note names no fix.	defect; prosperity-system.md §10.1 D9 (line 879)
prosperity-D10	The festival pin bag is flattened across all saves and keyed by the single fork id HBF-85. So one settlement's decree would fix every festival that opens that tick (inferred). Host `src/domain/worldPulse/pulseKernel.js:362`.	A local DM choice could change outcomes realm-wide.	Open question prosperity-Q-f: is the pin meant to be per settlement.	defect; prosperity-system.md §10.1 D10 (line 881)
prosperity-D11	The constants file says Subsistence is internal and kept for old saves; every writer can set it. Host `src/data/constants.js:85-87`.	Readers that ignore Subsistence will meet it in live play.	Correct the comment.	doc fix; prosperity-system.md §10.1 D11 (line 883)
prosperity-D12	The word "Thriving" still gates merchant dominance and two narrative lines, and the AI layer's prosperity slot holds the viability summary. Host `src/generators/power/rulingStructure.js:497-500`; `src/generators/narrativeGenerator.js:598`; `src/generators/aiLayer.js:140`; `docs/GENERATION_COHERENCE_AUDIT.md:678-680`, `:1126-1128`, `:1136-1137`.	Those gates and lines may never fire or read the wrong quantity.	Still open in the generation coherence audit.	defect; prosperity-system.md §10.1 D12 (line 885)
prosperity-C1	The upswing design §1 says both flags are absent and preset lighting is owner-gated; the presets light them in RR, DC, LR and FS. Host `docs/DESIGN_UPSWING_RELATIONSHIP_FLOWS.md:17-22`.	The design misstates what is live in the default preset.	Update the upswing design §1.	doc fix; prosperity-system.md §10.2 C1 (line 892)
prosperity-C2	The upswing design A2 says a buyer buys from the cheapest reachable surplus neighbour, rides shipments, credits the seller, and transfers are zero-sum. The code buys only after a refusal, instantly, with no route, credits a 0.4 band nudge, and is not zero-sum. Host `docs/DESIGN_UPSWING_RELATIONSHIP_FLOWS.md:48-57`, `:120-125`.	A map cannot draw purchases as trade along routes; there is no route.	The note names no remedy; design and code must be reconciled.	contradiction; prosperity-system.md §10.2 C2 (line 895)
prosperity-C3	The upswing design A3 calls credit a bounded prosperity transfer; it is a grain obligation. A4 says the overture reads the trade tally; it reads give-stream warmth. Host `src/domain/spatial/generosityReactions.js:425-449`.	Readers expect prosperity to move with credit; it does not.	The note names no remedy.	contradiction; prosperity-system.md §10.2 C3 (line 897)
prosperity-C4	Four places say the generated prosperity is a frozen baseline; four writers change it in play. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md:1675-1677`; `src/domain/spatial/tradeFlow.js:12-18`; `src/domain/worldPulse/supplyKernel.js:590-591`; `src/domain/causalState.js:776-779`.	Anyone reading those comments assumes the band never moves.	Bring the four statements into line; the atlas and EC S18/S39 already record the writers.	doc fix; prosperity-system.md §10.2 C4 (line 899)
prosperity-C5	The bust comment and the upswing design B2 say a bust drives people out. In code a bust writes no population, only raises trade pressure slightly, mints no emigration pulse and is not a crisis-flight type.	A map cannot show refugees fleeing a bust.	The note names no remedy.	contradiction; prosperity-system.md §10.2 C5 (line 902)
prosperity-C6	The spatial design §4i says granary capacity reads prosperity; the code reads tier and institutions. Host `docs/PHASE55_SPATIAL_ENGINE_DESIGN.md` §4i.	Design readers expect rich towns to store more grain.	Correct §4i (the map's own note is corrected under prosperity-9-16).	doc fix; prosperity-system.md §10.2 C6 (line 907)
prosperity-C7	EC rows M7, M8 and M11 say famine, harvest and plague move prosperity today; that writer is generation-only. Row M47 says the band is written each tick; festivals write yearly, purchases on refusals, ransoms on events and booms about every seven ticks. Host `docs/DESIGN_FP_ARCH_EC.md` rows M7, M8, M11, M47.	A lens designed from EC would expect a band that moves every tick.	Correct the EC rows.	doc fix; prosperity-system.md §10.2 C7 (line 908)
prosperity-C8	The traditions comment says its flag is dark; it is lit in four presets. Host `src/domain/worldPulse/traditionsKernel.js:31-35`.	Readers think festivals do not run.	Correct the comment.	doc fix; prosperity-system.md §10.2 C8 (line 911)
prosperity-C9	Resettlement says "nearby"; the donor pool is realm-wide. Host `src/domain/worldPulse/settlementLifecycleFirstClass.js:291`, `:319`, `:411-412`; `docs/DESIGN_SETTLEMENT_LIFECYCLE.md:83`. Same finding as population-10.1-6.	Resettlement can join far-apart towns while the text says neighbours.	The note names no remedy.	contradiction; prosperity-system.md §10.2 C9 (line 912)
prosperity-C10	The comment defines migration "richness" as relative to the richest reachable hub; the code is absolute. Host `src/domain/spatial/migration.js:237`.	Migration pulls are misdescribed.	Correct the comment.	doc fix; prosperity-system.md §10.2 C10 (line 914)
prosperity-C11	The town-map grammar says no 0..1 prosperity scalar exists; `prosperityRank01` is one. Host `src/domain/townMap/arch/conditionGrammar.js:74`.	Town-map work may build a duplicate.	Correct the comment.	doc fix; prosperity-system.md §10.2 C11 (line 915)
prosperity-C12	The treasury is promised lit in three presets; it is lit in none. Host `src/domain/worldPulse/simulationRules.js:470-473`.	Readers expect a treasury that does not run.	Correct the comment or light as promised (owner's call).	doc fix; prosperity-system.md §10.2 C12 (line 916)
prosperity-C13	Comments say "35%" and "30% luckier"; the code gives 40% to Struggling and 60% to Poor. Host `src/generators/economy/prosperity.js:128-131`.	Tuning readers misread the odds.	Correct the comments.	doc fix; prosperity-system.md §10.2 C13 (line 917)
prosperity-C14	The default preset's comment says a world born into it runs the stack from its first tick. A `createCampaign` birth infers the same preset id with no writer lit, so any surface showing "Realistic Regional" overstates what runs (inferred). Host `src/domain/worldPulse/simulationRules.js:886-892`.	The DM and the map are told the default simulation runs when it does not.	The note names no remedy; the map must read resolved rules.	contradiction; prosperity-system.md §10.2 C14 (line 918)
prosperity-C15	The fold's §2 is the status of record and has no prosperity entry; its requirement 21 (one word, one meaning) has not been applied to prosperity, which the host uses for three quantities. Host `docs/DESIGN_FP_COHERENCE_FOLD.md` §2, requirement 21.	No single place says what prosperity is or what moves it.	Add a prosperity entry and apply requirement 21.	doc fix; prosperity-system.md §10.2 C15 (line 921)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
prosperity-C17	Causal state names vassal extraction and market shock as moving economic capacity, which they do not, and omits war-cost, occupation and worked-out-vein conditions, which do. Host `src/domain/causalState.js:776-780`; `src/domain/activeConditions.js:91-97`.	Explanations of a capacity change can blame the wrong cause.	Correct the comment.	doc fix; prosperity-system.md §10.2 C17 (line 926)
prosperity-C18	The comment says festivals are cancelled by plague, famine, war or occupation; the set also holds vassal extraction and a cut trade route, and its "occupation" member is the occupier's burden. Host `src/domain/worldPulse/traditionsKernel.js:188`, `:197`.	Readers miss that vassalage freezes festivals.	Correct the comment.	doc fix; prosperity-system.md §10.2 C18 (line 930)
prosperity-C19	The traditions relations module describes vassals forced to adopt a tradition. The trigger is only the occupation ledger's "vassalized" rung; treaties, hegemony and graph vassal edges are never read. Host `src/domain/traditions/relations.js:25-26`; `src/domain/rulingPowerSeat.js:290-294`.	A DM who draws a vassal edge expects rite imposition that never happens.	The note names no remedy.	contradiction; prosperity-system.md §10.2 C19 (line 932)
prosperity-Q-a	An Edit-Mode re-derivation regenerates the band (note §3.1); whether it is written to the stored settlement was not traced.	A DM edit could silently undo a boom or bust.	Trace the store write.	open question; prosperity-system.md §10.3 (a) (line 938)
prosperity-Q-b	A band step changes the economic field; the v2 town plan's sensitivity to it was not traced.	Decides whether the town-map freeze (prosperity-9-15) is urgent.	Trace the plan's sensitivity.	open question; prosperity-system.md §10.3 (b) (line 940)
prosperity-Q-c	Should booms cool gracefully instead of only busting? Recorded as an owner question in code. Host `src/domain/worldPulse/upswingKernel.js:644-653`.	Busts are the only exit, which shapes how a map shows boom towns.	Owner ruling.	open question; prosperity-system.md §10.3 (c) (line 941)
prosperity-Q-d	Should the grain purchase debit carry its fraction across ticks, as the boom accrual does (prosperity-D3)?	Without it two bands buy for free.	Host decision.	open question; prosperity-system.md §10.3 (d) (line 942)
prosperity-Q-e	Festivals push good towns up and poor towns down year on year, with no realm-level brake (note §3.3).	Long campaigns split into rich and poor with nothing to pull them back.	Owner ruling on whether the ratchet is intended.	open question; prosperity-system.md §10.3 (e) (line 943)
prosperity-Q-f	Is the festival pin meant to be per settlement (prosperity-D10)?	If not, one decree affects the whole realm.	Host decision.	open question; prosperity-system.md §10.3 (f) (line 944)
prosperity-Q-i	Does any shipped screen mount the surveyor's autonomous runs, so a DM can stop a run on a band (note §3.10)?	If none does, there is no DM control over those runs.	Check the shipped surfaces.	open question; prosperity-system.md §10.3 (i) (line 947)

Plague

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
plague-9-9	The epidemic record keeps a source id while it lives, and the materialization names the source. Arrows are recoverable now, but only until the prune deletes the record.	Without a host change, plague arrows vanish from history once pruned; the map must snapshot them itself.	The host ask is in the note's §8.6 items 1-2 (parallel to faith's); see plague-10.2-5.	open question; plague-system.md §9 item 9 (line 842)
plague-10.1-1	A plague can create two records: a "condition" (the status most systems read) and a "stressor" (the live pressure). The DM's Apply-stressor makes both; plagues born on their own or arriving by the front make only the stressor; generation makes only the condition. Causal scores, capacity, factions, traditions, NPC agency, the threat profile, the town map and the population rate read only the condition. No document names this (inferred from the note's §3.8).	A plague that arrives by spreading barely affects the towns it reaches, and its causal mark appears only after it ends.	Open: plague-10.2-4 asks whether faction deltas and the causal scan should read live stressors; plague-10.2-1 asks whether the disease-floor signature comes first.	defect; plague-system.md §10.1 item 1 (line 872)
plague-10.1-2	The header says care shortens the record's active phase. In code care only speeds the recovering fade; the active phase ends on stressor resolution or a 26-tick floor. Host `src/domain/spatial/pestilence.js:44-47` against `:533-558`.	Healers look stronger in the text than in play.	Correct the header.	doc fix; plague-system.md §10.1 item 2 (line 877)
plague-10.1-3	Headers say materialization automatically feeds the "gods_abandonment" seam. That compound also needs famine and a conversion fracture in the same town, and the revival seam ignores the crisis type. Host `src/domain/spatial/pestilence.js:56-59`; `src/domain/worldPulse/pestilenceKernel.js:16-18`; `src/domain/worldPulse/realmEvents.js:26-37`.	Readers expect a faith event that rarely fires.	Correct the headers.	doc fix; plague-system.md §10.1 item 3 (line 879)
plague-10.1-4	The "plague_quarter" scar is unreachable (note §3.14).	A designed lasting mark of plague never shows in a town.	The note names no fix.	defect; plague-system.md §10.1 item 4 (line 882)
plague-10.1-7	The PG design header says "DRAFT - NOT YET LANDED" while the file sits on master, and the fold's bracket says "NOT on master". Host `docs/DESIGN_FP_ARCH_PG.md:4-11`.	Readers cannot tell what has landed.	Correct the PG header and the fold bracket to "no code on master", with fold §2 as the record.	doc fix; plague-system.md §10.1 item 7 (line 889)
plague-10.1-8	Fold §2 says refugee-carried plague waits on a fold of POP-4. The same fold already folds POP-4 into PG, and pg-1 depends only on landed landing seams. Host `docs/DESIGN_FP_COHERENCE_FOLD.md:349` against `:270`; `docs/DESIGN_FP_ARCH_PG.md:331-333`.	Refugee-carried plague looks blocked when it is not.	Correct fold §2.	contradiction; plague-system.md §10.1 item 8 (line 892)
plague-10.1-9	EC's M11 row says the front reads the trade tally "so trade spreads it". The tally weights only onset; spread rides channels and shipment links. Host `docs/DESIGN_FP_ARCH_EC.md:2244`; `src/domain/worldPulse/pestilenceKernel.js:166-200`.	A lens built from EC would draw spread along the wrong thing.	Correct the EC row.	doc fix; plague-system.md §10.1 item 9 (line 894)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
plague-10.1-10	The certification row cites `candidateEvents.js:442` (the gate is at `:520`) and says disease spread is suppressed whenever the canon marker is set; since FIX #2 only the trade-channel part is dropped. Host `src/domain/certification/subsystemRowsPeople.js:190`; `src/domain/worldPulse/candidateEvents.js:476-502`.	Certification describes plague behaviour that no longer holds.	Correct the row.	doc fix; plague-system.md §10.1 item 10 (line 896)
plague-10.1-11	The dispatch comment calls the plague front unbuilt and promises a plague term; the front is built and the term was never added. The generosity module claims a plague hazard read through that seam, which has none. Host `src/domain/spatial/dispatchEV.js:41-44`; `src/domain/spatial/generosityEV.js:310-316`.	Caravans route straight through plague towns; route danger on the map has no plague term for them.	Add the term or correct both comments.	contradiction; plague-system.md §10.1 item 11 (line 898)
plague-10.1-12	Arrival news and `channelType: trade_route` are stamped even for hops carried by shipment; pg-1's column hops would inherit the same words. Host `src/domain/worldPulse/pestilenceKernel.js:241-243`, `:290`, `:293`.	News and any map arrow mislabel how the plague arrived.	The note names no fix; matters before pg-1 lands.	defect; plague-system.md §10.1 item 12 (line 900)
plague-10.1-13	The display's weeks-away count ignores the season (note §5.2).	The DM sees the wrong arrival estimate in winter.	The note names no fix.	defect; plague-system.md §10.1 item 13 (line 902)
plague-10.1-14	The front's care list, the healing ledger's pattern and capacity's healing read disagree on what a healer is: a cathedral or druid grove is care for onset but not a healer for the gate; a temple is both; "Priest (resident)" is neither. The events module says its pattern matches the front's roster; it does not. Host `src/domain/events/mutateWorld.js:555-558`; `src/domain/spatial/pestilence.js:163-168`; `src/domain/healingLedger.js:31-32`; `docs/DESIGN_FP_ARCH_PG.md:237-241`.	The same town counts as having healers for one plague rule and not another.	The note names no fix; the impairment pattern now runs only on replayed PLAGUE events.	defect; plague-system.md §10.1 item 14 (line 903)
plague-10.1-15	The front's second draw is called a "clearance gate"; it is the onset draw, and clearance is deterministic. Host `src/domain/worldPulse/habitForkRegistry.js:523-526` (HBF-57).	Misleads tuning of plague onset.	Correct the registry text.	doc fix; plague-system.md §10.1 item 15 (line 908)
plague-10.1-16	The advance function's doc says a failed onset is pruned with a refractory window. The code keeps a recovering record at level 0 until the 8-tick refractory ends, which is what blocks re-seeding. Host `src/domain/spatial/pestilence.js:395-397`, `:510-517`.	Cosmetic, but it hides the mechanism that stops re-infection.	Correct the doc comment.	doc fix; plague-system.md §10.1 item 16 (line 910)
plague-10.1-17	Unmapped endpoints, and worlds with the marker but no digest, get one-week hops, the fastest possible (inferred; note §5.2).	Plague races into settlements founded after the canon.	Defect candidate; the note names no fix.	defect; plague-system.md §10.1 item 17 (line 913)
plague-10.1-18	Arrival severity reads the source's current severity and falls back to a flat 0.5 when the source recovered during incubation (inferred). Host `src/domain/worldPulse/pestilenceKernel.js:205`; `src/domain/spatial/pestilence.js:520-522`.	A plague can arrive stronger or weaker than its source was when it left.	Defect candidate; the note names no fix.	defect; plague-system.md §10.1 item 18 (line 915)
plague-10.1-19	Severity has a fixed point that does not decay along chains of dense towns (inferred; note §3.7).	A plague could cross a dense realm at undiminished strength.	Design question for the owner.	open question; plague-system.md §10.1 item 19 (line 917)
plague-10.1-20	In Static Campaign on a mapped realm a DM plague's front still runs (the kernel reads no rules flag) while stressors never age. The 26-tick floor sends towns to recovering. Once a record prunes, a still-live seed re-enters as patient zero, a cycle that ends only when the DM resolves it (inferred, untested).	A preset meant to be static keeps re-infecting towns indefinitely.	Open defect; untested, so confirm first.	defect; plague-system.md §10.1 item 20 (line 918)
plague-10.1-21	Front materializations split one outbreak into many records; where each came from survives only in prose (note §4).	A map cannot rebuild one outbreak's tree from the records.	The note names no fix (relates to the host history ask, plague-9-9).	defect; plague-system.md §10.1 item 21 (line 922)
plague-10.1-22	Realm and compound arcs are built before the front advances, so a tick's new cases count on the next tick; compound severities use the record's severity, not each town's weakened severity. Host `src/domain/worldPulse/realmEvents.js:155`.	Realm-level plague events lag and can overstate severity.	The note names no fix.	defect; plague-system.md §10.1 item 22 (line 923)
plague-10.1-23	The Table Ledger's `plague` obligation injects a roaming record typed `plague` with no catalog entry: no spread, no front, no counterforce, named "Realm-wide plague" instead of "The Plague", no power or economy rerun hooks, yet it still displaces refugees. Host `src/domain/tableLedger.js:131-133`, `:265`; `src/domain/stressorPicker.js:58-62`; `src/domain/worldPulse/stressorsCore.js:301-309`; `src/domain/worldPulse/flows.js:42-44`.	Two plagues behave differently and a lens cannot treat them as one.	Defect candidate; plague-10.2-7 asks whether this key is meant to be the one plague.	defect; plague-system.md §10.1 item 23 (line 925)
plague-10.1-24	The crisis lifecycle calls PLAGUE the DM's dedicated outbreak button. The verb cannot be authored and is folded into Apply-stressor, so healer impairment is unreachable from today's authoring. Host `src/domain/crisisLifecycle.js:96-102`; `src/domain/events/affordanceManifest.js:109-119`, `:645-649`; `src/domain/events/mutateWorld.js:553-575`.	A designed DM effect (healers impaired by plague) cannot be triggered.	The note names no remedy.	contradiction; plague-system.md §10.1 item 24 (line 930)
plague-10.1-25	The plague condition adds +0.14 disease pressure and a healing drain, feeding its own re-mint. Famine's identical loop is capped by a ledger ceiling; plague's is not (inferred; note §3.8).	Plague may self-sustain where famine cannot.	Design question for the owner.	open question; plague-system.md §10.1 item 25 (line 933)
plague-10.1-26	A front materialization overwrites a same-id resolved echo without the memory re-ignition an organic birth gets; rare on mapped worlds (note §3.14). Host `src/domain/worldPulse/pestilenceKernel.js:248-256`.	Re-infection strength depends on how the plague arrived.	The note names no fix.	defect; plague-system.md §10.1 item 26 (line 935)
plague-10.1-27	A moderate (0.6) DM plague never reaches the refugee-flow bar, because aging runs before flows read severity (note §3.15). No document says so.	A DM expecting refugees from a moderate plague gets none; the map has no refugee columns to draw.	plague-10.2-8 asks whether it should; at minimum document it.	defect; plague-system.md §10.1 item 27 (line 936)
plague-10.2-1	Should the end tuning review's signature on the disease floor (OD-FC-28) come before any plague lens work, since organic plague is otherwise invisible?	A plague lens built first would mostly show DM-authored outbreaks.	Owner ruling.	open question; plague-system.md §10.2 question 1 (line 942)
plague-10.2-2	Front materialization runs on its own even under the DM-only autonomy mode. Accept that, or adopt the queued proposal-gated alternative?	Decides whether the DM approves each plague spread.	Owner ruling.	open question; plague-system.md §10.2 question 2 (line 944)
plague-10.2-3	The front has no rules flag, so mapping a quiet or narrative campaign turns on inter-settlement plague its preset otherwise forbids (inferred: today it does).	Canonizing the map changes plague behaviour in presets meant to keep it local (ties to population-9.3-1).	Owner or host ruling.	open question; plague-system.md §10.2 question 3 (line 946)
plague-10.2-4	Should faction deltas and the causal scan read live plague stressors (plague-10.1-1)?	Without it spreading plague stays mechanically invisible.	Host decision.	open question; plague-system.md §10.2 question 4 (line 948)
plague-10.2-5	Where should a plague front's history live: in the host (a receipt per hop) or map-side (pulse snapshots)?	Decides whether the map can show past spread it did not watch.	Host and map decision (plague-9-9).	open question; plague-system.md §10.2 question 5 (line 949)

#	What the code does versus what its document says	What it costs	What the host should do	Kind and source
plague-10.2-6	Does pg-1's P2 arm wait on `demographicsEnabled` permanently (OD-FC-18), leaving only M4 columns to carry plague?	Decides which refugee movements can carry plague.	Owner ruling.	open question; plague-system.md §10.2 question 6 (line 950)
plague-10.2-7	Is the Table Ledger's `plague` meant to be the one plague (plague-10.1-23)?	Two plague types break "one plague truth".	Host decision.	open question; plague-system.md §10.2 question 7 (line 951)
plague-10.2-8	Should a moderate DM plague displace refugees (plague-10.1-27)?	Today the DM's default band sends nobody.	Owner ruling.	open question; plague-system.md §10.2 question 8 (line 952)
plague-R7	PG pg-4's quarantine or gate-closure verb is DESIGNED ONLY and PG names quarantine an owner value: is it to be built, and if so as the host's own verb with a stance input — never as a patrol fact the map draws (a patrol holds no quarantine line)?	Without the verb the DM has no plague-control action to author, and no border, patrol or march can stand in for it.	Owner ruling (PG names it an owner value).	open question; plague-system.md §5.1 row 14 (line 531), §7 (line 672); retrovalidation/plague ruling R7 (Fable, 2026-10-18)

Noted for the map, not for the host

Ninety further rows in the same note sections are the map's own rules, not host work; they are noted for the map, not for the host. They are the faith lens rules (faith-26, faith-map-1 to faith-map-6), the alignment lens rules (alignment-25a, alignment-25c, alignment-map-1 to alignment-map-9) and the trade lens rules (trade-41, trade-map-1 to trade-map-8). They are also the population lens rules (population-9.1-1 to 9.1-5, 9.1-7, 9.1-8, 9.2-1 to 9.2-4, 9.3-2 to 9.3-7, 10.3-2 to 10.3-5), the prosperity lens rules and note corrections (prosperity-9-1 to 9-11, 9-13, 9-14, 9-16, C16, Q-g, Q-h), and the plague lens rules and note corrections (plague-9-1 to 9-8 with 9-6b, 9-10 to 9-16, 10.1-5, 10.1-6, 10.3-1 to 10.3-7). Each has one line in `extract-defects-a.md` or `extract-defects-b.md` in this folder.

Patrols within a territory: what the map needs from the simulation

A territory on the map is a projection of the simulation's political facts over the terrain, never a fact of its own. Patrols derive from military enforcement, neighbour hostility and route class, offered back as coverage per road stretch the host uses only by its choice (docs/PROGRESS.md J377 (line 436), J378 (line 437); docs/volumes/V13-world.md §3.9 (line 1794)). By the owner's decision of 2026-10-18, patrols stay a map-only picture for now: the host is not asked to consume the coverage, the legend says patrols change nothing in the simulation, and the optional door (H16) stays open for the host's own programme to take later.

Words used here. *The host* or *the simulation* is SettlementForge, the program that runs the campaign week by week; *the map* is this module. A *seat* is a settlement the host counts as a political seat. The *seat territory* is the host's own kind of territory: every land cell goes to the seat it is cheapest to reach, with no politics (V13 §3.9 (line 1794)).

The *sway* is the map's picture of who politically holds which land, derived and redrawn, never the host's truth (V13 §3.9 (line 1794)). *Debatable land* or a *march* is land where two hostile reaches meet and neither holds it (V13 §3.9 (8) (line 1859)). A *cadence* is how often a power's patrol passes a road stretch: weekly, every four weeks, every thirteen weeks, or never (V13 §3.9 (14) (line 1976)).

A *window* is the bundle of facts the host sends after each advance; the *polity facts window* is its political bundle, `PolityFacts` (V13 §2.2 (line 822)). An *H-row* is a numbered host change in V13 §8.5 (line 3715; its table from line 3739). *Lit* means a host feature is switched on and sending; *dark* means it is not. *RRb* is the host's most common birth preset (research/sf/lens-implications.md §3.3 (line 373)).

The binding rules, in plain words

- **Two partitions, never one.** The host's seat territory and the map's sway stay separate; the sway never becomes the host's territory, never enters the canon, and the host's crossings and distances never come from it (V13 §3.9 (line 1794); §5.9 sway refusals (line 3283); J-V13.65 (line 3842)).
- **Borders move only on dated facts, after a wait.** Each changed fact waits its dwell (prosperity 13 weeks; strength, force and culture bands their own); a fact that rose and fell inside one window never reaches the border (V13 §3.9 (13) (line 1951); T-V13-41 (c) (line 3600); J377 (line 436)).
- **The map never decides a war.** Between hostile neighbours the border's push defers to the host's own strength order; only a host verdict moves a march's hold band; a raid with no outcome word only marks "fighting" (J384 (3), (5) (line 443); V13 §3.9 (8) (line 1859), (11) (line 1917)).
- **Military strength is the tie-breaker only.** It enters only land left debatable, never extends reach, never moves a decided line, never touches a seat's own cell; the per-tier ceiling holds (J378 (line 437); V13 §3.9 (10) (line 1904)).
- **Patrols by address and week, never a roll.** Cadence per road stretch and power is revised only with the sway; presence in a week is fixed by a hash of the stretch and the power, so every run agrees (V13 §3.9 (14) (line 1976); §1b.2 `patrolAt` (line 522); T-V13-42 (line 3601)).
- **Cadence never reaches the host as a time.** The offer sends only the stretch covered and its basis; the host prices any round trip with its own distance reader, and reads the offer only behind its own switch (V13 §3.9 (14) (line 1976); §3.11.6 (line 2360); §5.9 patrols refusals (line 3298)).
- **No invented effects.** A patrol is never drawn lowering danger or holding a quarantine line; a march is never a road hazard; no border or patrol slows a plague (V13 §5.8 (4) (line 3167); §5.9 (line 3298); lens-implications.md §2.2 (line 193)).
- **Reach is bounded but exponential; unclaimed land stays wild.** Reach grows with tier, size and relative prosperity, fades with travel time, and is capped per tier (J377 T2 (line 436); V13 §3.9 (7) (line 1844)).
- **Prosperity counted once, relative to the realm's median.** The rung enters as its distance from the realm's held median, clamped to plus or minus 3, and is left out of the force band (J384 (1), (4) (line 443); V13 §3.9 (7) (line 1844)).
- **Culture and water as the host has them.** Culture resists reach only by the host's plain-form culture distance; water carries reach only along the host's lit lanes, to at most a coastal foothold (J382 (line 441); J384 (2) (line 443); V13 §3.9 (6) (line 1827)).
- **Vassal chains in two passes; satellites never counted twice.** A vassal joined to its overlord's land is continuous and gets a capped bonus, else it is an exclave; a satellite is a claim point, never a chain member, never in its parent's size (V13 §3.9 (9) (line 1879), (12) (line 1934); J384 (11) (line 443)).
- **Occupation is an overlay, never ownership.** It hatches land at the host's rung and changes no holder until the host's vassalized rung; a razing transfers nothing (V13 §3.9 (12) (line 1934); docs/AMENDMENTS-2026-10-17m.md item 714 evidence (line 1171)).

- **The host reads the sway only between advances, and its live reads win.** Any host read is fetched at the start of an advance, fixed for it, at most one advance stale; the DM's sway locks never feed what is offered (V13 §3.9 (13) (line 1951); §3.11.6 (line 2382); §1b.3 `setSwayLock` (line 546)).
- **Honesty, banner and audience.** Each lens states its regime from what the host sends; an off gate shows as "still", never hidden or extrapolated; every lens opens DM-only, and a covert power's patrols fail closed (V13 §5.8 (2), (3), (7) (lines 3146, 3153, 3192); §5.9 (lines 3260, 3287); J384 (13) (line 443)).
- **No budget, cost vector or power key is asked of the host.** The sway's settings are the map's own; the host sends only its facts (V13 §3.9 (5) (line 1821); AMENDMENTS-2026-10-17m.md item 714 (line 968)).
- **One word, one meaning.** "Fighting", never "embattled"; a third realm in a cell is "outranked", never "claimant" (the host's word for an heir); "gate" is never written bare (V13 §3.9 (8) (line 1859); naming rule (line 2458); §5.8 (6) (line 3183)).
- **Patrols are per road stretch; water has none.** The cadence is set per road stretch and power, its inputs are garrisons, hostility and the road grade, and the sway crosses water only along the host's lit lanes to a coastal lodgement (V13 §3.9 (14) (line 1976), (6) (line 1827)); no sea patrol exists in the design, and the only water "watch" is the host's own naval blockade (W11, host-owned; research/sf/fp-physical-integration.md §2.4 (line 90)). The owner's directive of 2026-08-01 that water routes form and fade organically exactly as land routes do, one calibration for all movers (fp-physical-integration.md §6 row I20 (line 324)), means H12's route changes and grades per window must include the lived state of water lanes as well as roads; a host developer should not expect the patrol layer to cover them.

What the map reads from the simulation

Host paths below are as research/sf/lens-implications.md reports them (its `wp/` is `src/domain/worldPulse/`, its `sp/` is `src/domain/spatial/`, line 22); each file named was confirmed present in the host clone. "Must create" means the host has no such record today.

Fact the map needs	Where the host keeps it today, or must create it	How it reaches the map	Source
The list of seats and where each stands.	The host's own settlement records; a satellite, phantom, world site or dead remnant adds nothing.	The port's inbound `bindSeat` and `importSeats` messages (M-W1); with no seats the sway lens is "still".	V13 §3.9 (1) (line 1803); §5.9 sway regimes (line 3260)
The host's cost law (how hard land is to cross) and travel law (foot pace, army pace, weeks per cost unit).	The host's own laws; must be sent as data in the map's schema (H6) with calibration at every canon (H8).	`costLaw` and `travelLaw` messages (M-W1); the vassal reinforcement band comes from the travel law, "never authored here".	V13 §3.9 (2), (9) (lines 1803, 1879); §2.2 `TravelLaw` (line 814); §8.5 H6, H8 (lines 3744, 3746)
Each seat's tier (its rank label).	`settlement.tier` on the save row, on the host's tier order; never re-derived from population.	`PolitySeat.tier` in the polity window (H13); tier changes as dated facts asked in H25.	V13 §3.9 (5) (line 1821); §2.2 (line 827); lens-implications.md §3.2 (line 360), §5 ask 7 (line 519); §8.5 H25 (line 3763)
Each seat's size third (its own people only).	`settlement.population` within the tier's band; satellites' people left out so they are not counted twice.	`PolitySeat.sizeThird` (H13).	V13 §3.9 (7) (line 1844); §2.2 (line 827); J384 (11) (line 443); lens-implications.md §3.2 (line 361)
Each seat's prosperity rung and the week it last changed.	`economicState.prosperity`, the host's seven-rung ladder; no other wealth number. Frozen on RRb.	`PolitySeat.prosperity` with `since` (H13); a band-change receipt asked in H29.	V13 §3.9 (7) (line 1844); §2.2 (line 827); lens-implications.md §3.2 (line 362), §5 ask 6 (line 518); §8.5 H29 (line 3767)
Each seat's strength band (the host's "who is stronger").	`settlementStrength` in `src/domain/worldPulse/relationshipRuleHelpers.js` (line 312), the comparator its drift uses to pick a raider and an overlord; the band and its sending must be created.	`PolitySeat.strengthBand` (H13); optionally the stronger side per hostile pair (H25). Absent: the pair reads level, warned once per window.	V13 §3.9 (8) (line 1859); §2.2 (line 827); J384 (3) (line 443); lens-implications.md §3.2 (line 364), §5 ask 1 (line 513); §8.5 H13, H25 (lines 3751, 3763)
Each seat's force band and whether it has a garrison.	The capacity facets in `src/domain/worldPulse/militaryStrength.js` (manpower, institutions, materiel, logistics, will; economy left out); no garrison or standing band exists today, so it must be created.	`PolitySeat.military.forceBand` and `.garrisonPresent` (H13). Until sent, enforcement is dark and the patrol garrison input is a proxy.	V13 §3.9 (10), (14) (lines 1904, 1976); §2.2 (line 827); J384 (4) (line 443); lens-implications.md §3.2 (line 368), §5 ask 3 (line 515)
Every pair's relationship, whether at war, and since when.	The relationship-state labels its simulation reads (`src/domain/worldPulse/relationshipEvolution.js`); drift is on by default, so hostile pairs exist without the war layer.	`PolityRelation` (H13), sorted by the map into kin, respect, neutral, adversarial; `criminal_network` needs a class.	V13 §3.9 (8) (line 1859); §2.2 (line 836); lens-implications.md §2.1 (line 175), §3.2 (line 363), §5 ask 1 (line 513)
Vassal links, the four subordinating treaty terms, and forced tributes.	Graph `vassal` labels, treaty terms (tribute, compelled alliance, puppet seat, treaty-held occupation), drift's `hostile_forced_tribute` rule.	`PolityLink` rows with their weeks (H13). One host treaty term is drawn three ways on the sway: as a member link (a vassal, a puppet seat, an occupation at its vassalized rung), as a band and a relation (a tributary keeps its own sway in its overlord's band; a compelled ally is "respect"), and as an overlay (an occupation short of vassalage, a treaty-held one included); V13 records the divergence as J-V13.67, so the host developer should expect one term to appear in three places and not read it as three facts (V13 §3.9 (9) (line 1879); docs/AMENDMENTS-2026-10-17m.md item 714 (9) (line 1041)).	V13 §3.9 (9) (line 1879); §2.2 (line 833); lens-implications.md §2.1 (line 179), §3.2 (line 367)

Fact the map needs	Where the host keeps it today, or must create it	How it reaches the map	Source
Whether each vassal is connected to its overlord (a continuity verdict).	The host's reach verdict lives in its dark sovereignty lane; none is lit today.	`PolityLink.continuous`, optional (H13); until sent the map's own border-or-road test decides, labelled "map verdict".	V13 §3.9 (9) (line 1879); §2.2 (line 833); lens-implications.md §3.2 (line 367), §5 ask 5 (line 517)
Occupations: who occupies whom, at what rung, treaty-held or not.	`worldState.occupations`, the ledger `src/domain/worldPulse/occupation.js` keeps; `src/domain/spatial/embattlement.js` only mirrors its rung ladder (lines 115–139) and takes the occupation state as a caller-supplied input (line 189); the two records (the ledger's and the stressor's) resolved to one. Dark without the war layer.	`PolityLink` occupation rows with `occupationRung` and `heldByTreaty` (H13).	V13 §3.9 (12) (line 1934); §2.2 (line 833); lens-implications.md §2.1 (line 177), §3.2 (line 370)
Held sites (forts, watchposts) with holder and garrison band.	Must be created alongside the force bands; the host has no garrison bands today.	`PolitySite` (H13 "held sites"); a patrol's round trip is measured from the nearest garrisoned seat or site.	V13 §3.9 (10), (14) (lines 1904, 1976); §2.2 (line 837); §8.5 H13 (line 3751)
Satellites with parent, tier and fixed site point.	The satellites ledger; a satellite's site becomes a world point under H5. Never on RRb.	`PolityFacts.satellites` (H13); one with no placed point is neither drawn nor projected.	V13 §3.9 (12) (line 1934); §2.2 (line 838); lens-implications.md §2.1 (line 180); §8.5 H5 (line 3743)
Culture distance between seats, in fifths, with the week sampled.	`cultureDistance` in `src/domain/spatial/cultureDistance.js`, its plain form (no trade tie) over the whole vector.	`PolityFacts.culture` (H13); an unsent pair reads 0, warned once. H25 asks what "same governing name" means.	V13 §3.9 (6) (line 1827); §2.2 (line 822); J384 (2) (line 443); lens-implications.md §3.2 (line 365), §5 asks 2, 35 (lines 514, 547)
Whether the water-lane system is lit, and its lanes.	The frozen record's sea-lane slot (`src/domain/spatial/distanceRead.js`), present only on a new or re-canonized canon with two or more ports. Not the "Sea lanes" switch.	`PolityFacts.waterLanes` (H13); lanes rest on H11.	V13 §3.9 (6) (line 1827); §2.2 (line 822); lens-implications.md §3.2 (line 366), §5 ask 17 (line 529); §8.5 H11 (line 3749)
Road watches: closed roads with their closer, and wardens' terms.	Only the DM's "cut trade route" event (`config._cutRoutes`) exists, naming no closer and stopping no caravan; a warden's term is designed with no producer.	`PolityFacts.roadWatch` (H13); the closer named and caravans reading the cut asked in H25.	V13 §3.9 (14) (line 1976); §2.2 (line 822); lens-implications.md §2.2 (lines 191, 192), §5 asks 8, 9 (lines 520, 521); §8.5 H25 (line 3763)
Road grades and route changes per window.	The host's lived grades are dark today; the map's route classes stand in.	Route changes and grades per window (H12); a hidden road gets no patrol and carries no vassal continuity.	V13 §3.9 (9), (14) (lines 1879, 1976); lens-implications.md §2.2 (line 190), §3.2 (line 371); §8.5 H12 (line 3750)
Dated fight outcomes over the march (battle winner and place, clash, lifted siege, raids).	War-layer verdicts; drift's `hostile_raid` (`src/domain/worldPulse/relationshipRulesAdversarial.js`) has aggressor and victim but no outcome and no place.	The events window (H9); a raid's outcome word and place asked in H25.	V13 §3.9 (11) (line 1917); J384 (5) (line 443); lens-implications.md §3.2 (line 369), §5 ask 4 (line 516); §8.5 H9, H25 (lines 3747, 3763)
Dated life events of seats (death, resettlement, razing, transfer, satellite founded or dropped).	The host's committed events.	The events window (H9), each with its week.	V13 §3.9 (12), (13) (lines 1934, 1951); lens-implications.md §3.4 (line 382); §8.5 H9 (line 3747)
The week each fact began; optionally the dated changes inside a long advance.	Links, relations and occupations carry a since-week; seats' rung changes are kept only if the host chooses.	`since` on every row and optional `PolitySeat.changes` (H13); without the log the wait is measured at window resolution and the revision says so.	V13 §3.9 (13) (line 1951); §2.2 (line 827); lens-implications.md §5 ask 34 (line 546)
Lineage claims (one seat's standing claim on another).	The host's lineage lane, dark in every shipped preset.	`PolityFacts.claims` (H13); drawn as a DM-only outline, moves no cell.	V13 §3.9 (12) (line 1934); §2.2 (line 822); §5.9 sway claims row (line 3279)
The DM's realm names (christenings).	The DM's naming in the host.	`PolityFacts.christened` (H13); otherwise a descriptive label, never an invented name.	V13 §3.9 (17) (line 2004); §2.2 (line 822)

Fact the map needs	Where the host keeps it today, or must create it	How it reaches the map	Source
The resolved audience of every fact (DM, public, a party's beliefs).	The host resolves audience per fact; covert links and powers flagged.	Events with audience (H9); every lens fact with audience (H23).	V13 §5.8 (7) (line 3192); §2.2 `PolityLink.covert` (line 833); lens-implications.md §5 ask 32 (line 544); §8.5 H9, H23 (lines 3747, 3761)
The host's resolved regime (birth defaults or preset, canon marker, each lane lit or dark).	The host knows it; it is not exported today, so a lens would otherwise guess from a preset's name.	With the lens window (H23); without it every banner reads "the host's regime not sent".	V13 §5.8 (2) (line 3146); lens-implications.md §5 ask 33 (line 545); §8.5 H23 (line 3761)
Field armies and embattlement phases, as presentation only.	The host's movers and phases.	Presentation messages (H17); drawn through `hostOverlay`, never as a mover of a border.	V13 §3.9 (10), (11) (lines 1904, 1917); §8.5 H17 (line 3755)
A pair's dated trade stance (optional).	The host's trade-stance word; dark until sent.	`PolityRelation.stance` (H13); carried on the crossing-holder reading the map feed offers.	V13 §2.2 (line 836); §3.11.6 `mapFeed` (line 2374)
The host's adoption of a canon (`canonized`).	The host's canonize, at birth for new campaigns (H22) or a re-canonize.	The `canonized` receipt (M-W1); the seat territory changes only at a re-canonize.	V13 §3.9 (4) (line 1815); §8.5 H8, H22 (lines 3746, 3760)

What the map offers back

Derived thing	How it is offered	What the host may do with it	Source
Patrol coverage per road stretch and power.	`patrolOffer(world, revision)` in the port's simulator feed: the stretch covered, its basis (host watch, garrison or debatable) and the nearest garrisoned seat or site; never a time or cadence. Computed without the DM's presentation locks.	Read only behind its own switch, re-priced through its one distance reader, read by outcome rolls only (H16). No host consumer exists today, and none is asked for (the owner's decision, 2026-10-18): the route-wardenship slot is a pair-level commercial reason the host reserves for its own route estate, peacetime roads carry no danger to relieve, and the coverage is sway-derived, outside the canon; the offer stays an optional door the host's own programme may open (H16).	V13 §3.9 (14) (line 1976); §3.11.6 (lines 2360, 2382); J378 (line 437); lens-implications.md §3.2 (line 371), §5 ask 8 (line 520); §8.5 H16, H25 (lines 3754, 3763)
The sway as a start-of-advance overlay.	`mapFeed(world, week, audience)`: the picture at the revision in force, stamped with its revision.	Fetch it at the start of an advance, keep it fixed for that advance, at most one advance stale; the host's live occupation, relationship and closure reads override it (H16, optional).	V13 §3.9 (13) (line 1951); §3.11.6 (line 2374); §8.5 H16 (line 3754)
Who holds each border crossing (a reading, never a gate).	Per crossing in `mapFeed`: holder, status, and the pair's trade stance when sent.	The host's intermediary settlement keeps each gate until the sway passes its 300-year soak; then the holder is offered as a per-gate policy the host adopts gate by gate (H16).	V13 §3.9 (13) (line 1951); §3.11.6 (line 2374); J385 (line 444); §8.5 H16 (line 3754)
The seat territory and its crossings (the canon's travel slot).	`canon(world, request)`: every land cell to its cheapest-reached seat, and the cheapest crossing between neighbours with each side's cost (`TravelEdge`).	Adopt it only by its own re-canonize (H5), or at birth for new campaigns (H22); every sway revision leaves it byte-identical.	V13 §3.9 W8a (1)-(4) (lines 1803-1815); §1b.2 `WorldPort` (line 511); lens-implications.md §3.1 (line 345); §8.5 H5, H22 (lines 3743, 3760)
Contested and fighting marches, hatches and patrol tints for the host's own map surfaces.	`mapFeed`: the sway's borders and hatches (debatable, held by force, occupation, fighting) and the patrol tints, for the audience asked. The overlay word is "fighting", never "embattled".	Show the march without re-deriving it; presentation only.	V13 §3.11.6 (line 2374); §3.9 (11) (line 1917), Output (line 2011); naming rule (line 2458)
A place for a fight that names only a pair (`borderSegment`).	`borderSegment(world, a, b, week)`: the debatable or shared border stretch nearest the two seats' line, with point and status, labelled a map reading.	Use it as a hearing card's "where" (H16), lit only on the host owner's word; it lets a verdict land on a stretch and move that stretch's hold band.	V13 §3.9 (11) (line 1917); §3.11.6 (line 2389); §1b.2 (line 519); §8.5 H16 (line 3754)
Border-post suggestions (ghost pins).	At each revision, every road crossing of a realm border or contested band yields a suggested post (bridge fort, pass fort, watchpost, toll house) as a ghost pin the DM may accept.	Nothing unaccepted is kept; an accepted post projects nothing until the host names its holder and garrison.	V13 §3.9 (16) (line 1999); lens-implications.md §2.1 (line 183)
Exports of patrols and sway.	A DM export of patrol stretches with power and cadence (GeoJSON lines), and realms as shapes with status, allegiance and revision (M-W3).	Use in outside tools; derived on load from the revision.	V13 §1b.4 (line 568)

The host changes these lenses depend on

H-row	What the host changes, in plain words	Milestone	Source
H2	Measure its mover bands from the realm's own spectrum, the reinforcement band first; the vassal chain's reach is that band, never the map's.	Now	V13 §8.5 (line 3740); §3.9 (9) (line 1879)
H4	Put the world port's messages in the host client, with no port call inside a tick; every input and offer rides it.	With M-W1	V13 §8.5 (line 3742)
H5 and H22	Build its frozen record over the map's canon, a satellite's site as a world point (H5); canonize new campaigns at birth (H22, a product ruling, not an offer).	H5 with M-W1; H22 with M-W2, new campaigns only	V13 §8.5 (lines 3743, 3760); J384 (6) (line 443)
H6 and H8	Send its cost law in the map's schema (H6) and its travel law with calibration at every canon (H8).	With M-W1	V13 §8.5 (lines 3744, 3746)

H-row	What the host changes, in plain words	Milestone	Source
H9	Send the events window after every advance, undo and resume, with each event's resolved audience.	With M-W1; waits on H4	V13 §8.5 (line 3747)
H11	Lay its water lanes over the canon's candidates under its own law.	With M-W1 to M-W2	V13 §8.5 (line 3749)
H12	Lay its road network at birth over the canon; send route changes and grades per window.	With M-W2	V13 §8.5 (line 3750)
H13	Send the polity facts window: the whole input of the sway and of patrols (every row of the first table marked H13).	With M-W2; waits on H4, H9	V13 §8.5 (line 3751); lens-implications.md §5 ask 1 (line 513)
H16	The optional reads: patrol coverage, crossing holders, the sway overlay, the border segment. The only door through which patrols or the sway can ever touch the simulation. The soak that unlocks the per-gate offer is the map's own test T-V13-41 (d), 300 simulated years under the B-POLITY corpus with the polity cells measured; it cannot run before M-W2 and its measurement, so the gate offer is the last of H16's parts (V13 §11 (line 4317); §12 (line 4355)).	Each optional; waits on H12, H13	V13 §8.5 (line 3754)
H17	Presentation messages: movers and embattlement phases projected for the audience.	With M-W2; waits on H4	V13 §8.5 (line 3755)
H23	The lens window with the host's resolved regime and each row's audience.	With M-W5's first step	V13 §8.5 (line 3761)
H25	The sway and patrol asks: strength order; a raid's outcome word and place; dated tier changes; a host meaning for patrol coverage; a cut road's closer named; the sea-lanes switch copy; the "same governing name" term. The patrol-coverage meaning is not asked (the owner's decision, 2026-10-18): it is an open question to the host's own programme, tied to its route estate and OD-FC-2.	With M-W5's sway and patrols step; waits on H13, H16, H23	V13 §8.5 (line 3763); §5.9 patrols table (line 3294)

Until H25 lands, the closed-road legend reads "cut by the DM: the host names no closer and its caravans do not heed it (until H25)" (V13 §5.9 patrols table (line 3294)).

When a fact is dark in the default preset

On RRb the sway honestly shows reach by tier, size and wealth, split by hostility and resisted by culture (lens-implications.md §3.3 (line 373)). Military enforcement, occupation and fight verdicts are not yet host facts there (lens-implications.md §3.3 (line 373)). A dark fact never stops the picture: the sway runs "civil", patrols run "derived", and each banner says what the host has not sent (V13 §5.9 (lines 3260, 3287); §5.8 (3) (line 3153)).

Encounter maps and world sites: the FP-related items

FP is the host's family of settlement-simulation programs: the parts of SettlementForge that move trade, people, armies, news and danger across the realm week by week (`research/sf/fp-physical-integration.md` §1, line 13). Encounter maps (a framed, gridded crop of the map where a fight or a scene plays out) and world sites (a ruin, fort, camp or crypt standing outside a settlement) take four things from FP: where things are, how far they are, how dangerous the road is, and what the players have been told (`docs/volumes/V14-sites.md` §1b.5, lines 455–466; §2.3, line 770).

The research ledger (the row-by-row list of every physical fact the simulation reads) counts 80 rows: 30 supplied by the map, 17 partly, 7 missing, 26 where the host owes the map nothing (`research/sf/fp-physical-integration.md` §1, assembler's note, line 25). A dated note of 2026-10-17 moves row A1 (seat territory) from supplied to partly, so the working count is 29, 18, 7 and 26 (`fp-physical-integration.md` §8, lines 458–468). The ledger's own summary paragraph still says 62 rows; the assembler's note explains the drafts did not match and keeps both for review (line 25).

As of 2026-10-18 the owner has answered yes to V13 §11 Q4, so the world canon (the map's one fixed record of the world) is outside door 1, and H5 (the host's own build of its territory record over the map's) is lit. Several rows below lean on H5 and are marked where they do. The headings sort each fact by what the host has today, not by the ledger's verdict, so a "missing" ledger row the host already holds appears under "supplies"; the sorting rule is stated at the end of this section.

The host supplies today

Each row is a fact the host already holds or derives, with where it lives and the path by which the map reads it. "Ledger I/O" words: ONE-SHOT means read once when the world is captured (canonize); TICK means read every simulated week; OUT means a derived output other systems read.

Fact
The list of places the players know. While a SettlementForge world is connected, the party's known places are kept in the host's map record; the map keeps a shared copy, and the host's reveal writes both, so a site the players were told about shows in their view. The one admission rule that decides what a party view may draw, list, coun

Fact

Fact

Fact

Fact
Culture, and culture distance, for naming and for the sway. The world's culture map (used only for names) reads the host's own culture, and the culture term of the sway (the political-reach picture) reads the host's own culture distance; V14 never makes a culture map of its own. Keeps cultural facts host-owned. (ENC-HS-02)

Fact

Fact
How dangerous a region is (embattlement). The host keeps one danger number per settlement, 0 to 1, rising and falling slowly; it latches so it does not flicker. A region's danger is the danger of its roads. Any danger shown on an encounter map, a road or around a patrol is this host number drawn as stored; the map may not compute its

Fact

Fact
The travel-time clock and everyone who moves. The host turns route cost into weeks of travel and runs every mover; the map owes nothing for these rows. A site used as a destination gets its travel time from this host clock whenever the time is shown as a host fact; the map never prices travel itself (law 2). (ENC-HS-04)

Fact

Fact

Fact

Fact

Fact

Fact
Seasons, war, trade and capacity rules. These are host-owned rules or tables needing nothing from the map. Listed so the ledger is total: none may be re-derived by the map for an encounter or a site; where shown, they are drawn as the host stores or derives them (law 10). (ENC-HS-05)

Fact

Fact

Fact

Fact

Fact

Fact

Fact
The kind of ground along each stretch of a main road. The host records eight classes (water, desert, grassland, forest, wetland, tundra, glacier, mountain) per stretch so a DM can ask "why this road?". The map's world port (the one doorway through which the host reads map facts) gives each route a line, a total cost and a reason, but no

Fact

Fact
Several candidate routes per pair, re-scored when a carrier leaves. The host picks among a few routes by the danger the sender believes in, then rolls the outcome against the truth. The map's port returns one route per pair. An encounter on a road is only placeable on a road the host chose (an inference from the row). (ENC-HS-07)

Fact

Fact
Gates: the territories a route passes through. For each route from A to C the host notes which settlement B's territory it crosses; B is a "gate" that can toll, forbid or confiscate. The port as first read did not emit gates; a dated note says gates are now answered by the canon's crossings, which the map builds into the host's travel slot. "Holc

Fact

Fact
Sea-lane capacity, recorded but not enforced. The host records that a sea lane carries ten times a land route, and does nothing with it; the map's routes carry no capacity at all. Context only for encounters at sea or on rivers. (ENC-HS-09)

Fact

The host partly supplies

Each row is a fact where one side holds half. "What is there" names the half that exists; "what is missing" names the half still owed, and by whom.

Fact	What is there	What is missing	Source
A settlement founded on top of a world site. (ENC-HP-01)	The host makes the new settlement record; the map's world placement decides whether it lies over a free-standing site and folds the site in (adoption). Host record through V13's placement steps P1–P5 and V08's creation; read by G8 (e) adoption (V14 §3.9 (e)), raised by the application on V13's `placed` outcome.	Without the host's founding no site is adopted automatically; adoption is automatic only for a settlement generated from the world's own charter. When: M-SITE-F (V14 §8 SE6).	`docs/volumes/V14-sites.md` §1b.5 (line 465); §3.9 (line 1903)
Recent events near a point in the world: a battle, a plague, a fire. (ENC-HP-02)	The host has dated events; they reach the map's world record as stored facts through the port's `events(world, events)`. An encounter opened on the world map reads what happened nearby, through V13's world event index (`WorldEvent` via `worldFactsAt(world, wCm, sinceWeek)`, V13 §2.2, §2.4; SW-07), read by E1's `world` context.	The host does not yet hand them over by this route, because the world regime is not built. Lets a world-map encounter say "a battle was fought here three weeks ago" as a host fact. When: M-SITE-F, after V13 lands (V14 §8 SE6).	`docs/volumes/V14-sites.md` §1b.5 (line 466); `research/sf/fp-physical-integration.md` §8 (lines 371, 420)
Seat territory: whose area each place lies in, for travel. (ENC-HP-03)	The map builds each settlement's area and the crossings between areas (W8a) into the host's travel slot. Ledger row A1, first read SUPPLIED through `seats(world, seeds)`, re-ruled PARTLY by the dated bracket (amendments n; J-RR.3, J-RR.48).	The host still owes its own build of its frozen record over it: V13 §8.5's H5, now lit. The sway (W8b, the political picture) is never handed to the host as its territory. Patrols "within a territory" and site ownership lean on this (law 11). Ledger I/O: ONE-SHOT at canonize.	`research/sf/fp-physical-integration.md` §2.6 (line 111); §8 (lines 427, 458–468)
Cost of travel between every pair of settlements. (ENC-HP-04)	The map returns one route per pair with its total cost, under the cost rule the host hands in: port `routes(world)` with `costTotal` under `costLaw(world, law)`.	The host wants one whole-number cost for every pair under its own rule (host `distanceMatrix`, DIGEST); no full all-pairs table is returned. Every host travel time, so every "how far is the site" answer, derives from this matrix. Ledger row D1, PARTLY; the surface named is historical per the dated bracket. Ledger I/O: ONE-SHOT at canonize.	`research/sf/fp-physical-integration.md` §2.1 (line 36); §8 (line 377)
Neighbour tiers: near, middle and far neighbours. (ENC-HP-05)	The host ranks neighbours by how many gates lie between them (host `tiers`, DIGEST).	The map does not emit the gates the ranking counts. Context for which settlements count as a site's near neighbours in host terms. Ledger row D10, PARTLY. Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.1 (line 45); §8 (line 386)
The neighbour named at each town exit, and how long the trip is. (ENC-HP-06)	The map supplies the neighbour's name and direction at each road leaving a town (`roads[].toward`).	The travel time ("to Mossgate — three days") is the host's hopWeeks, a STUB on the host today. The same split binds any travel time printed on an encounter or site panel: direction is the map's, time is the host's. Ledger row D16, PARTLY. Ledger I/O: OUT.	`research/sf/fp-physical-integration.md` §2.1 (line 51); §8 (line 392)
The per-cell land cost field: the host's cost for crossing each patch of ground. (ENC-HP-07)	The map takes the host's cost rule (`costLaw(world, law)`) and returns routed costs.	The map does not emit the field itself (host `costField`, DIGEST, a port of FMG's `landCost`; FMG is the Fantasy Map Generator the host used before). Under the world proposal the host stops building one. Any site's reachability in host terms rests on this rule. Ledger row T1, PARTLY. Ledger I/O: ONE-SHOT; changing it is a re-canonize event.	`research/sf/fp-physical-integration.md` §2.2 (line 57); §8 (line 393)
A settlement's terrain word: plains, hills, forest, riverside, coastal, mountain or desert. (ENC-HP-08)	The map gives rich ground facts: biome, land use, geology, relief, water.	The host's seven-word vocabulary is not emitted; either the host re-versions its reading or the port emits the host's words (host `resolveConfig`, CONFIG, "genesis is sacred"). The host's word and the map's ground must agree for sites near settlements (laws 12 and 13). Ledger row T3, PARTLY. Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.2 (line 59); §8 (line 395)
Terrain along an army's route. (ENC-HP-09)	The host slows armies on bad ground (host `armyTransit.js`, DERIVED).	It needs the per-stretch terrain classes the map does not return ("needs T2"). Context for encounters with armies on the move. Ledger row T9, PARTLY. Ledger I/O: TICK.	`research/sf/fp-physical-integration.md` §2.2 (line 65); §8 (line 401)
Slope as a travel cost. (ENC-HP-10)	The map's height model can give slope.	The host's cost rule has no slope term (ABSENT, `spatialCost.js:385`); adding one is a cost-law version change on the host side. Context only. Ledger row E3, PARTLY. Ledger I/O: none today; a `costLawVersion` bump if added.	`research/sf/fp-physical-integration.md` §2.3 (line 74); §8 (line 405)
Whether a settlement is a port. (ENC-HP-11)	The map supplies the geography half (`SettlementMapRecord.port`).	The institution half stays the host's by the owner's rule (host `isPort`, seaLanes). Context for coastal sites and encounters at harbours (law 16). Ledger row W4, PARTLY by design. Ledger I/O: ONE-SHOT, re-derived on founding.	`research/sf/fp-physical-integration.md` §2.4 (line 84); §8 (line 410)

Fact	What is there	What is missing	Source
The first road network and its grades. (ENC-HP-12)	The map gives route classes: highway, road, track, trail, sea, river.	The host's grade ladder keys on its own access setting and must map the classes, keeping its own config ("genesis is sacred"); host `routeNetworkGenesis` is DERIVED and unmounted, OD-FC-18 mounts it at world birth. Road encounters sit on these roads. Ledger row R2, PARTLY. Ledger I/O: ONE-SHOT at world birth.	`research/sf/fp-physical-integration.md` §2.5 (line 98); §8 (line 419)
Roads that rise, decay and become hidden over the campaign. (ENC-HP-13)	The host can read the map's routes (`events(world, events)`).	No way was found for the host to write its lived road grades back; the host's road ledger (17 modules, dark) stays a side record over the map's routes. An encounter on an "overgrown road" depends on a grade only the host knows. Ledger row R3, PARTLY (read supplied; write-back not found). Ledger I/O: TICK, slow.	`research/sf/fp-physical-integration.md` §2.5 (line 99); §8 (line 420)
Roads a user draws between two chosen places. (ENC-HP-14)	Host `roads/userRoutes.js`.	Not in the port list read; V13's edit events were not read in that lane. Context. Ledger row R5, PARTLY (unverified). Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.5 (line 101); §8 (line 422)
"Why this road?" explanations. (ENC-HP-15)	The map gives each route a reason record (`WorldRoute.reason → ReasonRecord`).	No cost broken down by ground type (host `routeReceipts`). Context; parallels V14's per-unit reason records offered in the VTT vendor block (context item 9 below). Ledger row R6, PARTLY. Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.5 (line 102); §8 (line 423)
Being cut off: no land route and no port. (ENC-HP-16)	The map gives landmass ids.	The host derives "no viable land route and no port" (DERIVED), but genesis isolation is a config string today. Context for remote sites. Ledger row A4, PARTLY. Ledger I/O: OUT.	`research/sf/fp-physical-integration.md` §2.6 (line 114); §8 (line 430)
Seasonal slowing of land travel. (ENC-HP-17)	The map supplies the ground and the season phase.	The multiplier table stays the host's (`SEASON_TERRAIN_COST`, CONSTANT). Winter slows, never cuts (law 15). Ledger row S2, PARTLY. Ledger I/O: TICK.	`research/sf/fp-physical-integration.md` §2.7 (line 124); §8 (line 435)
Frozen harbours. (ENC-HP-18)	The map has frozen lakes and temperatures that would let one derive it.	No sea-ice fact on the map; no table row on the host (STUB). Context for coastal sites and encounters in winter. Ledger row S9, PARTLY. Ledger I/O: TICK.	`research/sf/fp-physical-integration.md` §2.7 (line 131); §8 (line 442)
Airship routes. (ENC-HP-19)	Straight-line distances can be derived from settlement positions.	No aerial travel field exists on either side; the host has an institution-name match only (STUB). Context. Ledger row M2, PARTLY. Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.10 (line 157); §8 (line 453)
Land or sea for each leg of a journey. (ENC-HP-20)	Each map route has one class; a mixed journey is a chain of routes.	Host `seaRoads` is dark. Context. Ledger row M3, PARTLY. Ledger I/O: TICK.	`research/sf/fp-physical-integration.md` §2.10 (line 158); §8 (line 454)

The host lacks

Each row is something the host must add before the map can treat the fact as host truth. Until then the map works without it: every site is map-tier (the map's own) until the host record lands (V14 §11 Q2, line 3073).

What the host must add	Why the map needs it	The V14 ask or ledger row	When	Source
A host record for world sites. SettlementForge has none today: whether a site exists, its kind, name, owner or faction, its status and its position. (ENC-HL-01)	It is the only route by which a site becomes a host fact rather than a map invention; the owner question names it a capability across the SettlementForge boundary. Until it lands every site is the map's own and nothing is lost.	V14 §1b.5: `WorldSiteRecord`, carried as a new row of the host's dossier (its generated record) through derivation row DR-V14-1 that V08 adds; read as `WorldSite.host`, fact tier `dossier`. §10.2's V08 row; owner question Q2 (record shape V14 §2.2's `WorldSiteHost` and V08 §2.8's custom-record form).	Q2 recommends yes, with V13's host changes (item 138 (e)), scheduled in SettlementForge's own program; the map reads it from M-SITE-A on.	`docs/volumes/V14-sites.md` §1b.5 (lines 455, 461); §10.2 (line 3059); §11 Q2 (line 3073)
A site's status changes, dated: garrisoned, abandoned, destroyed, cleared or restored, with the week of each. (ENC-HL-02)	Lets the host's campaign, not the map, decide a site's fate; the map then changes the site's drawing. The record's holder and garrison decide whether a fort near a village is its working keep or a ruin it robs, and whether it joins the village's sway.	V14 §1b.5: status and receipt week through DR-V14-1 and V08 §3.6's events, applied by G7 with cause `dossier`; §3.9 (f) (`host.status`, `host.holder`); §10.2 V08 row ("the status words' normalisation").	With the host record (row above); the map side lands across M-SITE-A and M-SITE-F.	`docs/volumes/V14-sites.md` §1b.5 (line 462); §3.9 (line 1930); §10.2 (line 3059)

What the host must add	Why the map needs it	The V14 ask or ledger row	When	Source
A site position the host fixes. (ENC-HL-03)	Host-fixed facts outrank the map's siting rules: the map keeps the site exactly there and never re-sites it.	V14 §1b.5: the record's world position through V13's `Placement`, DR-V14-1, read by G3 as a fixed placement.	With the host record; the map side in M-SITE-F.	`docs/volumes/V14-sites.md` §1b.5 (line 463)
An event when the party reaches a site, and a way to reveal a site to the party. (ENC-HL-04)	Without it the DM marks every reveal and every reach by hand; with it the campaign does. The map turns these into its three states "DM only", "revealed" and "discovered", and lists which parts the party has reached.	V14 §1b.5 (V08 §3.14's party knowledge: reveal; reached → `SecretState`); §2.3 (`SecretState`, `SiteKnowledge.players`, `SiteKnowledge.reached`); §10.2 V08 row; Q2.	With the host record (Q2); the map's reveal path is in M-SITE-A (V14 §8 SE3).	`docs/volumes/V14-sites.md` §1b.5 (line 464); §2.3 (lines 736–739, 756–757); §10.2 (line 3059); §11 Q2 (line 3073)
A site's capacity carried as a derived fact. (ENC-HL-05)	Gives the host a number it could use for a site (a garrison, a destination) without inventing one; it arrives as an offer, not a host truth (law 10). The lines read say no more about it.	V14 §10.2, the V08 row ("a site's derived capacity as a derived fact"); readers V14 §1b.5, §2.3, §3.8.	Not stated in the lines read.	`docs/volumes/V14-sites.md` §10.2 (line 3059)
Real-world scale for distances: kilometres per map unit. The host designed it so weeks of travel match real kilometres, but nothing writes it; every world still runs the old relative distances. (ENC-HL-06)	Until it is written, a site's or encounter's travel time can only be relative: "absent scale = legacy normalized behavior, byte-identical, forever" (law 3). The map's world is in whole centimetres, so the scale is a unit declaration; the host must still convert.	Ledger row D3: `spatialDigest.kmScale` STUB, `creationKmScale` has zero callers (FPQ-37); OD-FC-22; verdict SUPPLIED (host conversion owed): `costUnitsPerMapUnit(costLawVersion)` assumes FMG pixels and must be re-versioned.	Ledger I/O: ONE-SHOT at canonize; the host's creation flow.	`research/sf/fp-physical-integration.md` §1 (line 13); §2.1 (line 38); §8 (line 379)
Height thresholds restated for the map's height model. The host's land, hill and mountain cut-offs are on FMG's 0–100 scale; the map gives heights in centimetres. (ENC-HL-07)	Context: hills are "unknowable" to the host today (row E2), which the map's height model fixes.	Ledger row E1: `LAND_HEIGHT` 20, `MOUNTAIN_HEIGHT` 60, `HILL_HEIGHT` 40 (CONSTANT); verdict SUPPLIED (host re-version owed).	Ledger I/O: ONE-SHOT; a `costLawVersion` change.	`research/sf/fp-physical-integration.md` §2.3 (line 72); §8 (line 403)
Something in the host that reads the wind. The map carries a prevailing wind for each settlement; no wind exists anywhere in the host. (ENC-HL-08)	Context: smoke, fires and fog in an encounter crop are the map's own drawing; a host wind would matter only if the host wants downwind siting.	Ledger row S7: ABSENT in host; map `charter.climate.windCdeg`, `windStrength`; verdict SUPPLIED (host consumer owed).	Ledger I/O: OUT.	`research/sf/fp-physical-integration.md` §2.7 (line 129); §8 (line 440)
Climate bands restated for the map's climate record. The map gives temperature and rainfall per settlement. (ENC-HL-09)	Context: the host must re-version its climate-band reading, or the port emits the host's words.	Ledger row S8: `CLIMATE_BAND_VERSION` 1 under the virtual `climateTruthEnabled` flag; verdict SUPPLIED (host re-version owed).	Ledger I/O: ONE-SHOT.	`research/sf/fp-physical-integration.md` §2.7 (line 130); §8 (line 441)

What the host must add	Why the map needs it	The V14 ask or ledger row	When	Source
Detours around a dangerous place. The host designed roads that bend around danger and stay bent after it clears, but did not build it; the map offers no detour either. (ENC-HL-10)	An encounter on a worn bypass has no geometry to sit on until one side builds it.	Ledger row R4: STUB (DESIGN_ROUTE_LIFECYCLE §5c); verdict MISSING.	Ledger I/O: OUT.	`research/sf/fp-physical-integration.md` §2.5 (line 100); §8 (line 421)
Line of sight: what can be seen from where. Nothing in the host or its designs computes it; the map offers none at world scale either. (ENC-HL-11)	A host-side watchtower or ambush rule has nothing to read. The encounter crop's VTT export (VTT: a virtual tabletop program such as Foundry) does write sight-blocking lines from the map's own walls, cliffs and items; that is map geometry, and the map must not present it as host truth.	Ledger row L1: ABSENT (zero hits); verdict MISSING; V14 §5.3's `line_of_sight` lists.	Nothing planned on the host side.	`research/sf/fp-physical-integration.md` §2.8 (line 137); §8 (line 443); `docs/volumes/V14-sites.md` §5.3 (line 2535)
A "hidden" road class: paths only smugglers and wanderers use, never armies. The host's road ladder ends in "hidden", designed but dark; the map's classes end at "trail". (ENC-HL-12)	Smuggler and wanderer encounters would sit on these paths.	Ledger row M4: STUB; verdict MISSING.	Ledger I/O: TICK.	`research/sf/fp-physical-integration.md` §2.10 (line 159); §8 (line 455)
The hearing week: the host's designed pause at a one-week advance where every engagement between settlements is docketed with who, upon whom, where and the default verdict, and the DM may leave, tilt or give the verdict. (ENC-HL-13)	H16's "border segment as a hearing card's where" and law 7 below assume it; `borderSegment` has no host consumer until it lands. Every engagement kind needs a place the map can name, so the host's engagement registry is where the map's typed places and border segments will be read.	The fold's requirement 24 (`research/sf/fp-simulator.md` §2.1 row 24, line 474): "dark-identical behind `hearingWeekEnabled`"; the flag appears only in the host's design documents (`docs/DESIGN_FP_GRAMMAR.md:13`, `DESIGN_FP_FAITH.md`, `DESIGN_FP_ARCH_POP.md`), in no code at 81f1cd96.	Not scheduled; nothing is scheduled on the host side yet (law 9).	`research/sf/fp-simulator.md` §2.1 (line 474); `docs/volumes/V13-world.md` §8.5 row H16 (line 3754)

FP laws the map obeys

These are the host's standing rules that bind every encounter map and world site. "Always" means the rule holds in every build stage.

1. **The host's geography is frozen; the campaign only adds side records.** FP reads the world's geography as a record taken once and never changes it. Everything the campaign adds (roads lived, foundings, deaths, danger, occupation, columns, caravans) is kept in side records keyed by the host's settlement ids. A site's status change or an adoption must reach the host as a dated event or a receipted re-capture, never as a silent rewrite. Always. (ENC-FL-01; `research/sf/fp-simulator.md` §0 (line 40); §8 item 2 (line 1081); DESIGN_ROUTE_LIFECYCLE §1 laws 1–2; DESIGN_FP_ARCHITECTURE L10(e))
2. **One distance law only.** All FP travel times come from one rule, hopWeeks and its calibration over the frozen record. New terrain costs or travel modes enter that rule or the one graph as edge types, never a second pathfinder or another distance price. A site panel that shows a travel time as a host fact takes it from the host's rule, not the map's route costs. Always. (ENC-FL-02; `research/sf/fp-simulator.md` §8 item 4 (line 1087); `research/sf/fp-physical-integration.md` §1 (line 13), §2.1 row D2 (line 37))
3. **No absolute week numbers until the real scale is written.** A realm's span is 2–8 march-weeks under the per-world normalisation, so a fixed number of weeks in a rule is a trap. Until the real-distance scale is supplied, every world runs the relative reading, unchanged. The map must not label a site "two weeks away" in absolute terms on a world without the scale. Holds until the host's creation flow writes `kmScale` (lacks row 6). (ENC-FL-03; `research/sf/fp-physical-integration.md` §1 (line 13); §2.1 row D3 (line 38))
4. **The contract is the frozen record plus its scale.** FP reads positions, adjacency and gates, a whole-number cost table and, when supplied, the scale; it never writes them. Everything the map's world gives the host arrives at that one capture point, canonisation. Note: the simulator note says the map must export its distance scale and unit (OD-FC-22); the ledger's row D3 says the map's declared centimetre unit replaces that read. (ENC-FL-04; `research/sf/fp-simulator.md` §8 item 1 (line 1078); `research/sf/fp-physical-integration.md` §8 row D3 (line 379))
5. **Terrain is not part of FP's law.** FP's governing rulings never name terrain; it lives in generation, calamities, the edit-mode fact card and the town-fabric program. A richer terrain model reaches FP only as a new record value, off until present, with a named consumer stage. Encounter-crop water, depth and difficult terrain are map facts the map may not expect FP to read. Always. (ENC-FL-05; `research/sf/fp-simulator.md` §8 item 5 (line 1090))

6. **Show presence, not intention.** The map shows where things are, not where they mean to go; secret movers stay hidden unless the viewer may see them ("fail closed"); one way of drawing each kind of thing. An encounter crop or site shows a host mover only where the host says it is, and hides covert ones from the party. The town map is the town-fabric program, outside FP and the map family's own. Always. (ENC-FL-06; `research/sf/fp-simulator.md` §8 item 6 (line 1094); WY §3b–§3e)
7. **Every engagement has a place.** In the host's designed weekly "hearing" (the fold's requirement 24, behind the flag `hearingWeekEnabled`, which exists only in the host's design documents, `docs/DESIGN_FP_GRAMMAR.md:13`, and in no host code at 81f1cd96), road contacts and besieged towns would become the subjects of the DM's weekly hearings; these are natural places for an encounter crop once the hearing is built (an inference). Route geometry used for forecasts must give the same answer throughout one advance, so the map's road geometry must be stable within one advance. Always. (ENC-FL-07; `research/sf/fp-simulator.md` §8 item 7 (line 1096))
8. **Most of FP is switched off in a default world.** In the default setting only pact formation and a patron's fall run. Belief exists only when the world's geography has been captured and the information mode is not all-knowing, which three of seven presets set. The map's site knowledge (DM only, revealed, discovered) must stand on its own; it cannot count on host belief being on (an inference). Always. (ENC-FL-08; `research/sf/fp-simulator.md` §0 (line 44); `simulationRules.js:789–802`; `beliefMap.js:204–209`)
9. **Nothing is scheduled to be built on the host side yet.** The governing fold schedules no build; the owner's standing order (UI and prose release first; the EC gate "pushed and deployed") still binds. Every "host lacks" row above waits on that order, and the map must work without them, as V14 does. Binds the timing of every host ask. (ENC-FL-09; `research/sf/fp-simulator.md` §0 (line 49); fold lines 24–28)
10. **No invented fact: four kinds of mark, and only four.** A map lens may draw only: facts the host stores, drawn as stored; facts recomputed by the host's own exported function on the same snapshot, labelled so, never a re-implementation; map geometry used only to place a host fact (a territory fill, a "likely road", a site's centre), labelled when it is a guess; and the two owner-ruled map derivations, the sway and the patrol cadence, drawn as the map's own and reaching the host only as offers. Everything else is refused: quantities the host discards, fields interpolated from points, motion without a host clock, effects the host does not apply ("patrols lowering danger", "borders stopping plague") and meanings the host does not compute. This binds every number or label an encounter map or site card shows as host truth. Always. (ENC-FL-10; `research/sf/lens-implications.md` §4.3 (lines 437–449); FILE-M item 714 (13)–(14); T1–T3, J382)
11. **Patrols within a territory: the map's own, offered, never host truth.** The patrol cadence (how often a patrol passes a spot) is one of only two map-made pictures the owner allowed; it is drawn as the map's projection and reaches the host only as an offer it may take. The host does not lower danger for patrols, so the map may not show that effect; danger is the host's embattlement (supplies row 3). The territory the host uses is the seat territory (W8a); the sway (W8b) is never handed to the host as its territory. So for the owner's question about patrols: the map supplies the patrol picture and the territory geometry, the host supplies danger, and nothing joins them unless the host consumes the offer; the owner decided on 2026-10-18 that nothing joins them for now — patrols stay map-only and the host is not asked to consume the offer. Always; the territory geometry with V13's seat territory. (ENC-FL-11; `research/sf/lens-implications.md` §4.3 class 4 and the refused "patrols lowering danger" (lines 445–449); `research/sf/fp-physical-integration.md` §8 dated bracket (lines 458–468; J-RR.3, J-RR.48), §2.6 row A6 (line 116))
12. **A settlement's generated facts are sacred.** The same seed gives the same settlement; re-terraining happens only as an explicit, labelled regeneration. For new campaigns terrain inputs may come from the world; for existing ones the world is fitted to the frozen config. Adopting a site into an existing settlement must never alter the host's generated settlement; V14 obeys this, since a settlement made before its placement changes only by the DM's word. Always. (ENC-FL-12; `research/sf/fp-physical-integration.md` §2.2 row T3 (line 59; J-D1, proposal §6.6), §2.5 row R2 (line 98; route lifecycle Law 2 "GENESIS IS SACRED"); `docs/volumes/V14-sites.md` §3.9 (line 1906))
13. **The dossier is the only fixed truth.** The settlement's dossier (its generated record) cannot be contradicted by the map: a placement that contradicts it is refused, naming the fact and the nearest lawful spot. A site placed in or near a settlement must agree with that settlement's dossier. ONE-SHOT at placement. (ENC-FL-13; `research/sf/fp-physical-integration.md` §2.2 row T4 (line 60); §8 (line 396); V13's `dossier` coherence term at weight ∞, `refuse.charter.dossier`)
14. **Named people never jump across the map.** No path may move a named person between settlements instantly; every leg takes at least one week and the road is a real place along the way. An encounter may not place a named host person who the host says is elsewhere or on the road (an inference). Always. (ENC-FL-14; `research/sf/fp-physical-integration.md` §2.1 row D8 (line 43); realm directives amendment M, "IMPLEMENTATION PROHIBITION")
15. **Weather slows travel, never cuts it.** Winter and storm season make travel slower, capped, never impossible. Host travel to a site is never closed by season; V14's own "not discoverable" (every entrance flooded deep or bricked) is the map's physics of the site, not a host travel rule (an inference). Always. (ENC-FL-15; `research/sf/fp-physical-integration.md` §2.7 row S2 (line 124; "WINTER TRAVEL — SLOW, NOT SEVER", cap `SLOW_NOT_SEVER_MAX` 6), §2.4 row W7 (line 87))
16. **A port needs both geography and institutions.** The map may say a place could be a port; only the host's institutions make it one. A harbour encounter or coastal site may not claim a working port the host has not made. Always. (ENC-FL-16; `research/sf/fp-physical-integration.md` §2.4 row W4 (line 84); the owner's port rule; ports totality, route lifecycle Law 6)

Where the map's design and the host's own fabric charter part

The host's settlement-map fabric charter (its `DESIGN_FP_ARCH_MF.md`, ruled out of its simulation program as a map-family document, OD-FC-38) is evidence for this design, not its decision rule; V13 keeps a crosswalk row per law (`docs/volumes/V13-world.md` §10 BCR-V13-32, lines 4179–4199). Four rows are conflicts ruled against the charter: the forced-fact law (J-V13.75), the water mode (BCR-V13-33: a frame whose dossier is not riverside is never bisected), relief (BCR-V13-34) and reroll (BCR-V13-35: every generate and reroll mints entropy, recorded); two are owed (terraforming and visible work; the wall trace). Three facts in the charter's own text are the host's to settle: no typed founding record exists (`settlementReason` is prose; MF-P1), so the "planned geometry" morphology band is an authored override, never a derivation (`research/sf/fp-physical-integration.md` §3.3, lines 207–213); the fabric draws in a y-down frame while the world is +Y north, and the flip is owed by the first wave that publishes a world-framed artifact (ODQ §303.6; fp-physical-integration.md §6 row I35, line 339), the same adapter as H6's; and the realm-coherence law of the host's older settlement-map design ("the town site is a zoom-in of its realm position") is the one host sentence the charter closes (fp-physical-integration.md §3.4, last bullet, lines 225–228).

Map-side items, for context

These are the map's own, listed so the host developer can see where the host facts above land and what is not being asked of the host.

Item	What it is	Why it is here	When	Source
1. Every other site value is the map's own; gaps are filled by rules, never by asking the host. (ENC-MS-01)	Beyond the host record, everything a site holds is derived by the map. A host change arrives through the change envelope with least disruption; a removed host record leaves the site on the map with its host fields cleared and reported, never torn down. (item 20 (h); §7.7 item 5; item 11; J-V14.25)	Guarantees the map never stalls on the host.	M-SITE-A onward	`docs/volumes/V14-sites.md` §1b.5 (line 468)
2. Three secrecy states and the one reading rule. (ENC-MS-02)	A site anchor (where a site hangs in a town, cave or building) is "DM only", "revealed" or "discovered". While DM only, its entrance is drawn in no party plan, exported in no party scene and listed nowhere below "secret"; the DM toggles it by one edit, and lowering it again is allowed and recorded. (`SiteAnchor`, `SecretState`, `effectiveKnownTo`, C14-18, BCR-V14-2)	The host's reveal and reach (lacks row 4) feed these states; the rest is map-only.	M-SITE-A	`docs/volumes/V14-sites.md` §2.3 (lines 723–750, 762–769)
3. The site's knowledge layers. (ENC-MS-03)	Each site records: it exists; the DM knows it; what the players know (none, revealed, discovered); which parts they reached; whether it can be found at all (false when every entrance is flooded deep or bricked); and its default secrecy. (`SiteKnowledge`)	The "players" and "reached" layers are where host facts would land.	M-SITE-A	`docs/volumes/V14-sites.md` §2.3 (lines 753–759)

Item	What it is	Why it is here	When	Source
4. The one way in to an attached site. (ENC-MS-04)	An attached site (a crypt under a church, a cellar complex) gets exactly one entrance, on the anchor's floor, as a trapdoor or hidden door, descending by a new hidden stair kind V03 does not hold today (a request, designed as granted). Where no entrance fits, the anchor is moved up to 5 m, else reported. The host map gains no geometry. (§3.9 (a), (b); §10.2 V03 row; SW-36; J-V14.10)	None of this is a host fact.	M-SITE-A	`docs/volumes/V14-sites.md` §3.9 (lines 1846–1875); §10.2 (line 3054)
5. Other site forms. (ENC-MS-05)	A surface site becomes a building with the site's rooms as its programme; a site dropped inside a settlement's frame becomes that settlement's content; camps, battlefields and graveyards are marker-like cards on the ground; a free-standing site's world mark is made from its own document and posted to the world. (§3.9 (c), (d), (g), (h))	Context.	M-SITE-A and M-SITE-F	`docs/volumes/V14-sites.md` §3.9 (lines 1877–1954)
6. Adoption: a settlement placed over a site takes it in. (ENC-MS-06)	Automatic only when the settlement was generated from the world's own charter; otherwise the DM is offered adopt, move the site, or cancel. The site's walls, buildings, plots and ways enter the settlement as drawn, locked constraints; its levels become the settlement's; the site document is retired; a dying settlement keeps its sites. The world model never calls V14 directly. (§3.9 (e), (f); C14-16; SW-10; V13 §3.11.5 (1))	The trigger is the host's founding (partly row 1); the rest is map-side.	M-SITE-F	`docs/volumes/V14-sites.md` §3.9 (lines 1903–1933)
7. The encounter crop view. (ENC-MS-07)	An encounter opens as a framed, gridded part of the same map, the rest dimmed, with a switch between ground, storeys and levels. Small figure marks show how crowded a room is by count, never by name; temporary items (barricade, cart, fire, smoke, camp, numbered pin); water by depth from V13; daylight by default with the DM narrating the hour. (§5.3 "The crop view"; J-V14.5; C14-21; E2 (c))	None of the crop's contents are host facts, except where the vendor block or events carry them.	M-ENC, after M1; interiors with V03's P1	`docs/volumes/V14-sites.md` §5.3 (lines 2516–2529)
8. Encounter export to virtual tabletops. (ENC-MS-08)	The crop exports to the Universal VTT format: the grid's origin comes from the place, not the view; sight-blocking lines go in both lists because Foundry's importer reads only one; one image and one file per floor, numbered; plus a Foundry scene file with every floor as a level. The same crop exported twice gives identical files. (§5.3 "The VTT crop scope"; BCR-V14-8; C14-06; §11.1)	Context.	M-ENC	`docs/volumes/V14-sites.md` §5.3 (lines 2531–2569)
9. Optional "why" block in exports (owner question Q4 of V14). (ENC-MS-09)	An optional block, off by default, can carry each part's reason record and the anchor's secrecy state in a namespaced corner of the export, admitted by the viewer's key like any fact. (`ExportOpts.vtt.vendorBlock`, `plat.sites`; J-V14.47; recommended yes, off by default)	The secrecy state shown there must follow the same reading rule as the map.	M-ENC (V14 Q4)	`docs/volumes/V14-sites.md` §5.3 (line 2559); §11 Q4 (line 3075)
10. Order and cost of the work (owner questions Q1, Q3 of V14). (ENC-MS-10)	Recommended: ship the in-town encounter crop right after the viewable town, interiors later; total 25.75–36.25 person-weeks (crop 4.25–5.75, sites 13–18.5, 2.5D view 7.75–11). Milestones: M-ENC, M-SITE-A, M-SITE-F, M-25D.	Sets the "when" for every map-side item above.	Owner decisions Q1, Q3	`docs/volumes/V14-sites.md` §11 (lines 3072, 3074); §8 (line 2876)
11. World-map encounters lean on V13, the world volume. (ENC-MS-11)	An encounter at a bare point in the world asks V13 for a temporary 100 m frame with no recorded placement (10 ms or less); the crop's cache and export stamps follow V13's water changes; V13's placement emits the "placed" outcome that triggers adoption. (§10.2 V13 requests (2), (3), (4), (9), (14))	These are map-to-map requests, not host asks.	M-SITE-F and M-ENC	`docs/volumes/V14-sites.md` §10.2 (line 3053)
12. What the map already gives the host at the settlement, owed nothing back. (ENC-MS-12)	Ledger verdict SUPPLIED, no host debt: D11 seat positions; D12 a real frame for instant worlds; D13 stored placement coordinates; D14 true neighbour bearings at the town's edge; D15 the town's scale bar; T4 the hard dossier-terrain check; T5 ground and resources at a satellite; T6 a siting score with no randomness; T7 the town's ground consistent with its realm position; T8 terrain, water and slope at the realm position; E2 hills; W1 coast and river adjacency and distances; W2 river navigability; W3 sea vs lake, fresh vs salt; W5 sea lanes with cost; W6 river lanes along the course; W8 water closeness in real distance; W9 the facts a town's water relationship needs; R1 road lines; R8 road classes at the frame's edge; R9 road entry points and bearings; A3 territory outlines; A7 the bearing to a named neighbour; S3 a climate band; L2 sun direction.	Listed so the ledger is total. The source says the map supplies these "in richer form than the host's design asked for".	Ledger I/O: mostly ONE-SHOT or OUT	`research/sf/fp-physical-integration.md` §8 (lines 387–391, 396–400, 404, 407–409, 411–412, 414–415, 418, 425–426, 429, 433, 436, 444, 472)

How the ledger rows were sorted into these tables

The ledger's verdicts say what the map gives the host, which is the reverse of this section's headings. So rows where the host owes nothing, and "missing" rows the host already holds today (T2, R7, A2, C2), sit under "supplies". "Partly" rows, plus A1, sit under "partly supplies". Rows where the map supplies the fact but the host still owes a conversion or a reader (D3, E1, S7, S8), and "missing" rows nobody has (R4, L1, M4), sit under "lacks". Plain map-supplies rows sit under map-side context. All 80 rows appear exactly once (`research/sf/fp-physical-integration.md` §2 and §8; the extract at the scratchpad file `extract-encounters.md`).

Open points

- The host's site record carries two names in V14: `WorldSiteRecord` in §1b.5 (line 461) and `WorldSiteHost` in owner question Q2 (line 3073). One name is needed before the host developer builds it.
- The ledger's summary says 62 rows; the assembler's note counts 80 (`fp-physical-integration.md` lines 23–25). The note keeps both for retrovalidation.
- The world scale is described two ways: export the scale and unit (fp-simulator.md line 1078, OD-FC-22) or declare centimetres (fp-physical-integration.md line 379, row D3). One ruling is needed.
- The patrol question has no host patrol rule in any source read. If the owner wants patrols to change danger, that is a new host rule to request, not a map change (law 11). Decided 2026-10-18: not requested now; patrols stay map-only, the door open.
- The capacity of a site "as a derived fact" (lacks row 5) is named once in V14 §10.2 (line 3059) with no further detail in the lines read; its shape and consumer are unknown.
- Both research notes are marked "pending Fable review" by their authors (`research/sf/fp-physical-integration.md` line 3; `research/sf/fp-simulator.md` line 3).
- The owner's answer to V13 §11 Q4 ("1. yes", 2026-10-18, file r item 926) is recorded in V13 itself (§11 Q4, lines 4309–4315) and in the H5 and H22 rows (lines 3743, 3760); the H5-dependent rows are marked accordingly.

Data contracts: receipts, exported functions, snapshots and the no-invented-fact rule

The map draws only what SettlementForge stores, what SettlementForge's own law works out and sends, what the map places on its geometry, or what the owner ruled; nothing is invented (docs/volumes/V13-world.md §5.8 (4), line 3167). A "lens" is one drawn layer over one simulated system, such as plague or territory (V13 §5.8, line 3118). Every mark on a lens is one of four classes, and the map refuses any other:

1. **Host-stored** — a fact the host keeps, drawn exactly as kept (V13 §5.8 (4)(a), line 3167).
2. **Host-derived** — a value the host computes by its own law and sends, labelled "by the host's law" (V13 §5.8 (4)(b), line 3168).
3. **Placement** — the map's geometry used only to put a host fact somewhere, marked `placed` where it is a guess (V13 §5.8 (4)(c), line 3171).
4. **Owner-ruled map derivation** — the territory projection (the "sway") and the patrol cadence, labelled "projection", reaching the host only as offers (V13 §5.8 (4)(d), line 3173).

A world site the DM placed by hand is the map's own fact, class `dmPlaced` (V13 §5.8 (4), line 3174). Anything else is refused with a reason, `refuse.lens.invented` (V13 §5.8 (4), line 3175).

Receipts: what the host sends, and when

A "receipt" here is a dated message the host sends the map after something happened. An "advance" is one turn of the simulation. A "window" is the bundle of facts the host sends after each advance.

Receipt	What it carries	When the host emits it	Who reads it	Source
The lens window, with the host's regime (`HostRegime`)	One bundle of facts per drawn system, plus the rules the campaign actually runs: birth defaults or an applied preset, whether the world is frozen into the host's record (the "canon"), which sea-lane, teleport and seasonal slots the record holds, which lanes are on, any deity seated, any plague live, whether beliefs are recorded, and the intensity. A window with no regime raises `W_LENS_REGIME_ABSENT` and every banner reads "the host's regime not sent".	After every advance, undo and resume (H23).	Every lens's legend banner, first; then each lens's elements.	V13 §5.8 (2) (line 3146); V13 §8.5 row H23 (line 3761)
The events window	The dated events of the turn (an occupation, a conversion, a boom, a raid): committed facts only, never a forecast. Keyed by the advance's one-time number (its "nonce") when `advanceEpochEnabled` is lit, else by the pulse-history sequence. Each row carries its resolved audience: the host's `gm` means DM-only; `hidden` is never sent. The map files them in one shared store, each tagged with the lenses it moves, so one event moves every lens at the same week.	After every advance, undo and resume (H9). The advance epoch's flag `advanceEpochEnabled` is already in the host's gated-key manifest (`src/domain/worldPulse/simulationRules.js:241`) and read in five modules, but it is lit in no preset until OD-FC-31/32 are built (research/sf/fp-simulator.md §1.12, line 419); until then the key is the pulse-history sequence. When the host lights the epoch, THE PROMISE changes shape, "a seed is a starting world, forever; a lived history, once lived, forever" (fp-simulator.md §2.5, lines 590–596): the window key switches from the sequence to the nonce mid-campaign as a declared shift, and the map's plates must tolerate both keys on one campaign.	The shared event store; every lens's time scrub.	V13 §8.5 row H9 (line 3747); V13 §5.8 (1) (line 3131); research/sf/fp-realm-review.md §5 row H9 (line 656)
The travel law and the time-delta receipt	The travel law as data with its calibration: weeks per unit of travel cost, as two whole numbers (anchor weeks over the median nearest-neighbour hop cost); a `canonized` message; a stored receipt of how journey times changed; a migrated campaign's frozen leg weeks.	At every canon the host adopts: the birth canon and every re-canonize, the additive one included, normalized or absolute (H8).	The map's one travel-time read (`TravelLaw`); every journey the map shows.	V13 §8.5 row H8 (line 3746); fp-realm-review.md §5 row H8 (line 655); research/sf/lens-implications.md §4.7 (line 494)
The reframe receipt and the canonize receipt's causes	Readings carried forward only inside one geometry version, with a "reframe" receipt across a change (H6). A geometry shift or new road lay adopted only by a receipted re-canonize, after a dry run listing what would change: membership, edges, reach-band crossings (H15). Optionally the canon's cause list as `causedBy` parents of a canonize receipt (H16).	Across a geometry version change; at each receipted re-canonize.	The canon's version slots; the clerk's trace of why the frozen world changed.	V13 §8.5 rows H6, H15, H16 (lines 3744, 3753, 3754); fp-realm-review.md §5 rows H6, H15, H16 (lines 653, 662, 663)
The dated events that let the territory map revise	From populations: a tier change at its mint, a death, a resettlement, a steading founded or dropped. From prosperity: a rung held 13 weeks, a boom or bust (a boom enters at throughput 3.0 with centrality 0.3 after a 3-tick dwell; a bust below 1.0 or on embattlement — 1.5 is the building-phase floor, not a boom exit, `src/domain/worldPulse/upswingKernel.js:644-653` [Fable-reviewed 2026-10-18]). From alignment and faith: a culture band crossing held 13 weeks. From relations: a class change, a link made or broken, a subjugation accepted, a vassal rebellion. Plague and trade supply nothing territorial.	Each such event, dated at its week, inside the events window.	The sway (the map's territory projection) and the patrols lens, as the only moments a border may move.	lens-implications.md §3.4 (line 382; marked INFERRED by its author); host `src/domain/worldPulse/relationshipRulesCore.js` lines 762–770 for the rebellion
Host-dated fields on rows the host already keeps	Where the host already stamps a week on a fact, that week travels with it: news ticks; condition ids carrying ticks (boom, bust); ledger tick fields (moral drift, an epidemic, a column's departure and arrival, a steading's founding, an occupation's start).	With every lens window and events window.	The time scrub, which reads these before any snapshot of the map's own.	V13 §5.8 (8) (line 3201); lens-implications.md §4.6 (line 479)
A rewind, sent as a window replace	The host's rewind arrives as a replacement of the window; the map then drops every snapshot after the rewound window, pinned ones included.	On every host rewind or undo.	The plate cache (the map's snapshots); the scrub.	V13 §5.8 (8) (line 3214)

Receipt	What it carries	When the host emits it	Who reads it	Source
The `consumed` acknowledgement, and a physical change applied at the next tick	The map never writes into the simulation; it makes offers (a DM-placed site, a `patrolOffer`, a `physicalChange` with its durability). The host answers each with `consumed`, holds world edits while an advance runs or is parked, and applies a `physicalChange` at the next tick's start as a first-class cause carrying the DM's name.	On each offer; the change lands at the next tick head (H15).	The map's offer ledger; the host's cause list.	V13 §8.5 rows H4, H15 (lines 3742, 3753); V13 §5.8 preamble (line 3128); fp-realm-review.md §5 rows H4, H15 (lines 651, 662)
Optional: political changes inside a long advance	At the host's choice, the dated rung changes of tier, prosperity, force band and strength band during a multi-week advance, captured into the window; or a compact per-week delta log.	Optional; sent with that advance's window (H13, H23).	The scrub, which otherwise declares the weeks inside an advance as a gap.	V13 §8.5 rows H13, H23 (lines 3751, 3761); V13 §5.8 (8) (line 3216)

Open: a consumed offer after an undo. When the host undoes an advance whose tick head applied a `physicalChange`, or whose outcome rolls read a `patrolOffer` or the sway overlay, nothing yet says whether the `consumed` acknowledgement is revoked or re-issued. H15 says `consumed` is sent and H9 re-sends the events window after undo (V13 §8.5 rows H9, H15, lines 3747, 3753); the map's offer ledger would otherwise show a change the simulation no longer holds. A day-one question for the host, answerable from its undo path (`src/domain/events/undoEvent.js`).

Values the host computes and sends (the brief's "exported functions")

One correction to the brief's wording. The map does not call host functions; the host runs its own law on the same window and sends the result, labelled "by the host's law" (V13 §5.8 (4)(b), line 3168). The older note's phrase "the host's own exported pure function" (lens-implications.md §4.3, line 438) is superseded on this point. The reason is plain: the host already has three settlement-character derivations and four archetype tables that disagree, so any copy on the map side would disagree with at least one (lens-implications.md §4.3, line 441).

The host's law	What it sends	Why the host computes it, not the map	Source
Culture distance between two settlements (`cultureDistance` with `buildCultureVector`)	How different two settlements' cultures are, in its plain form over the whole culture vector with no trade tie, for every pair within the reinforcement band or at neighbour tier 2 or nearer, banded in fifths with the week each band was sampled; plus whether the sea-lane system is lit and which lanes are adopted.	Culture distance is a term of the territory reading, and faith enters territory only through culture; only the host's own number keeps the map in step with the simulation. Host paths, located by this seat: `src/domain/spatial/cultureDistance.js:165`, `src/domain/worldPulse/migrationKernel.js:154`.	V13 §8.5 row H13 (line 3751); lens-implications.md §4.3 (line 438), §3.5 (line 393)
The settlement-strength comparator (a vassal against its overlord)	The host's strength order, banded in fifths by its own strength banding (the one its envoys read); where it chooses, per adversarial pair the stronger side or a tie. The comparator feeds a vassal's independence pressure, so a vassal rebellion reaches the map as a dated event.	It is the host's law for when a vassal breaks away; moral drift and belief must never be read as strength. Host paths, located by this seat: call at `src/domain/worldPulse/relationshipRulesCore.js:764`, defined at `src/domain/worldPulse/relationshipRuleHelpers.js:312`.	V13 §8.5 rows H13, H25 (lines 3751, 3763); lens-implications.md §3.4 (line 382), §3.5 (line 393)
The force band from the capacity facets	A settlement's force band from manpower, institutions, materiel, logistics and will, the economy facet left out, renormalised on the host's existing five floors; plus the garrison institution bucket for garrison presence. No new banding.	Prosperity must count once in territory; the military economy facet would count it again. The five-floor table appears to be `STRENGTH_BANDS` at `src/domain/display/armyStrength.js:48` (floors 78, 60, 42, 26, 0); the host has several tables of that name, so the host developer should confirm.	V13 §8.5 row H13 (line 3751); fp-realm-review.md §5 row H13 (line 660); lens-implications.md §3.5 (line 393)
The other host-derived values the lenses need	The alignment selector, the faith reach functions, the plague onset and care functions, route choice (`chooseRoute`); in the window as the host chooses: each live plague node's next-hop chance per carrier link, each carrier link's faith pressure per deity, each settlement's population trend sign, ruler-and-people tension and crisis disorder, each grand rite's pilgrim draw, each pair of patrons' quadrant class, each labelled with the law that computed it.	One law, the host's.	V13 §8.5 row H23 (line 3761); lens-implications.md §4.3 (line 438)
Verdicts over the canon, sent view-only	The port verdict over the canon's water access, returned as a view attribute; per-settlement `hostRequires` from the frozen canon; the per-link continuity verdict whenever the sovereignty market is lit.	The map shows the host's verdicts as the host's, never as its own.	V13 §8.5 rows H11, H13 (lines 3749, 3751); fp-realm-review.md §5 rows H11, H13 (lines 658, 660)

History and the time scrub: why the map keeps its own snapshots

Every simulated system keeps its facts per settlement or per pair, and the host keeps no history of most of them: bands, shares, phases, axes, stocks, front states (V13 §5.8 preamble, line 3120; lens-implications.md §4.6, line 479). So the time scrub — dragging the view back through past weeks — would have nothing to show but the present. The map therefore keeps the history it scrubs itself (V13 §5.8 preamble, line 3127).

The scrub reads, in order: host-dated facts; the sway's own revision history; the map's snapshots; and, before the first snapshot, the host's own pulse records (V13 §5.8 (8), line 3201). A "plate" is one snapshot per host advance, written when a lens window arrives: the window's facts for both audiences plus the polity facts' seat rungs, stored as deltas against the previous plate with a whole plate every `plates.keyEvery` (V13 §5.8 (8), lines 3206–3209). Plates keep full resolution for 520 weeks behind the present; older plates thin to one per 13 weeks, the last of each host quarter kept; a plate the DM pins with `pinPlate` is never thinned (V13 §5.8 (8), lines 3210–3213).

The plates live in `Document.lensPlates`, outside the model digest, the records digest, the canon and every export, kept across loads with their own storage budget (`WORLD-14`; V13 §2.7, lines 1292–1294; V13 §5.8 (8), line 3208). They are a cache: dropping them loses only the scrub's past (V13 §5.8 (8), line 3218). The map writes them; the host never reads them.

Before the map's first plate, the scrub reads the host's own per-advance pulse records, sent once when the map first binds (`LensFacts.pulses`): their population deltas and outcome severities, at most 80 advances, labelled with their horizon (V13 §5.8 (8), lines 3204–3206; V13 §8.5 row H23, line 3761). The host's horizons are 80 pulse records with interior weeks dropped, 240 news items and 12 population rows (lens-implications.md §4.6, line 489). The older note adds that population deltas are arrays, and that the host's own track reads

them as a map and probably never fires (lens-implications.md §4.6, line 484).

The territory and patrols lenses scrub from the sway's own revision log, on its own version axis (`swayRevision` with the polity law's version), never the canon's and never a geometry change (V13 §2.7, line 1283; V13 §5.8 (8), line 3203). The scrub says where it is partial: before the first plate, beyond what the pulse records hold, the interior weeks of a multi-week advance the host did not send, a store the host's quota dropped (`W_LENS_PLATE_GAP`) (V13 §5.8 (8), lines 3215–3217). A quantity that decays, such as moral drift, is replayed only by the host's arithmetic as sent, never by a designed curve (V13 §5.8 (8), line 3217; lens-implications.md §4.6, line 492).

Two contracts the window stream still needs. First, catching up. The host's pulse-history ring keeps 80 advances (`src/domain/worldPulse/advanceInterval.js:130`, `MAX_HISTORY=80`), its news 240 items and its population rows 12 (lens-implications.md §4.6, line 489), and the events window is keyed by that sequence while the epoch is dark (H9). A map unbound for more than 80 advances, a DM who never opens the world view, cannot be caught up from the host. The contract must say whether the host keeps a per-map cursor and replays from its ring, or whether the map declares the missed span a gap (`W_LENS_PLATE_GAP`, V13 §5.8 (8), lines 3215–3217); the retention law is the host's requirement 23 (research/sf/fp-simulator.md §2.1 row 23, line 473). Second, order after undo. The host's undo discards a suffix of its pulse history, and a pulse-history entry once written and committed is never rewritten (fp-simulator.md §2.4 item 9, line 584), so sequence numbers are reused after a rewind; the rewind-replace (V13 §5.8 (8), line 3214) must reach the map before the first post-rewind window, or a new plate lands on a key a dropped plate used.

The canon, versions and digests

A "digest" is a fingerprint: a short number that changes if any byte it covers changes. The "canon" is the world as frozen into the host's record.

Every world document names every version it depends on. The version table (`WorldRecipe.snapshot`) holds the generator's output version, the file schema (`schemaVersion`), the table versions W0–W9 read, the charter schema's version (`FrameCharter.version`), the adapter's pinned import version and the export schema's version (V13 §2.7, lines 1273–1277). An older charter is migrated by its own migration chain, never by re-generating the frame (V13 §2.7, line 1279).

The canon's versions are per slot, never the file format alone. The slots are the geometry (the generator and a hash over the physical layers and the member seats' points), the cost law as the host handed it in (version and hash), the travel law's version, the adapter folder's vocabulary tables, and the water and sites slots' input hashes; a slot whose inputs did not change is byte-identical (V13 §2.7, lines 1280–1283). So a change in one part never forces the host to re-read the others.

The host already has three version axes; the canon's slots must map onto them. The host's digest carries a geometry version, a cost-law version and a seasonal-overlay version (`src/store/campaignSpatialCanonize.js:17-21` imports `SPATIAL_GEOMETRY_VERSION`, `COST_LAW_VERSION`, `SEASONAL_OVERLAY_VERSION`) and an `overlayVersion` for its live overlays (`src/domain/spatial/spatialDigest.js:37, 539-541`); the owner's round-16 ruling of 2026-07-11 named frozen geometry, live overlays and three version axes (research/sf/fp-physical-integration.md §6 row I9, line 313). The canon is versioned per slot, and the additive re-canonize steps the version of each slot it moves while the header's `geometryVersion` and the scale stamp stay (V13 §3.11.6, lines 2344–2346; H6 "carried only within one geometry version", line 3744). The sway overlay is stamped on its overlay version axis (H16, line 3754): that is the host's `overlayVersion`. The host's own counter `spatialCanonVersion` bumps on every re-canonize (`campaignSpatialCanonize.js:40, 184-186`) and must bump on the additive one too (J388's version step). A day-one question: does `spatialCanonVersion` become the canon's `geometryVersion`, or stay the host's counter with the slot hashes stored beside it?

The birth canon and the second digest builder. A new world from the map module is canonized at the campaign's birth, the product ruling the host implements, never gated by its premium canonize entitlement; it waits only on H5, H7, H8 and H12, and V13 §11 Q4, which the owner answered on 2026-10-18 (V13 §8.5 preamble, lines 3728–3732; row H22, line 3760). Until it lands, a lens opens in its uncanonized regime (V13 §8.5 preamble, line 3733). H5 builds a second builder of the host's frozen record over the canon, its reader-facing shape unchanged, with a conformance fixture: one fixture, two builders, identical results (V13 §8.5 row H5, line 3743; fp-realm-review.md §5 row H5, line 652).

Who stores the world. On a host-bound world the host stores the world document per campaign beside its settlement map records — snapshot, log and backups under the frames' store pattern and lifecycle (V13 §2.7, lines 1284–1286; V13 §8.5 row H10, line 3748). A settlement map opened without its world falls back to its recorded charter values and refuses placement edits (V13 §2.7, line 1286). Budgets: a continent world at most 24 MB gzipped, a realm 11 MB, a province 3 MB, with at most 10^5 reason records (V13 §2.7, lines 1309–1311 and 1304–1305).

What the world's digest covers, and what it does not. The model digest covers the raster chunks, surfaces, reaches, watersheds, routes, territories, proposals, placements and traces; glyphs and lens plates are caches outside it (V13 §2.7, lines 1289–1294). A placement is in the world's digest and in no settlement map's: moving a settlement on the world map leaves that settlement's own model digest byte-equal, only its records digest changing by the charter event (T-V13-2; V13 §2.7, lines 1294–1301). The same holds for a free-standing site (SW-11), and `surfaceSetDigest(doc, week)` fingerprints the water surfaces in force at a week (SW-25; V13 §2.7, lines 1301–1308).

Reading the host's own stored maps. The map reads two pinned formats of the reference generator's full JSON export: 1.114, the host's own fork, through which each stored snapshot is read once; and 1.152, users' own files, with a `.map` file of 1.152 or later read through `mapFile: true` and an older one refused with the nearest supported version named (V13 §2.8, lines 1325–1333). The host supplies the export through one read-only bridge command for the loaded snapshot (V13 §8.5 row H1, line 3739; fp-realm-review.md §5 row H1, line 648). On the host's own snapshot (`origin: 'host-embedded'`) the file's settlements, states, provinces and cultures are dropped as inputs; the campaign's settlements arrive as the host's `importSeats` request, a legacy settlement matching a placement folded into it, any other kept as a hint (`W_IMPORT_LEGACY_BURG`), each host gate carried as a `HostGateRef`, and no settlement map is generated by the import (V13 §3.15 (3), lines 2744–2763; V13 §2.8 settlement row). Religions, military, economy and the like are dropped as host facts (V13 §2.8, last two rows).

One coordinate transform, recorded for the host. Every imported point uses one rule: cmPerPx = units per pixel × metres per unit × 100; x in cm = round(xPx × cmPerPx); y in cm = round((heightPx − yPx) × cmPerPx), because the file's y runs down and the world's runs north (V13 §3.15 (2), lines 2729–2734). The rule is written into the import report so the host can map its own pixel-keyed facts the same way, and the host keeps one coordinate adapter in both directions (V13 §8.5 row H6, line 3744). The report (`ImportReport`) lists scale, transform, hemisphere, dropped and unmatched items, and placements converted, folded, sited unverified or unresolved, kept on the world as a reason record (V13 §3.15 (4), lines 2769–2773).

Who writes what. The lens record (`LensLayer`) is the map module's own, with no host shape (lens-implications.md §4.1, line 406). The host's side is the world port's messages: outbound `canon`, `derivedFacts`, `mapFeed`, `routes`, `routeFacts`, `patrolOffer`, `pathGeometry`, `borderSegment`, `placeOf`; inbound `bindSeat`, `unbindSeat`, `importSeats`, `events`, `costLaw`, `travelLaw`, `hostRequires`, `canonized`, `consumed`, `genesis`, `network`, `polity`; presentation `clock`, `hostOverlay`, `beliefView`, `audienceKnowledge`; the offer `physicalChange` (fp-realm-review.md §5 row H4, line 651; V13 §8.5 row H4, line 3742). No port call happens inside a tick; answers are baked with their stamps (V13 §8.5 row H4, line 3742). Every port message carries the canon envelope's version vector: the world header, the per-slot versions and input hashes, the cost law as handed in, the causes since the prior canon and the skipped list; V08's host client carries each message so versioned (V13 §3.11.6, lines 2318–2324; docs/ARCHITECTURE.md line 12898; BCR-V13-11). The map tests its side against a mock host (T-V13-37, V13 §8.4, line 3712); the host has no mirror. Recommendation: publish V08 §2.2's message unions first and let the host build a mock map from them, so H4 to H17 can be built and golden-tested, with the dormancy proofs the host-change list asks for, before M-W1 lands rather than after.

Host exports change no simulated byte. Every optional host read lands switched off and data-gated, absent meaning the prior bytes; within H23–H30 the exports change no simulated byte, and the clauses that would change behaviour are offers the host may decline, no lens waiting on any of them (V13 §8.5 preamble, lines 3718–3727). Patrol coverage and the sway, if the host consumes them, reach it only through its own readers: coverage re-priced by its one distance reader and read by outcome rolls only, the sway as an advance-start overlay at most one advance stale, stamped `swayRevision` (V13 §8.5 row H16, line 3754). Never a time or a cadence from the map (V13 §8.5 row H16, line 3754).

The two closing rules

Every lens names the campaign's actual rules before it draws, from the regime the host resolved and never from a preset's name, and an element whose switch is off is drawn as "still" with its reason — listed greyed but openable, never hidden, never an empty map, never extrapolated (V13 §5.8 (2)–(3), lines 3146–3158; lens-implications.md §4.2, line 420). Every lens opens DM-only, a row sent with no resolved audience is folded DM-only, covert links, columns and patrols are omitted rather than greyed, and a player sees only rows the host resolved public or, where belief is recorded, the party's believed rows (V13 §5.8 (7), lines 3192–3200; lens-implications.md §4.5, line 467).

The DM-secret layers: reuse the host's projector, never a second one. The host already projects audience in one place, `src/domain/display/worldSnapshotPublic.js` (it hard-denies `religionStates` at line 96 and strips every covert, seed and prose key by `COVERT_KEY_RE` at line 147), with `publicSafe.js` beside it, under the war program's law 6, premium isolation plus audience projection (`includeCovert`, `includeGroundTruth`; research/sf/fp-simulator.md §2.2, line 497); its channels carry `visibility: 'gm'` and `'hidden'` (`src/domain/region/graph.js:90, 683-718`), and belief reaches anything only through the `infoMode` gate (truth to belief, through one door; fp-simulator.md §2.4 item 3). The per-row audience of the events and lens windows (H9, H23) and the believed view (H17) must come from that one projector, `hidden` never sent, `gm` folded DM-only, the party's rows from the belief map, not from a new allowlist in the port client; the faith read records that published maps already drop the DM-default faith fronts and reject a snapshot carrying a latent pantheon server-side (research/sf/faith-system.md §9.1, lines 1145–1148; §9.4, line 1192). A single writer for audience is the host's own rule.

Everything else worth knowing

The one decision the owner makes next is when the host's track runs and whether replacing the realm map counts as post-launch work (V13 Q2; docs/volumes/V13-world.md §11 Q2 (line 4286)). The one risk that matters most is the host's map itself: it is a built copy of the generator, rebuilt from source that is not in the repository, with no integrity pin, and the host's own upgrade step would break it (research/fmg/notes/sf-host-fmg-integration-audit.md §2.5 (lines 78–80)). Freeze that copy now and do not upgrade it until the port lands.

Three words recur below. The "port" is the one-way channel that carries world facts from the map to the host. The "canon" is the host's frozen record of a world; to "canonize" is to freeze it. A "lens" is a dated map overlay, such as plague spread or patrols.

Sequencing

Each step pairs a map milestone with the host batch that rides on it. The gate is what must be true before the host batch starts.

1. **Now, before the host launches: H1–H3.** On its current copy of the generator, the host stamps its map capture with frame, seed, scale and unit, adds an export command, runs a per-map census of travel-time bands, and adds its calibration rows. Map side, this is the M0 gate. Gate: the brief's bar T (18). The band census also ends the travel-distance deferral (see RK-05). Source: docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (lines 199, 206); docs/volumes/V13-world.md §11 Q2 (line 4286).
2. **With the first world milestone (M-W1): H4–H11.** M-W1 delivers terrain, water, climate, biomes, siting, the charter and the port. The host builds its side of the port and a second builder for its spatial digest, the frozen integer summary of territories, gates and distances. H5 is that builder; the owner's YES on Q4 lit it, so it waits only on M-W1. Gate: M-W1 lands. Source: docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (lines 205–206); docs/volumes/V13-world.md §11 Q4 (line 4309).
3. **Also with M-W1 and M-W2: the kernel's random-source rows, including patrols.** The map engine's core keeps append-only lists of named random sources and stages. The world adds its rows, among them `world.patrol`, which fixes a patrol's timing from the road segment's lineage (a stretch of route, not a border) and the ruling power's host key, never from the week. The "power's host key" is the allegiance root seat's own `hostKey`, the settlement key `bindSeat` writes and `unbindSeat` clears (V13 §2.2, line 746); the map asks no separate power, faction or state key of the host (V13 §3.9 (5), line 1821), so the host's one obligation is that a seat's key stays the same across advances, undo and import. Gate: after M0, as a reviewed change under base request BCR-V13-8. Source: research/sf/fp-realm-review.md §7 (lines 700, 708).
4. **With the second world milestone (M-W2): H12–H17 and H22.** M-W2 delivers routes on the host's road genesis, each seat's territory, the political sway "with patrols", and siting stage 2. The host sends its road genesis, its political facts, the decree law for world edits and the presentation messages. The map draws territory, sway and patrols only from these facts, so W2 has nothing to draw until they arrive. Source: docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (lines 205–206).
5. **Inside M-W2: every new campaign canonized at birth (H22).** This happens whatever the campaign's tier. It comes after H5, H7, H8 and H12; the brief lists H5, H7 and H12, so plan to the longer list. Gate: those four host items. Source: docs/volumes/V13-world.md §11 Q4 (line 4309); docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (line 206); docs/PROGRESS.md row J384 (6) (line 443).
6. **After M-W1, alongside M-W2: open-country encounters and free-standing sites.** Encounters set outside a town and free-standing sites are built on the world's facts in V14 step SE6 (milestone M-SITE-F). Until M-W1 lands, placing a free-standing site just takes a point. Gate: M-W1. Source: docs/volumes/V14-sites.md §8 (lines 2895, 2904).
7. **With the lens milestone (M-W5, after W2): H23–H30.** The host opens its lens window, the dated facts the lenses read, and answers the lenses' asks in order: plague; territory and patrols; trade and populations; religion; prosperity and alignment. Gate: the follow-up design run that carries the thirteen lens rulings into V13 and the host-change list must have happened first (see SQ-13 in the open points). Source: docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (line 206); docs/PROGRESS.md row J384 (8) and NEXT (line 443).
8. **After the third world milestone (M-W3): H18–H20.** M-W3 delivers labels, exports and the adapter that reads generator files. The host then freezes the old capture to a deadline, offers the elective rebase (open question OQ-01), and retires its generator fork. Gate: OQ-01 ruled, and RK-07 settled. Source: docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (line 206); docs/volumes/V13-world.md §11 Q2 (line 4286).
9. **The other site milestones.** The encounter crop (M-ENC) comes after M1; attached sites (M-SITE-A) after the underground milestone M6-U; the 2.5D view (M-25D) after M4. Two parts of site step SE3 wait for M-W5. Source: docs/volumes/V14-sites.md §8 (lines 2875, 2892, 2904).
10. **The fallback that protects the map's dates.** The world generator ships without the host; a stand-in "mock host" (test T-V13-37) plays the host's side of the port. The host track sets the timing of host features, not of the map. Source: docs/volumes/V13-world.md §8.4 (line 3712).
11. **Smaller base changes, all after M0.** The realm review also asks for forced gate picks, a water mode, frame growth, a skeleton term and a relief default (BCR-V13-30 to -36). None touches the kernel or blocks the host. Source: research/sf/fp-realm-review.md §7 (line 729).
12. **The lighting-order trap.** The host lights its own flags in dependency order as separate lighting units, only when a layer's instrument reads satisfied; a flag lit out of order is an invalid configuration its convergence walker rejects, and flag-minting commits serialize across the host's volumes. So a host-change row that mints a flag cannot be built in the same worktree at the same time as one of the host's own flag waves. H12's genesis mount at world birth comes before the host's `routeLifecycleEnabled` and

`demographicsEnabled` (its ruling OD-FC-18), and H22's birth canon lights every canon-gated lane at once on the host's birth defaults, a behaviour switch the host's population and plague reads say to record with the owner. Source: research/sf/fp-simulator.md §3.3 (lines 688–706), §3.1 CQ5 (line 660), §3.4 item 7 (line 730); research/sf/fp-physical-integration.md §6 row I41 (line 345).

Risks on the host side

Risk	What it would break	The spike or fallback	Source
The host's map copy is rebuilt from source that is not in the repository, is not hash-pinned, and its main script sits outside every check. The host runbook's own upgrade step would drop a rename that 28 call sites depend on.	Any upgrade or unnoticed change to the fork breaks the map or silently changes what the canon reads.	No spike. Freeze the fork and do not upgrade it until H20 retires it.	research/fmg/notes/sf-host-fmg-integration-audit.md §2.5 (lines 78–80)
The host's schedule slips past M-W1.	Campaigns get no host seats on a native world until the host catches up.	The host's side of the port is mocked (T-V13-37) and the standalone world ships without host seats.	docs/volumes/V13-world.md §8.4 (line 3712)
The band census (H2) slips.	The owner's 2026-08-05 directive (absolute distance for scaled worlds) stays unmet; native worlds keep the host's normalized travel arm. Extending the deferral needs the owner's word.	None; it ends only when H2 lands.	docs/volumes/V13-world.md §12 (line 4371)

Risk	What it would break	The spike or fallback	Source
Retiring the fork strands old map files. The host's snapshots are version 1.114.2; V13's adapter reads `.map` files of 1.152 or later and older ones only through the fork's open-and-resave, while the brief says 1.114 and 1.152.	Old campaigns' worlds could become unreadable after H20.	Confirm which reader opens 1.114.2 snapshots before H20; see open point 2.	docs/volumes/V13-world.md §12 parity table (line 4400); docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (line 205); research/fmg/notes/sf-host-fmg-integration-audit.md §2.1 (line 20), §7.4 (line 270)
The importer's matching rules (T-V13-26) are tested only on a made-up file.	Importing existing campaigns' worlds on request could fail on real files.	A real generator file "is the owner's to supply, never captured by this lane".	docs/volumes/V13-world.md §12 (lines 4359–4360)

Risk	What it would break	The spike or fallback	Source
Freezing the canon is too slow at continent size.	The canon misses its 2 s time budget (WORLD-12).	Spike SP-W8 over five seeds; fallback: crossings refined only for the settlement pairs the host asks for, and the water slot's K cut to the host's minimum. The host must then name those pairs and that minimum.	docs/volumes/V13-world.md §8.4 (line 3713)
Export, import and export again loses keys other generator tools read.	The host's export command and third-party tools misread files.	Spike SP-W4 must be byte-identical on shared keys; fallback: the schema gains the missing keys, and a key with no world fact exports its documented default.	docs/volumes/V13-world.md §8.4 (line 3709)

Risk	What it would break	The spike or fallback	Source
No host record for world sites.	The simulation never sees a dungeon or fort, so it cannot react to the party reaching one. Separately, V13's siting score may lack two terms sites need.	Every site stays "map-tier" (exists only in the map, losing nothing) until OQ-05 lands; the two terms fall back to 1 minus overlook and one approach from the route count.	docs/volumes/V14-sites.md §8 Risks (5), (8) (line 2928); docs/volumes/V14-sites.md §11 Q2 (line 3073)
Instant worlds may never have canonized. They place settlements in a nominal 1000×600 box not known to match the generator's frame, and the host refuses to invent a transform.	Such a world cannot canonize until one placement is moved through the map frame.	H1's frame stamp and native coordinates remove this by M-W1. The host should confirm whether any instant world has ever canonized.	research/fmg/notes/sf-host-fmg-integration-audit.md §4.6 (line 174), §7.4 (line 267)

Risk	What it would break	The spike or fallback	Source
Neighbouring town frames agree only at their edges, not inside, until base request BCR-V13-29 lands. This is the one deviation from an owner rule (item 135 (c)) the design carries.	Two neighbouring towns could disagree in their interiors.	Test T-V13-13 is marked expected-fail until BCR-V13-29 is applied. No host action.	docs/volumes/V13-world.md §12 (line 4364)
Map-side compute budgets at continent size (no host action).	A fallback changes resolution or route detail the host later sees through the canon, but no host contract.	Water flow: SP-W1, 1 800 ms, or the fine cell grows to 1 250 m. Terrain noise: SP-W6, 3 200 ms, or 4 octaves with detail only near towns. Route search: SP-W7, 520 ms, or the lower half of routes routed coarse then refined.	docs/volumes/V13-world.md §8.4 (lines 3704–3706)

Risk	What it would break	The spike or fallback	Source
Other map-side checks (no host action).	None of these changes the host's work.	Live drag: SP-W2, 40 ms per check, or 10° steps refined on release. Siting stage 2 oscillating: SP-W3, or one iteration fixed. Ground water step: SP-W5, 30 ms, or run lazily. A town's edge relief above the world's: the difference is a cost, and the DM takes an offer or lets the world win.	docs/volumes/V13-world.md §8.4 (lines 3707, 3708, 3710, 3711)

Risk	What it would break	The spike or fallback	Source
The world record per campaign is far larger than anything the host stores for a map today. A continent world is at most 24 MB gzipped, a realm 11 MB, a province 3 MB, with up to 100,000 reason records; the host's opaque snapshot is 100 KB to 1 MB and its frozen digest is capped at 400,000 bytes.	H5 keeps the digest's reader-facing shape and cap; the world document under H10 is a separate store with its own quota, backups and lifecycle. The host must size its per-campaign quota and backups before H10, and decide how the public gallery's shared map blobs (`supabase/migrations/045_gallery_maps.sql`, `046`, `088`) carry or omit a world record, the audit's open unknown.	The map's own budget: `plates.storeMbGz` 4 at the realm rung after 300 years (`WORLD-14`).	docs/volumes/V13-world.md §2.7 (lines 1309–1311), §6 (line 1133), §8.5 row H10 (line 3748); research/fmg/notes/sf-host-fmg-integration-audit.md §3.4 (line 130), §4.2 (lines 149–151), §7.4 (line 269)

Risk	What it would break	The spike or fallback	Source
Every row must stay off the host's first screen. The host's canonize body is already a lazy dynamic import kept out of the first-paint entry closure, verified by an entry-closure ratchet test.	The port client, the lens-window assembly and the events window must follow that pattern, every row landing as a lazy leaf under the host's first-paint ratchet.	The lens window per advance is bounded by the host's rings (80 pulses, 240 news, 12 population rows) and the map's `WORLD-14` and `WORLD-15` budgets; the SP-W8 fallback needs the host to name the settlement pairs it wants refined and its water K (`CanonRequest.waterK`).	docs/volumes/V13-world.md §8.5 preamble (line 3721), §2.2 (line 804), §8.4 (line 3713); research/sf/fp-simulator.md §2.2 law 4 (line 494); host `src/store/campaignSpatialCanonize.js:5-12`, `tests/build/vendorPdfLazy.test.js`

Risk	What it would break	The spike or fallback	Source
The port client must live where the host's purity lint reaches. The host bans `Math.random`, `Date.now`, locale reads and unsorted key iteration in its engine paths, with entropy only at the store layer.	A port client written outside the linted folders, or a key minted from the clock, breaks "same seed and inputs give byte-identical output, forever". The map-slice ids the host mints from the clock must never become `hostKey`s; the map never asks the host to recompute `patrolAt`'s phase, so the two hash families never have to agree.	Every row that moves output on an existing campaign is a declared same-seed shift recorded with its cause; phases 0 to 3 move none; H22 is a declared shift for new campaigns only.	research/sf/fp-simulator.md §2.2 law 3 (line 492), §2.5 (lines 601, 607); research/fmg/notes/sf-host-fmg-integration-audit.md §4.1 (line 141); docs/volumes/V13-world.md §8.5 preamble

The "spike" in each row is a timed test run before the milestone; "p95" means the slowest run in twenty.

Open owner questions

All but Q4 stand as the session's recommendation, which the owner can still overrule. Q4 is answered; it appears under the rulings.

Question in plain words	Recommendation	What waits on it	Source
When does the host's track run, and is replacing the realm map part of the map module's post-launch work? (V13 Q2)	Class it as post-launch work. The host launches on its fork with H1–H3 done; H4–H11 with M-W1; H12–H17 and H22 with M-W2; H23–H30 with M-W5; H18–H20 after M-W3. The transition is per campaign, opt-in and data-preserving.	Scheduling every H batch.	docs/volumes/V13-world.md §11 Q2 (line 4286); docs/volumes/INDEX.md §6 (line 222)
May a living campaign be moved onto a world from this module, which changes its future events? (V13 Q1)	Offer it per campaign, as a premium feature, defaulting to keep, with the full forecast difference shown before the DM confirms. Never automatic.	The elective rebase step in H18–H20, after M-W3.	docs/volumes/V13-world.md §11 Q1 (line 4277); docs/volumes/INDEX.md §6 (line 221); research/sf/fp-realm-review.md §6 Q-RR-3 (line 692)
Does a placed settlement's share code carry its own charter, so it replays alone? (V13 Q3)	Embed the charter. The code becomes a few kilobytes instead of one line, but a settlement replays from its own code.	The share-code format at M-W1.	docs/volumes/V13-world.md §11 Q3 (line 4294); docs/volumes/INDEX.md §6 (line 223)

Question in plain words	Recommendation	What waits on it	Source
Does the in-town encounter crop ship after M1 with plain building shells? (V14 Q1)	Yes. A DM gets a gridded, walled crop of the town at once; interiors appear when V03's P1 lands, with no rework.	The encounter crop milestone (M-ENC).	docs/volumes/V14-sites.md §11 Q1 (line 3072); docs/volumes/INDEX.md §6 (line 225)
Should SettlementForge keep a record for world sites: existence, kind, name, owner, status, position, the party reaching it, and the reveal of an anchor? (V14 Q2)	Yes, scheduled with V13's host changes. Its shape is V14 §2.2's `WorldSiteHost` and V08 §2.8's custom-record form. Until then every site is map-tier.	The host side of site steps SE3 and SE6 (M-SITE-A and M-SITE-F).	docs/volumes/V14-sites.md §11 Q2 (line 3073); docs/volumes/INDEX.md §6 (line 226)
Is the sites program's cost accepted: 25.75–36.25 person-weeks over three milestones? (V14 Q3)	Accept. The split is encounter crop 4.25–5.75, sites 13–18.5, the 2.5D view 7.75–11, content 0.75–1.	The start of M-ENC.	docs/volumes/V14-sites.md §11 Q3 (line 3074); docs/volumes/V14-sites.md §8 total row (line 2902)
Does the virtual-tabletop export carry an optional "why" block with each unit's reason and the anchor's knowledge state? (V14 Q4)	Yes, as an optional namespaced block that is off by default (`ExportOpts.vtt.vendorBlock`). Every importer still reads the common fields unchanged.	Site step SE2, the VTT crop scope.	docs/volumes/V14-sites.md §11 Q4 (line 3075); docs/volumes/INDEX.md §6 (line 228)

The questions the host team will ask on day one

Answerable only by the host or the owner; none is settled by the sources read.

1. Which key is a seat's `hostKey`: the settlement's save id, the generator's burg id, or another, and does it survive gallery import and copy? (V13 §2.2, line 746; audit §4.1, line 154.)
2. Where does the port client run, the page or a worker, and does the map package ship inside the host bundle under the first-paint ratchet? (V13 §8.5 preamble, line 3721; host `campaignSpatialCanonize.js:5-12`.)
3. Does `spatialCanonVersion` become the canon's `geometryVersion`, or stay the host's counter beside the slot hashes? (Data contracts, the version axes.)
4. What happens to `consumed` when the consuming advance is undone? (Data contracts, the open note after the receipts.)
5. How is a map caught up after more than 80 advances unbound? (Data contracts, the two window-stream contracts.)
6. Is the lens window a premium surface, and is the holy-site adoption command gated like DM authoring? (The host-change list, the note on H22's gate.)
7. Which reader opens the host's own 1.114.2 snapshots after the fork retires? (Open point 2 below.)
8. Has any instant world ever canonized, and what does `cells.g` mean on older packs? (Audit §7.4, lines 265–271.)
9. What replaces the generated-map gate for a canon from the map module? (The host-change list, the note on H22's gate; trade-OQ1 in the defect list.)
10. Who writes the shared `party` key when both the host's reveal and a DM edit touch it? V14 makes it a mirror of `MapRecord.party` when a host is bound, written by the owner and editor roles, two writers unless the host's reveal is the single writer while bound (V14 §10.1 BCR-V14-19, line 3041; §2.3, lines 771–773).
11. Does the host's week unit line up with the map's: `INTERVAL_WEEKS` and the 4-4-5 calendar on the host (`src/domain/worldPulse/advanceInterval.js:102-106`) against the map's `Week` and the plates' "last of each host quarter" thinning (V13 §5.8 (8), lines 3197–3200)? With a year-long advance the 520-week full-resolution horizon is ten plates.
12. Does the host's `foundry-module/` (present in the clone; unread by the audit, §8, line 296) read the generator's Full JSON, and will it read the map's ecosystem-shaped export unchanged? The wider ecosystem's Foundry importer needs the Full JSON plus a picture (research/fmg/notes/fmg-roadmap-and-ecosystem.md §7, lines 229–258).

Rulings the host work rests on

Each ruling is vetoable by the owner unless marked as the owner's own word.

Ruling in plain words	J-row	Source
World facts may reach the host; a town's own geometry may not. The owner's own word on 2026-10-18, answering V13 Q4 "1. yes". The host's builder over the canon (H5) and the birth canon (H22) are lit.	Owner's word (file r item 926); J388	docs/volumes/V13-world.md §11 Q4 (line 4309); docs/volumes/INDEX.md §6 (lines 188, 224)
Culture slows the sway by the host's own measure, and the sea is cheap carriage only along lit coastal lanes. The owner's own word.	J382; J-RR.49, J-RR.52	research/sf/fp-realm-review.md §6 (line 677)
Prosperity widens a seat's territory relative to its realm's median, clamped to ±3, read only at dated events with a 13-week dwell.	J384 (1)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q1 (line 555)
Culture resists the sway through the host's plain `cultureDistance`, the whole of it, banded with a 13-week dwell.	J384 (2)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q2 (line 557)
Who is stronger is decided by the host's own settlement-strength comparator; the map sets only reach, floor and ceiling. The host must expose that comparator's result.	J384 (3)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q3 (line 559)
Military bands read the host's manpower, institutions, materiel, logistics and will, without the economy facet.	J384 (4)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q4 (line 561)

Ruling in plain words	J-row	Source
A drift raid is not a verdict: it marks the march "fighting" (J384's own word was "embattled"; V13 reserves "embattled" for the host's danger ledger and writes "fighting" for the overlay) and moves no line. Only a host fight verdict moves a march line, in every regime.	J384 (5)	docs/PROGRESS.md row J384 (line 443); docs/volumes/V13-world.md §11 (line 4336)
A native world is canonized at birth for new campaigns, with the route network; existing campaigns are untouched.	J384 (6); OD-FC-18	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q6 (line 563)
Players never see derived alignment; the alignment lens is DM-only.	J384 (7)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q7 (line 566)
Lens build order: plague, territory and patrols, trade and populations, religion, then prosperity and alignment. This fixes the order of the host's lens asks (H23–H30).	J384 (8)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q8 (line 567)
Lens snapshots are a derived cache in the world document, one frame per host advance, kept in full for 520 weeks, then one per 13 weeks; DM-pinned frames never compacted.	J384 (9)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q9 (line 568)
Holy places and hidden cult sites are only DM-placed, tagged with a deity, DM-only by default, and offered to the host for adoption. Never generated.	J384 (10)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q10 (line 570)
Steadings are drawn at their frozen site cells and project a small reach as members of the parent's realm; nothing is counted twice.	J384 (11)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q11 (line 572)
Port and teleport columns the host's overlay hides are shown in the native lens, DM-only, styled by carriage.	J384 (12)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q12 (line 573)
A new campaign's lenses show "still" with a banner and a reason, collapsed in the picker; never hidden, never extrapolated.	J384 (13)	docs/PROGRESS.md row J384 (line 443); research/sf/lens-implications.md §6 Q13 (line 574)
The settlement whose land a path crosses keeps the gate, with its tolls, interception and contraband, until the sway passes its 300-year soak. Then the political holder is offered to the host as a per-gate policy (H16).	J385; J-V13.96	docs/volumes/V13-world.md §11 (line 4317)
The birth canon happens whatever the tier; the premium canonize entitlement never gates it.	J386; J387; J-V13.100	docs/volumes/V13-world.md §11 (line 4326)
A seat added later joins the canon through the host's own additive re-canonize, never premium-gated, covering only the added seats and what they re-derive. Any other geometry-shifting re-canonize stays the premium DM's act.	J388; J-V13.101	docs/volumes/V13-world.md §11 (line 4326)
Native worlds use the host's normalized travel arm until the band census (H2) lands. The owner's 2026-08-05 directive is deferred, bounded and visible.	J385	docs/volumes/V13-world.md §12 (line 4371)
Existing campaigns keep their frozen records forever; their worlds are imported on request and drawn without writing to simulated state.	J-V13.81	docs/volumes/V13-world.md §11 Q1 (line 4277); docs/volumes/INDEX.md §6 (line 221)
Four generator features are deliberately not carried: coastal chaining of short sea legs (built only if the host asks, D-V13.6); island and colony provinces (D-V13.7); GeoJSON and elevation-model import (D-V13.5); `.map` files older than 1.152, read only through the fork.	D-V13.5, D-V13.6, D-V13.7	docs/volumes/V13-world.md §12 parity table (lines 4396, 4398, 4400, 4401)

What the FMG audit found in the host's fork

The audit read the host's copy of the Fantasy Map Generator (FMG) and how SettlementForge drives it. Its file is research/fmg/notes/sf-host-fmg-integration-audit.md; section and line follow each item.

- The host's map is a built copy of FMG version 1.114.2 from April 2026 (upstream commit bb5039d7, host revision `sfdrop16`). It predates the minimap, the fill brush, eroded terrain and the economy layer. §1 (line 10), §2.1 (line 20).
- Inside the host the generator makes geography only; states, towns, religions and provinces are zeroed. So today nothing political comes from the map, and the host's political facts (H12–H17) are the only source. §2.4 (lines 72, 75).
- The host never reads the map file. It stores the file as opaque text (about 100 KB to 1 MB), a seed, and placement rows in the generator's pixel space. Old worlds can only be re-read by the adapter or the fork. §1 (line 10), §3.4 (line 130).
- Today's canon is a one-time copy of nine arrays, frozen into an integer digest, premium-only, for generated maps only, capped at 400,000 bytes, with no cell coordinates kept. H5's second builder replaces exactly this. §3.2 (line 96), §4.2 (lines 149–151).
- That copy leaves out lake-versus-ocean ids, distance to coast, harbour and border cells, river and lake entities, map size and the distance scale. The capture's frame fields come back empty every time, because the generator never sends them. H1's stamps close part of this gap. §3.4 (line 132).
- Because of those gaps the host guesses lake versus ocean by a bounding box, cannot call a lake salt, and routes sea lanes by a height rule. Native water classes and real rivers through the port fix this. §4.2 (line 152).
- No settlement input comes from the map. Terrain, coast, river, trade access and threat are chosen from settings or random pools, and any later disagreement with the map is recorded, never reconciled: "genesis is sacred". This is why the map bends to the dossier. §6 (lines 221, 224).

- The map bundle was rebuilt from modified source that is not in the repository (it exports `Settlements` instead of `Burgs`, with 28 dependent call sites), has no hash pin, and its main script is outside every check. A known upstream lake bug is kept on purpose so the same seed keeps giving the same map. §2.3 (line 37), §2.5 (lines 78–81).
- Terrain edits are under-reported: the "terrain changed" signal fires when a tool opens, not when terrain changes, and double-click edits are not reported at all. Native edits go through the decree law with fingerprinted outputs. §3.2 (line 112), §7.3 (line 261).
- Roads are routed inside the map frame by a path search capped at 25,000 steps, order-dependent, and never saved. The host's own design calls this "in the wrong layer". On native worlds the map computes and saves routes (W7); the host should consume them. §3.2 (line 105), §4.8 (line 180), §7.3 (line 256).
- The default scale constant is 3 per pixel and the host's comment calls it kilometres, but the host's own coherence fold (OD-FC-22, 2026-10-16) records that the generator's default unit is miles, so the default is 3 miles (about 4.83 km) per pixel and the comment in `src/domain/spatial/modeSpeeds.js:70-77` is wrong (research/sf/fp-physical-integration.md §6 row I37, line 341); the datum is set once at first canonize. Mode speeds are foot 150, cart 110, river 280, coastal 450 and sea 900 km per week; without a datum, travel falls back to about one week per hop. This is the normalized arm native worlds ride until H2. §4.2 (line 154), §7.2 (line 251).
- Town and annotation ids are minted from the clock and random numbers; they are interface state and must not leak into the canon or share codes. Four map events have no host listener. The generator's own SVG, PNG and GeoJSON exporters are not connected, and the map frame is mounted without a sandbox, deliberately, by the host's own comment ("Deliberately not sandboxed", `src/components/map/WorldMapStage.jsx:239`), both postMessage directions pinning the opposite origin instead. §4.1 (line 141), §3.3 (line 127), §4.9 (lines 183–184).
- The host's town map has no link to the realm map ("no FMG iframe, no bridge"); it draws its river and coast from the dossier. So door 1 staying closed matches the host's own design. §4.10 (line 189).
- No host document plans to replace the generator; the host calls the fork an open structural risk ("either commit to hard-fork or upstream the bridge hooks"). Two host design records are missing from the clone: `DESIGN_FMG_WEAVE` and the owner decision queue, which lives on a ledger branch. §5 (lines 204–206, 214).
- Six host facts the new world must treat as canon: the frozen digest and its version; placements and the move-lock; the frozen settlement settings; share codes and instant-world plan law 1; the port, river and coast contract; the `kmScale` datum and mode speeds. §7.2 (lines 246–251).
- What the host must send: a per-campaign map charter beside the seed (frame size, datum, which edges are sea, elevation range) and river requirements per settlement. What it must receive: a one-way derived-facts channel returning each settlement's cell, coast, river and port truth, biome, climate band and elevation, plus territory, gates and distances, versioned like the digest. The host must also emit or re-version the generator's biome ids 0–12 and 0–255 rainfall scale, and make the instant-world box the real frame. §7.3 (lines 254–262).
- Eight questions only the host developer can settle: where the modified source lives; what the two missing records say; the real production map size; whether any instant world has ever canonized; whether gallery import of map blobs still works and how blobs migrate; what `cells.g` means on older packs; whether any `.map` was saved under a version other than 1.114.2; where the town-map 3D viewer lives. Several gate H1 and H18–H20. §7.4 (lines 265–271).

Open points for this section

1. H22's prerequisites differ between sources: the brief lists H5, H7 and H12 (docs/OWNER-BRIEF-WORLD-SITES-HYDROLOGY-DRAFT-2026-10-16.md §9 (line 206)); V13's Q4 answer lists H5, H7, H8 and H12 (docs/volumes/V13-world.md §11 Q4 (line 4309)). This draft plans to the longer list.
2. Old map files: the brief's W3 row says the adapter reads 1.114 and 1.152 (line 205); V13's parity row says 1.152 and later, older files only through the fork (line 4400); the host's snapshots are 1.114.2 (audit §2.1, line 20). Which reader opens the host's own snapshots must be settled before H20.
3. research/sf/fp-realm-review.md §6 Q-RR-1 (line 679) and the J384 row text still call Q4 pending, and Q-RR-2 (line 685) gives a host schedule without H22 and H23–H30. Both predate the owner's answer; V13's form is used here.
4. Whether the follow-up design run named in J384's NEXT clause (carrying the lens rulings into V13 and the host-change list) has happened is not stated in the sources read. The host developer should check before scheduling H23–H30.
5. The full patrol asks are in the section "Patrols within a territory" and the encounter-map asks in "Encounter maps and world sites"; this section carries only their sequencing (steps 3, 4 and 7) and the rulings J384 (3), (5) and (8).
6. The audit's eight unknowns (last bullet above) are the host developer's to answer; none can be settled from the map side.